FIRST EDITION

The Financial Logic of Supply Chains

Cash, Credit, and Risk Across Firms

An Introduction to Supply Chain Finance,
from Inventory and Invoices to AI Infrastructure

Fasheng Xu

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Cash, Credit, and Risk Across Firms

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Preface

Picture a supply chain director who has finally been invited to present to the executive committee. The news is good: inventory turns are up, purchase costs are down, forecasts are more accurate, and a new long-term supply contract is in place. The committee listens politely and moves on. Its members think in another vocabulary, one of cash flow, margins, return on capital and credit ratings, and nothing on the slides was translated into it. Robert Trent, who tells a version of this story, draws the lesson in one line: "Everything we do within a supply chain ends up somewhere on a financial statement."[1]

Supply chain managers and finance managers describe the same pallets, invoices and payments in two different languages. This book is written for people who need to speak both. Its subject, supply chain finance, is the place where the two languages meet: the decisions about inventory, payment terms, financing and risk that determine how much cash a supply chain needs, who provides it, and at what price.

1 Trent, R. J. (2016). *Supply Chain Financial Management: Best Practices, Tools, and Applications for Improved Performance*. J. Ross Publishing, chapter 1.

Why this book, and why now

Why should a manager who already knows operations, or already knows finance, spend a semester on the space between them? Four developments make the question pressing.

The first is the amount of cash that supply chains hold. J.P. Morgan's 2024 Working Capital Index, which follows 964 companies in the S&P 1500, estimates that about $707 billion of working capital could be released if every company in its sample moved up one performance quartile within its industry, and that inventory accounts for $353 billion of that amount.[2] That cash is not lost. It sits in warehouses, in unpaid invoices and in payment terms, and the decisions that would release it are made by purchasing, operations, sales and treasury, often without talking to each other.

2 Source: J.P. Morgan, *Working Capital Index 2024* (August 2024), data through 2023.

The second is the tug-of-war over payment terms. Large buyers have learned that paying suppliers later is a cheap way to finance themselves. When a buyer stretches its terms, its suppliers, which are usually smaller and borrow at higher rates, finance the difference. Supply chain finance

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programs promise a better bargain: a bank or a platform lends the buyer's strong credit rating to the supplier, which is paid early at a low rate while the buyer keeps its longer terms. Done well, every party gains. Done badly, the arrangement squeezes suppliers and hides debt.

The third is the end of that opacity. In March 2021 Greensill Capital, then one of the best-known providers of supply chain finance, filed for insolvency after its credit insurance lapsed and Credit Suisse suspended the investment funds that bought its notes. Coming after the collapses of Abengoa in 2016 and Carillion in 2018, the failure convinced standard setters that investors could not see how much supplier financing sat inside accounts payable. Since 2023, companies reporting under U.S. GAAP have disclosed their supplier finance program obligations under the FASB's ASU 2022-04, and since 2024 companies reporting under IFRS have followed amended versions of IAS 7 and IFRS 7. Coca-Cola, for one, reported $1,421 million of confirmed obligations under its program at the end of 2023, with 120-day payment terms for most of its suppliers.[3] Reading such a note, and judging whether a payable has quietly become debt, is now part of every analyst's job.

3 Source: The Coca-Cola Company, Form 10-K for 2023.

The fourth is the infrastructure behind artificial intelligence, a buildout that its chief chip supplier, NVIDIA, describes as the largest in human history. On one consistent series, the five largest spenders (Amazon, Alphabet, Meta, Microsoft and Oracle) spent about $245 billion on capital expenditures in 2024 and about $415 billion in 2025.[4] In the first half of 2026, Amazon, Alphabet and Meta alone spent $228 billion on property and equipment, more than the $220 billion of cash their operations produced. Much of the money flows through structures that keep obligations off the balance sheet. In October 2025, for example, Beignet Investor LLC, a joint venture owned 80 percent by funds affiliated with Blue Owl and 20 percent by Meta, issued $27.3 billion of bonds to finance Meta's Hyperion data center campus in Louisiana. A year later the notes traded near 91 cents on the dollar. Leases, residual value guarantees, GPU-backed securities and vendor financing raise the same questions this book asks of payables finance: who bears the risk, who is paid first, and what does the balance sheet show? Part VI follows that money.

4 Source: Van Nieuwerburgh, *Data Centers: Financing the AI Buildout* (March 2026), from FactSet, Apollo, company releases and Bank of America research. First-half 2026: company Forms 10-Q, six months to June 30, 2026.

These threads are usually taught in different courses, if they are taught at all: working capital in corporate finance, inventory and sourcing in operations, letters of credit in international business, disclosure rules in financial reporting, hedging in risk management. This book pulls them into one sequence organized around the cash conversion cycle and the parties who finance it. It grew out of OPIM 5115 Supply Chain Finance, a course I teach at the University of Connecticut, and it gives students one self-contained text in place of a stack of lecture notes, articles and industry reports.

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Who this book is for

Who needs both languages? The book is written for MBA students and for graduate students in business analytics, finance, supply chain management and financial risk management. It also serves practitioners who work at the seams of the firm: buyers and category managers who negotiate payment terms, treasurers and credit analysts who fund them, and operations managers whose inventory decisions show up as working capital.

I assume one course in financial accounting, one in corporate finance, and fluency with spreadsheets. There is no calculus. Two chapters, Chapter 10 on hedging and Chapter 14 on capacity commitments, use the normal distribution, which Appendix A reviews along with the rest of the financial mathematics the book relies on.

What makes the book different

What should a reader expect on a typical page? Seven features set the book apart.

Real companies, dated facts. Procter & Gamble, Coca-Cola, Walmart, Greensill Capital, Credem, Unilever, Meta, NVIDIA, Micron and CoreWeave appear with facts taken from their filings, from published cases and from industry reports, and every time-sensitive figure carries its date. When a chapter needs a complete data set to compute with, it uses a fictitious running company, such as the Nutmeg Juice Company in Chapter 1 or the neocloud Bayou Compute in Chapter 16, and says so.

Numbers first, then algebra. Every model is worked with concrete numbers before its general formula appears, and every formula is evaluated with the running company's numbers as soon as it is written down. You will see why a result holds before you see it in symbols.

How-to exhibits. Each non-trivial calculation, from the cash conversion cycle to the price of an early payment and the spend at risk of a commodity portfolio, is codified as a numbered procedure that you can reuse at work and on an exam.

Cases in point. Short vignettes show each idea at work in a real company. The six cases used in the course, on David Berman, Credem, Grupo Pão de Açúcar, Procter & Gamble, Greensill and Unilever, are summarized in my own words and cited, never reproduced, and Appendix C offers guides for assigning them.

An integrative simulation chapter. Chapter 11 follows two teams through a simulated turnaround in The Cool Connection, a web-based business

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simulation of a personal care products manufacturer, and shows how the levers of the earlier chapters interact when sales, purchasing, supply chain and finance decide at once.

The frontier. Part V covers blockchain and deep-tier finance, sustainability-linked finance and carbon markets, areas in which practice is moving faster than textbooks.

Financing the AI supply chain. Part VI gives three chapters to the money behind artificial intelligence: the commitments and deposits that secure scarce chips and memory, the bonds, leases and power contracts that pay for data centers, and the GPU-backed loans and vendor guarantees that decide who holds the risk. Every figure in it carries its date, because these numbers change every quarter.

How the book is organized

How do the sixteen chapters fit together? They are grouped into six parts that follow a dollar from the warehouse to the frontier of finance. Two short pieces follow this preface: a guide to the anatomy of a chapter and the book's conventions, and a note on using AI tools for the kind of analysis the book teaches.

Part I, Foundations. Every supply chain decision eventually lands on a financial statement, and Part I builds the vocabulary for reading it. Chapter 1 introduces the flows of goods, money and information, the tug-of-war over payment terms, and a map of supply chain finance that organizes the rest of the book. Chapter 2 measures working capital and the cash conversion cycle and links them to return on capital and to the financial health of suppliers. Chapter 3 shows how inventory is valued on the books and what that valuation tells lenders and investors.

Part II, Financing Inventory and Trade. Goods in a warehouse, an invoice waiting to be paid and a container on the ocean are each cash in disguise. Chapter 4 studies lending against inventory, from borrowing bases to a bank that ages cheese. Chapter 5 treats trade credit, the bank inside the supply chain, and the true cost of a forgone discount. Chapter 6 covers letters of credit, documentary collections, Incoterms and the long shift of world trade toward open account.

Part III, Supply Chain Finance Programs. Here a strong buyer's credit rating is lent to its suppliers, and the question is when that creates value, who captures it, and when it quietly becomes debt. Chapter 7 sets out the standard market definitions of the techniques and a single engine for choosing among them. Chapter 8 examines payables finance, also called reverse factoring, through Procter & Gamble and one of its suppliers. Chapter 9 asks when

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payables become debt, explains the new disclosure rules, and tells the story of Greensill.

Part IV, Risk and Integration. Chapter 10 asks why firms hedge at all and how to measure and manage commodity and disruption risk, with spend at risk as its central tool and Unilever as its case in point. Chapter 11 then runs one company through a multi-quarter simulation and lets every lever in the book move at once.

Part V, The Frontier. Chapter 12 shows how shared ledgers can carry an anchor buyer's credit to the deepest tiers of a supply chain. Chapter 13 prices sustainability into supply chain finance and examines carbon credits and their integrity.

Part VI, Financing the AI Supply Chain. Chapter 14 treats the AI factory as a supply chain and asks how much capacity to commit before demand shows up, and who finances the deposits and take-or-pay contracts that secure it. Chapter 15 follows the money into the buildings, through capital expenditure, bonds, leases, joint ventures and power contracts, and asks which obligations stay off the balance sheet. Chapter 16 turns to the chips: GPU-backed loans and securitizations, vendor financing, and the paths by which a shortfall in AI revenue could travel to lenders and savers. An epilogue draws the threads together into a view of where supply chain finance is heading.

Four appendices support the chapters: financial mathematics (Appendix A), a spreadsheet toolkit (Appendix B), case assignment guides (Appendix C), and answers to the odd-numbered problems (Appendix D).

Using the book in a course

How much can a course cover? The chapters follow the order of a fourteen-session course that meets once a week for three hours, with six case discussions and a simulation that runs from the sixth session to the tenth (Table 1). Chapter 6 is a reference chapter in that course; programs with an international focus can assign it with the fifth or seventh session. Programs with a finance focus can assign Chapter 16 in place of one of the case sessions. The final session is reserved for the examination.

The book also supports shorter formats (Table 2). A seven-week module pairs chapters, keeps five of the six cases, and fits the six simulation rounds into four weeks. A two-day executive workshop concentrates on cash, payables finance, hidden leverage and risk, and uses the simulation chapter as a case rather than playing the game. Whatever the format, assign the chapter before the session and spend class time on the case, the simulation and the judgment calls; the examples and exhibits are designed to teach the arithmetic on their own.

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TABLE 1 A fourteen-session course built on the book. Each session lasts three hours. Simulation rounds are quarters of The Cool Connection; the team write-up is due at session 12.

Session	Topic	Chapters	Case	Simulation
1	Introduction to supply chain finance	1		
2	Supply chain and financial performance; working capital	2		
3	Accounting for inventory and inventory valuation	3	David Berman	
4	Inventory-based lending	4	Credem	
5	Trade credit: theory and practice	5	Grupo Pão de Açúcar	
6	The simulation: introduction and kickoff	11		Kickoff; Round 1
7	Supply chain finance solutions	7		Round 2
8	Payables finance (reverse factoring)	8	Procter & Gamble	Round 3
9	Hidden risks and new accounting rules	9	Greensill	Round 4
10	The simulation: grand finale	11		Rounds 5 and 6; debrief
11	Risk management and financial hedging	10	Unilever	
12	Blockchain and deep-tier finance; sustainable supply chain finance	12, 13		Write-up due
13	The AI supply chain; final remarks	14, 15, Epilogue		
14	Final examination			

Supplements

What comes with the book? Instructors who adopt it receive an Instructor's Solutions Manual with three parts: fully worked solutions to every end-of-chapter problem, each ending in its final answer; teaching guides for the six cases, with discussion plans and model answers; and a debrief guide for The Cool Connection, organized by round and by role. Students find the answers to the odd-numbered problems in Appendix D. The spreadsheets and short scripts that generated the book's charts and verify its numbers are provided as companion code, chapter by chapter. The cases themselves are available from their publishers, and The Cool Connection is licensed separately from its developer, Inchainge.

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TABLE 2 Two alternative paths through the book.

Unit	Chapters	Focus	Case or activity
Seven-week module, one three-hour session a week			
Week 1	1, 2	The two languages; working capital and the cash conversion cycle	Cycle of a public company
Week 2	3, 4	Inventory on the books and lending against it	Credem
Week 3	5, 6	Trade credit; financing cross-border trade	Grupo Pão de Açúcar; simulation kickoff and Round 1
Week 4	7, 8	The toolkit; payables finance	Procter & Gamble; Rounds 2 and 3
Week 5	9	Hidden leverage and disclosure	Greensill; Rounds 4 and 5
Week 6	10, 11	Hedging; integrated decisions	Unilever; Round 6 and debrief
Week 7	12 to 16	The frontier; the AI supply chain; the epilogue as synthesis	Final examination
Two-day executive workshop, four half-day blocks			
Day 1, morning	1, 2	Cash in the supply chain	Participants' own cash conversion cycles
Day 1, afternoon	5, 8	Trade credit and payables finance	Procter & Gamble
Day 2, morning	9, 10	Hidden leverage; spend at risk	Greensill
Day 2, afternoon	11, 15	Integrated decisions; financing the AI buildout	Team A and Team B decisions, discussed by role

Acknowledgments

This book owes most to the students of OPIM 5115. Their questions in class and in the chat, their case presentations, and their decisions in the simulation, including the ones that went wrong, showed me which ideas were hard and which explanations worked. Many of the pitfalls in these pages were first discovered by a team in the middle of a round. I thank my colleagues at the University of Connecticut School of Business for many conversations about teaching at the boundary of operations and finance, and the researchers and practitioners whose cases, reports and papers are cited throughout. The errors that remain are mine.

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How to Use This Book

A book on supply chain finance asks you to do two things at once: follow an argument about how goods and money move through a chain of firms, and carry out calculations precise enough to survive a meeting with a treasurer or a lender. How can you tell, on any page, which of the two you are doing and where to find what you need? This short chapter is a key to the page. It walks through the anatomy of a chapter, shows a live specimen of every kind of box, collects the notation and the day-count conventions in one place, and explains where the answers are.

The anatomy of a chapter

What should you expect when you open a chapter? Every chapter moves through the same four stages (Figure 1). It opens with a subtitle, sometimes an epigraph, and a short lead that drops you into a concrete situation at a company, followed by numbered learning objectives. A running company then carries the chapter. It is real when public facts suffice, and fictitious when the numbers must be complete enough to compute with, and the chapter returns to it section after section.

Each section begins with the question it answers, not with the answer. The answer arrives as numbers first: we work the running company's case, and only then write the general formula and codify the procedure. The chapter closes with a summary, a short list of managerial takeaways, the key terms, discussion questions, problems, and suggestions for further reading.

The boxes, one specimen each

Why so many boxes? Each kind does one job, and each has its own shape, so you can tell them apart at a glance, even in grayscale. The specimens below share one small story. Quinnipiac Tool Works, a fictitious Connecticut maker of hand tools, buys $73 million of steel, handles and packaging a year

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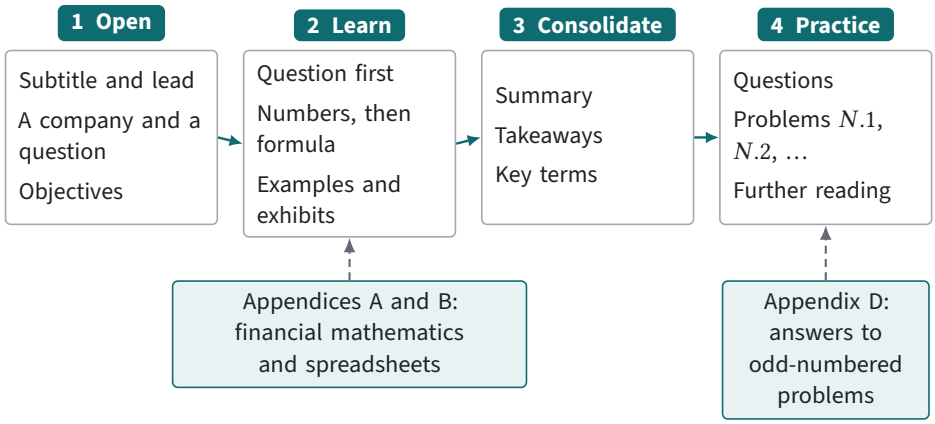


FIGURE 1
The anatomy of a chapter. Every chapter moves through the same four stages, and the appendices support the second and the fourth.

and pays its suppliers in 45 days. Its treasurer wants to know what moving to 60-day terms would be worth.

A *worked example* carries the numbers. Examples are numbered by chapter, so Example 5.3 is the third worked example in Chapter 5; the specimens in this unnumbered chapter carry the number 0.

EXAMPLE 0.1 What fifteen more days of payables are worth

Quinnipiac buys \$73,000,000 a year, or \$73,000,000/365 = \$200,000 a day. At 45 days its accounts payable are \$200,000 × 45 = \$9,000,000; at 60 days they would be \$200,000 × 60 = \$12,000,000. The extension releases \$3,000,000 of cash once, and keeps it released for as long as the new terms hold. Its suppliers' receivables rise by the same \$3,000,000.

A *key formula* isolates a relationship you will use again, after the numbers have shown where it comes from.

KEY FORMULA From days to dollars

$$\mathrm{AP} = P \times \frac{\mathrm{DPO}}{365}, \qquad \Delta\mathrm{AP} = P \times \frac{\Delta\mathrm{DPO}}{365},$$

where AP is accounts payable, P is annual purchases, DPO is days payables outstanding and Δ denotes a change.

An *exhibit* codifies a procedure as numbered steps that you can follow at work or in an examination. It states its day-count convention and may end with a spreadsheet formula.

EXHIBIT 0.1 A procedure to convert a change in payment terms into cash

Step 1. Find annual purchases P. If only cost of goods sold (COGS) is reported, use purchases = COGS + ending inventory − beginning inventory.

Step 2. Divide by 365 to get daily purchases (working capital days use a 365-day year).

Step 3. Multiply daily purchases by the change in days payables outstanding, ΔDPO. The product is the one-time change in accounts payable, and in cash.

Step 4. Name who finances the change: the suppliers' days sales outstanding on these sales moves by the same number of days.

```
=B2/365*(B4-B3)
```

A *case in point* shows the idea at work in a real company. Its facts come from public filings, published cases and industry reports, and they are dated.

CASE IN POINT 0.1 Walmart discloses what it owes the banks

Since the FASB's ASU 2022-04 took effect, buyers must report how much of their payables sits in supplier finance programs. Walmart's 10-K for the fiscal year ended January 31, 2024, reports $5.3 billion of outstanding payment obligations to financial institutions under such programs, against $5.2 billion a year earlier. Payment terms with participating suppliers generally range from 30 to 90 days, participation is optional for the supplier, and the obligations are classified as accounts payable.[5]

5 Source: Walmart Inc., Form 10-K for the fiscal year ended January 31, 2024. Chapter 9 shows how to read such a note.

An *insight* states a managerial lesson in a sentence or two.

INSIGHT Stretching terms moves cash; it does not create it

Quinnipiac's $3 million is its suppliers' $3 million. The extension creates value for the chain only if the extra days end up financed by whichever firm borrows more cheaply, which is the idea behind payables finance in Chapter 8.

A *pitfall* flags a mistake that careful people make.

PITFALL Mixing day-count conventions

Working capital days use a 365-day year, but financing charges use actual/360. A bank that finances a $100,000 invoice for 60 days at 5.4 percent charges $100{,}000 \times 0.054 \times 60/360 = \$900$. Divide by 365 instead and you get $887.67, a 1.4 percent error that makes your model disagree with the bank's statement.

A *spreadsheet tip* shows how to build a calculation so that it stays correct when the inputs change.

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SPREADSHEET TIP **Give the day count its own cell**

Put the basis, 360 or 365, in a labeled cell and refer to it instead of typing the number into formulas. For charges that run between two dates, `=B2*B3*YEARFRAC(B4,B5,2)` multiplies face value by the rate and by the actual/360 fraction of a year between the dates; basis 3 gives actual/365.

A *think-about* prompt, marked by an icon in the margin, asks you to pause before reading on.

THINK ABOUT IT

Quinnipiac borrows at 5 percent and its suppliers borrow at 9 percent. Who should finance the extra fifteen days, and what would each firm need to agree to before both are better off?

A *quotation* appears only when the words are verbatim in a published source and add something that a paraphrase cannot. It always carries its attribution. Here an accounting standard setter explains why the new supplier finance disclosures arrived so quickly.[6]

> *The cat is out of the bag on the topic because of what's happened in the marketplace.*
>
> LINDA MEZON-HUTTER, VICE CHAIR OF THE IASB, AS QUOTED IN THE WALL STREET JOURNAL

6 The quotation is reported in Gelsomino, L. (2023), *Supply Chain Finance Current Practices for Disclosure, Reclassification and Governance*, Supply Chain Finance Community.

Notation and conventions

Why fix the notation for a whole book? A symbol that means two things in two chapters is the fastest way to lose a reader, and a rate quoted without its day count is not yet a number. Table 3 lists the symbols used throughout the book; each chapter defines them again where they first appear, and a chapter may add symbols of its own, defined locally. Table 4 collects the conventions for days and rates, and every exhibit that needs one states it.

Money follows three more rules. A dollar sign means U.S. dollars unless the text says otherwise; euros are written €, Brazilian reais R$ and pounds £. Days are reported to one decimal and rates to one decimal of a percent, with two decimals or basis points when differences are small. Intermediate results are shown, so that every number in the book can be reproduced from the inputs given.

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TABLE 3
Symbols used throughout the book.

Symbol	Meaning
DIO, DSO, DPO	Days inventory outstanding, days sales outstanding, days payables outstanding
CCC	Cash conversion cycle, DIO + DSO − DPO
NWC	Net (operating) working capital: receivables + inventory − payables
F	Invoice face value
t	Days financed (tenor)
$d; t_d; t_n$	Cash discount as a fraction; discount period; net period, in terms "d/t_d net t_n"
r	Annual interest rate; r_B the buyer's and r_S the supplier's cost of funds; r_{SCF} the all-in rate of a supply chain finance program
r_{ref}, s	Reference rate (SOFR, EURIBOR, CDI; historically LIBOR) and credit spread
δ	Discount charged on an early payment, as a fraction of face value: $\delta = r_{\text{SCF}}\, t/360$
$P; S$	Annual purchases (spend); annual sales (revenue)
α	Advance rate: loan or advance as a fraction of collateral or receivable value
LTV	Loan to value
μ, σ	Mean, standard deviation
$z, \Phi(z), \phi(z)$	Standard normal value, its cumulative distribution and its density (in Excel, `NORM.S.DIST(z,TRUE)` and `NORM.S.DIST(z,FALSE)`)
Δ	Change in a quantity

TABLE 4
Day-count and annualization conventions. For terms of 2/10 net 30 the cost of skipping the discount is 2.04 percent per 20 days, 36.7 percent a year simplified, and 44.6 percent as an effective annual rate.

Quantity	Convention	Formula
Working capital days	365-day year	DIO = Inventory/COGS × 365 DSO = Receivables/Revenue × 365 DPO = Payables/COGS × 365 (or purchases, when stated) Average balances when two balance sheets are available
Financing charges	Actual/360	Interest or discount = $F \times r \times t/360$
Early-payment discount, simplified	360-day year, simple	$\dfrac{d}{1-d} \times \dfrac{360}{t_n - t_d}$
Early-payment discount, effective annual rate (EAR)	365-day year, compounded	$\left(1 + \dfrac{d}{1-d}\right)^{365/(t_n - t_d)} - 1$
Balance and flow	365-day year	AP = $P \times$ DPO$/365$

Problems, answers and solutions

How do you know that you have understood a chapter? Work its problems. Each chapter ends with two kinds of practice. *Discussion questions*, numbered 1, 2, 3, and so on, have no single right answer and are meant for class or a study group. *Problems* carry the chapter number, so Problem 5.3 is the third problem of Chapter 5. They run from short drills to multi-step analyses, each is set in a named (and invented) company, and each has one correct numerical answer, sometimes in several parts. Worked examples, exhibits, cases in point, figures and tables are numbered the same way: Exhibit 8.2 is the second exhibit of Chapter 8.

Answers to the odd-numbered problems are collected in Appendix D, so you can check your work; complete worked solutions to every problem are in the Instructor's Solutions Manual. Appendix D gives the answer, not the path, so work each problem to the end before you look. Appendix A reviews the financial mathematics, Appendix B collects the spreadsheet functions, and Appendix C offers guides to the six cases used in the course on which the book is based.

Margin notes and source lines

Where does a number come from, and how old is it? Look to the margin. The wide outer column holds side notes, numbered in sequence, that carry sources, dates and short asides that would interrupt the main text.[7] Figure and table captions also sit in the margin when the figure or table is as wide as the text; wider ones hang into the margin, with the caption below a figure and above a table.

7 Like this one. A side note never carries a step of the argument; you can skip every one of them and still follow the chapter.

Every fact about a real company is attributed in the text, as in "Walmart's 10-K for the fiscal year ended January 31, 2024, reports …", and its precise source appears in a side note or as the last sentence of a caption, beginning "Source:". Figures that change over time carry an *as of* date. Fictitious companies are flagged where they first appear, and their data sets are complete and consistent, so every number can be reproduced. Published cases are summarized in our own words and cited; their exhibits are never reproduced. Full references for further study close each chapter under Further Reading and are collected in the References at the end of the book.

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PART

I

Foundations

Every supply chain decision eventually lands on a financial statement. Part I builds the vocabulary: the flows of goods, money and information; working capital and the cash conversion cycle; and how inventory is valued and read by lenders and investors.

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CHAPTER 1

Where Goods Meet Money

An Introduction to Supply Chain Finance

> *Everything we do within a supply chain ends up somewhere on a financial statement.*
>
> Robert J. Trent, *Supply Chain Financial Management* (2016)

On a Monday morning in February 2026, the chief financial officer of Nutmeg Juice Company in Hartford, Connecticut, opened three messages. Shoreline Markets, the grocery chain that buys almost three quarters of Nutmeg's juice, announced that it would pay suppliers in 90 days instead of 60, and that its bank would offer an early-payment program to soften the change. Quinnipiac Bottle Works sent a $25,000 invoice for bottles on terms of 2/10 net 30. And Riverbend Bank asked for the latest balance sheet before renewing Nutmeg's credit line. Not one bottle of juice had moved, yet Nutmeg's cash position had just changed.

These are the everyday decisions of supply chain finance: managing the money that travels through a supply chain alongside the goods. Five years earlier, Greensill Capital had financed billions of dollars of supply chain transactions with programs that looked much like Shoreline's, and then collapsed. This chapter shows how goods and money move together, how to measure the time cash spends inside a business, and why the tools that can lower everyone's costs can also hide risk.

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LEARNING OBJECTIVES

1. Describe a supply chain as three flows of goods, money and information, and translate an operational improvement into the cash, profit and working capital terms that executives use.
2. Decode trade credit terms such as 2/10 net 30, estimate the rough annual cost of skipping the discount, and place the common payment methods on a risk spectrum.
3. Compute working capital and the cash conversion cycle, and read the cycle as a game board on which every financing instrument makes a move.
4. Quantify how a payment-term extension shifts working capital and financing cost up the chain, and how reverse factoring uses a strong buyer's credit to lower the cost for the whole chain.
5. Define supply chain finance, map its instruments to the chapters of this book, and explain the hidden risks and disclosure duties that the collapse of Greensill brought to light.

1.1 Nutmeg Juice Company

Why should three routine messages worry a company that makes money? To answer, we first need to know the company and the numbers it lives by.

Nutmeg Juice Company bottles apple and cranberry juice in a plant on the north side of Hartford. It buys juice concentrate from New England growers and plastic (PET) bottles from Quinnipiac Bottle Works in New Haven, and ships cases to grocery chains and convenience stores across southern New England. Its largest customer by far is Shoreline Markets, a regional grocery chain that buys $3,650,000 of juice a year ($10,000 a day), about 73 percent of Nutmeg's sales, and pays 60 days after delivery. Quinnipiac sells Nutmeg about $1,095,000 of bottles a year on terms of 2/10 net 30, and Nutmeg pays on day 30. Riverbend Bank provides a revolving line of credit at 10 percent a year.[1]

Table 1.1 collects the numbers we will use throughout the chapter. Last year Nutmeg sold $5,000,000 of juice, all of it on credit. Its cost of goods sold (COGS), which covers concentrate, bottles, labels and the plant, was $3,000,000, leaving a gross profit of $2,000,000 and, after $1,550,000 of selling, general and administrative expenses, an operating income of $450,000. On an average day the balance sheet showed $500,000 of inventory, $800,000 owed by customers (accounts receivable) and $400,000 owed to suppliers (accounts payable).

Here is the puzzle. Nutmeg earns $450,000 a year, yet its credit line at Riverbend is almost never paid down. Where does the money go? It waits: in the warehouse as concentrate, empty bottles and finished cases, and in Shoreline's accounts payable department as invoices that are not yet due.

1 Nutmeg, Shoreline, Quinnipiac and Riverbend are fictitious. The base figures adapt the cash conversion cycle exercise of the course on which this book is built.

TABLE 1.1

Nutmeg Juice Company: fiscal 2025 figures and trading terms. All figures are fictitious.

Item	Value
Income statement, fiscal 2025	
Net sales (all on credit)	$5,000,000
Cost of goods sold (COGS)	$3,000,000
Gross profit	$2,000,000
Selling, general and administrative	$1,550,000
Operating income	$450,000
Balance sheet, average balances	
Inventory	$500,000
Accounts receivable	$800,000
of which Shoreline Markets	$600,000
Accounts payable	$400,000
of which Quinnipiac Bottle Works	$90,000
Partners and terms	
Shoreline's annual purchases (net 60)	$3,650,000
Quinnipiac's annual sales to Nutmeg (2/10 net 30)	$1,095,000
Riverbend Bank revolving line, annual rate	10%
Shoreline's short-term cost of funds	4%

Part of the waiting is done by Nutmeg's own suppliers, who have shipped goods that Nutmeg has not yet paid for. The money travels through the supply chain with the goods, and someone has to finance every day of the trip.

Seen this way, each of the CFO's messages is about the length of that trip. Shoreline's new terms would make Nutmeg wait 30 more days for its cash: who pays for those days? Quinnipiac offers 2 percent for paying 20 days early: is that a good deal for a company that borrows at 10 percent? Riverbend wants to know how much cash the business ties up, and for how long. These sound like treasury questions, but each one is caused by a supply chain decision: the terms that sales and purchasing negotiate and the inventory that operations holds.

You might argue that a supply chain manager can leave such questions to the finance department. That would be a mistake. The people who set terms and hold inventory create the financing need, and they are best placed to reduce it, provided they can measure it and explain it in the language that the rest of the company uses.

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1.2 What Flows Through a Supply Chain

If money waits at several places between Quinnipiac's loading dock and Shoreline's checkout lanes, we need a precise picture of that path before we can measure the wait. What exactly travels between the firms, and in which direction?

A **supply chain** consists of everyone who takes part, directly or indirectly, in filling a customer's order. For a bottle of Nutmeg cranberry juice that includes the growers, the concentrate processor, the resin maker that supplies Quinnipiac, Quinnipiac, Nutmeg, Shoreline's distribution center and stores, and the shopper. It also includes parties that never own the juice, such as the trucking firm, or never touch it, such as the bank. A household detergent shows the same structure: paper, packaging, chemical and plastics firms supply a manufacturer such as Procter & Gamble, whose product moves through a retailer's distribution center to a Walmart store and a customer.[2]

2 Source: OPIM 5115 Lecture 1, slide 10.

Supply chain management is the management of the two-way flows that run through such a chain: materials and services moving toward the customer, funds moving mainly back toward the suppliers and information moving in both directions. Its domain runs from purchasing, planning and forecasting through transportation, distribution and inventory management. Supply chain management is a much younger discipline than finance, still maturing in three areas: financial management, risk management and data analytics.[3] This book is about the first two.

3 R. J. Trent, *Supply Chain Financial Management* (2016), chapter 1.

For our purposes it helps to group the flows into three, which we will call the **three flows** of a supply chain (Figure 1.1). *Goods* move downstream, from Quinnipiac to Nutmeg to Shoreline to the shopper. *Money* moves upstream, from the shopper to Shoreline to Nutmeg to Quinnipiac. *Information* moves both ways: orders and forecasts travel upstream, invoices and shipping notices travel downstream.

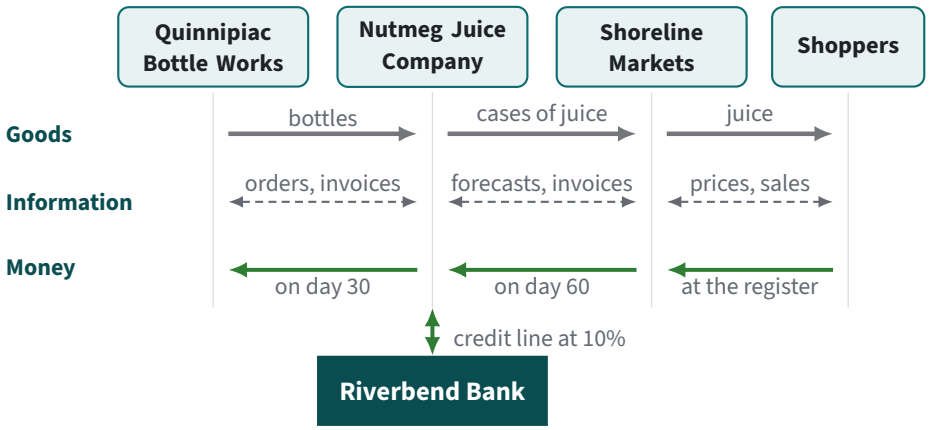


FIGURE 1.1
The three flows of Nutmeg's supply chain. Goods move downstream, money moves upstream, and information moves both ways. The bank finances the gap between the moment goods move and the moment money follows.

The three flows do not move at the same speed, and that is the heart of this book. A truckload of bottles reaches Nutmeg the day after it leaves New

Haven, but the money for those bottles goes back to Quinnipiac 30 days later. A case of juice reaches Shoreline's shelves within a day or two, but the money comes back to Nutmeg 60 days later. Every gap between the moment goods move and the moment money follows has to be financed by someone. The seller can wait, which means it finances the buyer. The buyer can pay before the goods even ship, which means it finances the seller. Or a third party, usually a bank, can step into the gap for a price.

Does the picture change when the product is not a bottle of juice but an answer on a screen? Follow a single message typed into an AI chatbot.

CASE IN POINT 1.1 One chat message, three flows

Type a question into an AI chatbot and an answer appears within seconds. Trace that message backward, as two economists at the Bank for International Settlements did in March 2025, and a supply chain appears. In their example the application, ChatGPT, ran on a foundation model, GPT. The model was trained on text stored in Microsoft's Azure cloud; Azure relied on graphics processing units (GPUs) from specialized makers such as NVIDIA; and NVIDIA's chips were produced by one foundry, Taiwan Semiconductor Manufacturing Company (TSMC).[4]

All three flows of Figure 1.1 run through this chain. Information moves both ways: the prompt travels upstream to the model and the answer travels back, the cloud meters the computing each customer uses and bills for it, and forecasts and purchase orders travel upstream to chip and memory makers long before the chips exist. Goods move downstream: silicon wafers, memory and racks of servers travel toward the data center, and electricity flows into the building to run them. What leaves at the end is computation, delivered as an answer. Money moves upstream: a subscription paid to the app, the app's bills from the cloud, the cloud's payments for servers, and the chip maker's payments to its own suppliers.

What differs from Nutmeg's chain is the timing of the money. Far upstream, buyers pay first. When Micron Technology, a memory maker, reported its results on September 30, 2026, its customers had made $32 billion of financial commitments under multiyear supply agreements, some running into 2031, the vast majority of them cash deposits; $12.3 billion of deposits had arrived in the quarter just ended. Micron returns the deposits in the later years of each agreement if the customer has bought its minimum volumes, and it reports the cash as a financing inflow, as it would a loan: in effect, the customer lends to its supplier.[5]

In the middle of the chain, the seller waits. NVIDIA's receivables reached $63.1 billion at July 26, 2026, or 60 days of sales, up from $40.7 billion and 45 days a quarter earlier. The company tied the rise to extended payment terms on large purchases by certain investment-grade customers, to be shipped over several quarters, and its quarterly report puts such terms at anywhere from 90 days to one year. A customer that pays in a year borrows from its supplier, just as Shoreline borrows from Nutmeg for 60 days, only at a scale of tens of billions

4 Source: L. Gambacorta and V. Shreeti, "The AI supply chain," BIS Papers No 154, Bank for International Settlements, March 2025, section 2 and footnote 4.

5 Sources: Micron Technology, earnings release for the fourth quarter of fiscal 2026 (Form 8-K) and prepared remarks for the earnings call, September 30, 2026; amounts as the company reported them on that date.

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CASE IN POINT 1.1 *continued*

of dollars. Yet NVIDIA also pays ahead: its annual report for fiscal 2026 says it has "paid premiums, provided deposits, and entered into long-term supply agreements and capacity commitments" to secure its own supply.[6]

So the gaps here run both ways at once, sometimes through the same company: customers finance scarce suppliers years ahead, while the chip maker finances its largest customers for months. Someone finances every gap in this chain too. Part VI, beginning in Chapter 14, asks who does, at what price, and who is left holding the gap if the answers on the screen earn less than the chain has committed to spend.

6 Sources: NVIDIA Corporation, CFO commentary, earnings call and Form 10-Q for the quarter ended July 26, 2026 (results released August 26, 2026); Form 10-K for the fiscal year ended January 25, 2026.

You might argue that the money flow is simply the mirror image of the goods flow. In value it is: every bottle shipped is eventually paid for. In time it is not, and time is what costs money. The information flow matters for a related reason. A bank can finance a gap only if it can see what is in it: which goods were shipped, which invoices were approved, which payments are due. As Section 1.7 shows, that visibility is part of the very definition of supply chain finance.

1.3 Why Supply Chain Managers Must Speak Finance

The three flows show where money waits, but they do not tell an executive why the waiting matters. Supply chain professionals often deliver real improvements (higher inventory turns, lower purchase costs, longer supply contracts, better forecasts) and then watch the leadership team react with polite indifference. Why?

The two groups speak different languages, a gap this book calls the **communication challenge**, following Trent's description of it. Executives think in strategy, corporate indicators, financial results and the bottom line. Supply chain managers report tactical operations, activity metrics and process improvements. News reports of a company's results rarely mention inventory turns; they mention gross margin, cash flow, net income and return on assets. Unless someone translates, a supply chain achievement never reaches the vocabulary in which the company keeps score (Figure 1.2).

Trent puts the remedy bluntly:

> *As you think about your professional development, learn to speak a foreign language called finance.*
>
> ROBERT J. TRENT, SUPPLY CHAIN FINANCIAL MANAGEMENT (2016)

The grammar of that language is three financial statements. The *balance sheet* is a snapshot in which assets equal liabilities plus equity; Nutmeg's

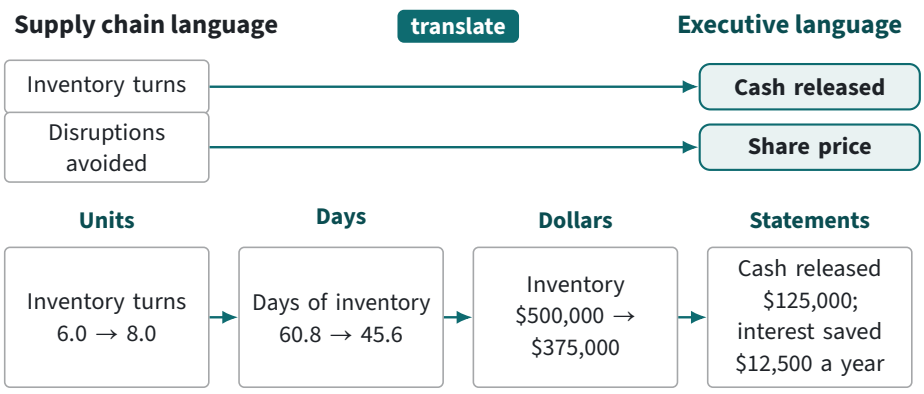


FIGURE 1.2

The communication challenge. Supply chain metrics have financial counterparts (top; the pairings are illustrative). The translation runs from units through days and dollars to a line on a financial statement, shown here for Nutmeg's inventory improvement of Example 1.1 (bottom).

inventory, receivables and payables live there. The *income statement* reports performance over a period, from revenue to net income. Purchase prices and inbound freight reach it as cost of goods sold when the goods are sold (until then they sit in inventory on the balance sheet), and spoilage reaches it as a write-down. The *cash flow statement* reports the money that actually moved. When Nutmeg builds inventory, its operating cash flow falls even if its profit does not. Chapter 2 reads each statement from an operations point of view.

Is the translation worth the effort? Do investors and boards actually reward supply chain performance, or is it a back-office matter that the market ignores? Two kinds of evidence suggest the former. The first, a BusinessWeek special advertising section, reports that companies that excel in supply chain operations perform better on almost every financial measure of success. Where supply chain excellence improves demand-forecast accuracy, those companies outperform their peers (Table 1.2): profit margins 5 percent higher, inventories 15 percent lower, perfect order ratings up to 17 percent stronger and cash-to-cash cycles 35 percent shorter. A **perfect order** is one delivered complete, on time and undamaged, and the study reported roughly 1 percent of extra profit margin for every 3 percent improvement in perfect orders.

Financial measure	Leaders versus peers
Profit margin	5% higher
Inventory levels	15% lower
Perfect order ratings	up to 17% stronger
Cash-to-cash cycle time	35% shorter

TABLE 1.2

Companies whose supply chain excellence improves demand-forecast accuracy, compared with their peers. Source: BusinessWeek special advertising section, "Supply Chain Excellence: Today's Best Driver of Bottom-Line Performance" (undated), as reproduced in OPIM 5115 Lecture 1.

A skeptic would say that this proves little. Well-run companies may be good at everything, so supply chain excellence and profitability could both be symptoms of good management. And the comparison comes from

an advertising section, not a peer-reviewed study. Both objections are fair, which is why the second kind of evidence matters more. Vinod Singhal and Kevin Hendricks studied what happens to a company's stock price when it announces a **supply chain glitch**, a disruption in matching supply with demand such as a production delay or a shipment failure. Because the announcement is a surprise, the price move at the announcement isolates the market's verdict on the glitch itself: on the published chart, returns fall by roughly 8 percentage points around the announcement day (Figure 1.3). Returns also drift down in the months before the news and do not recover over the following 120 trading days, so over the whole window the damage to shareholder value reaches up to 18 percent.

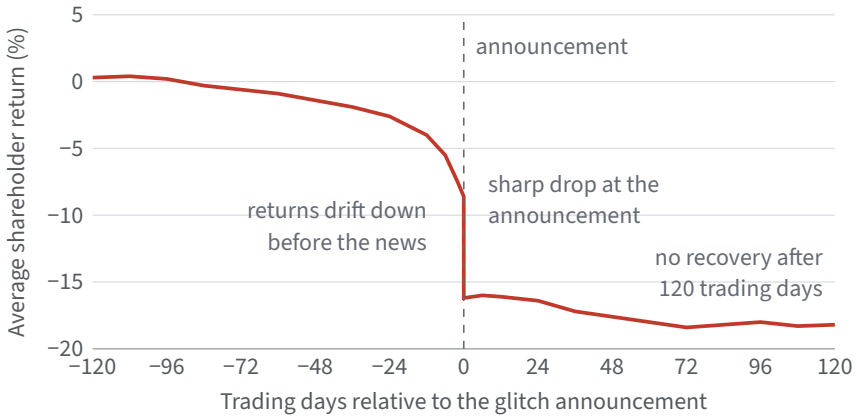


FIGURE 1.3

Average shareholder return around the announcement of a supply chain glitch. Schematic redrawn from the published chart; the path is approximate. Source: V. R. Singhal and K. B. Hendricks, "How Supply Chain Glitches Torpedo Shareholder Value," *Supply Chain Management Review* (2002), as reproduced in OPIM 5115 Lecture 1.

Supply chain performance, then, is not just operational. It shows up in margins, in cash and, through the stock price, in the value of the company. So how should a supply chain manager make the case for an improvement in terms an executive will act on? Let us try it on Nutmeg.

EXAMPLE 1.1 Translating Nutmeg's inventory improvement

Nutmeg's plant manager reports that a new production schedule has raised inventory turns from 6.0 to 8.0. The CFO does not react. Let us translate.

Step 1. Days of inventory fall from $365/6.0 = 60.8$ days to $365/8.0 = 45.6$ days, a reduction of 15.2 days.

Step 2. Inventory falls to COGS divided by the new turns, $\$3{,}000{,}000/8.0 = \$375{,}000$, a reduction of $\$125{,}000$ from $\$500{,}000$.

Step 3. The reduction releases $\$125{,}000$ of cash once, which Nutmeg can use to pay down its Riverbend line.

Step 4. At Riverbend's 10 percent rate, Nutmeg avoids $\$125{,}000 \times 0.10 = \$12{,}500$ of interest a year, before counting the storage space and spoilage it also saves.

EXAMPLE 1.1 *continued*

Step 5. The report the CFO will read: "The new schedule released $125,000 of cash and adds at least $12,500 a year to pretax profit, an amount equal to 2.8 percent of our $450,000 operating income."

Exhibit 1.1 codifies the five steps so that you can reuse them on any improvement.

EXHIBIT 1.1 A procedure to translate a supply chain improvement into the language of finance

Step 1. State the improvement in its own units and convert it into days of the flow it affects. For inventory, days of inventory equal 365 divided by inventory turns.

Step 2. Convert the change in days into dollars on the balance sheet: change in balance = annual flow × change in days / 365. Use COGS as the flow for inventory and payables, and sales for receivables. In a spreadsheet, put the annual flow in B2 and the old and new days in B3 and B4.

Step 3. Record the cash flow effect. A lower inventory or receivables balance, or a higher payables balance, releases that amount of cash once, through operating cash flow, in the year of the change.

Step 4. Record the income statement effect: interest avoided on the released cash at the firm's borrowing rate, plus any storage, spoilage or insurance costs avoided and any change in margin.

Step 5. Report cash released and the annual profit improvement, and compare the improvement with operating income to show its scale.

```
=B2*(B3-B4)/365
```

The translation produced two kinds of number, a one-time stock of cash and a recurring flow of profit, and confusing them is a common way to lose an argument. We will meet the pair again whenever terms or inventory change.

1.4 How Trade Is Paid For

The money flow in Figure 1.1 runs on terms that buyers and sellers negotiate. When Quinnipiac ships bottles to Nutmeg, when does Nutmeg pay, and what does it cost to wait?

Between businesses, most goods are not paid for on delivery. They are sold on **trade credit**: a short-term financing arrangement in which the seller lets

the buyer take the goods now and pay later. Also called vendor credit, trade credit is the foundation of business-to-business commerce, and its terms are printed on the invoice in a compact code. Quinnipiac's invoice reads "2/10 net 30," which Figure 1.4 decodes: Nutmeg may deduct a 2 percent **early-payment discount** if it pays within 10 days of the invoice date; otherwise the full amount is due in 30 days.

Is the discount worth taking? That depends on what the extra 20 days of credit cost compared with Nutmeg's other ways of financing itself.

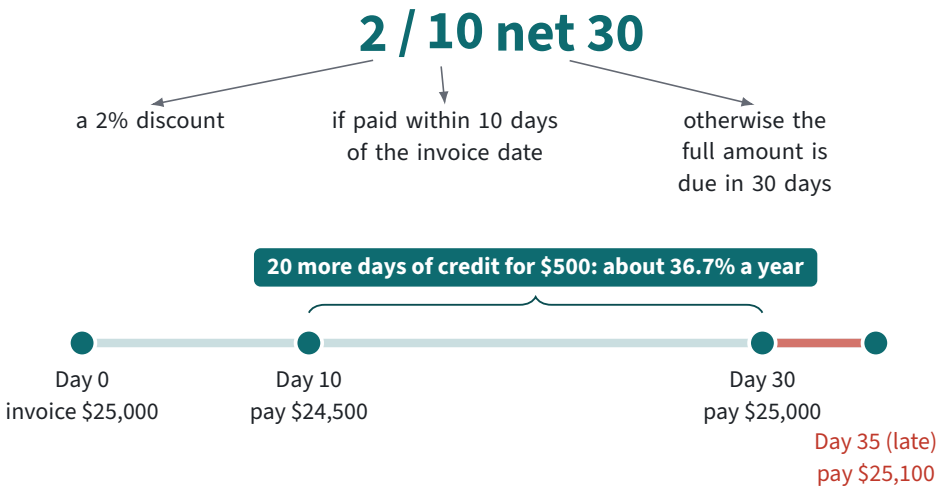


FIGURE 1.4
Anatomy of 2/10 net 30 (top) and the payment options on Quinnipiac's $25,000 invoice (bottom). Paying on day 30 instead of day 10 buys 20 days of credit for $500; paying after day 30 adds a late fee of $20 a day.

EXAMPLE 1.2 Nutmeg's bottle order under 2/10 net 30

Quinnipiac invoices 100,000 bottles at $0.25, or $25,000, on terms of 2/10 net 30, with a late fee of $20 a day. Paying by day 10, Nutmeg sends $25,000 × 0.98 = $24,500. Paying on day 30, it sends $25,000. Paying on day 35, five days late, it sends $25,000 + 5 × $20 = $25,100.

Waiting from day 10 to day 30 means paying $500 more to keep $24,500 for 20 more days, a rate of 500/24,500 = 2.04 percent for 20 days. A 360-day year contains 360/20 = 18 such periods, so the rough annual cost of skipping the discount is

$$\frac{d}{1-d} \times \frac{360}{t_n - t_d} = \frac{0.02}{0.98} \times \frac{360}{30-10} = 0.0204 \times 18 = 36.7\%,$$

where d is the discount, t_d the discount period and t_n the net period. If Nutmeg borrows $24,500 on its 10 percent line for those 20 days, the interest is $24,500 × 0.10 × 20/360 = $136, against a discount of $500. Taking the discount leaves Nutmeg about $364 better off on this one invoice.

Paying late is dearer still: the $100 fee buys 5 days of credit on $25,000, 0.4 percent for 5 days, or about 29 percent a year on a 360-day basis, before counting the damage to the relationship.

This is the hidden cost of "free" credit. Thirty days of credit look free, because no interest appears on the invoice, yet a buyer who skips the discount borrows from its supplier at about 37 percent a year, and with compounding the cost is higher still. Chapter 5 develops both measures and asks why suppliers offer such expensive credit. For now the intuition is enough: *2 percent for 20 days is roughly 37 percent a year*, and a firm that can borrow for less should pay early.

Trade credit smooths the buyer's cash flow and wins the supplier loyalty and volume, but it delays the supplier's cash and exposes it to bad debts. The result is a standing tension: buyers want to pay late to preserve cash, while suppliers want to collect early to stay liquid. Supply chain finance exists to bridge that gap.

How long is typical credit? A study of 2,137 U.S. supply agreements found that net 30 is the most common term in almost every industry, but that the average contractual payment period varies widely (Figure 1.5): under 20 days in transportation and public utilities, where net 10 is most common, and almost 53 in petroleum, rubber and plastics, where net 90 leads. Late-payment interest in the same agreements averages between 7.6 and 18.0 percent a year, depending on the industry.

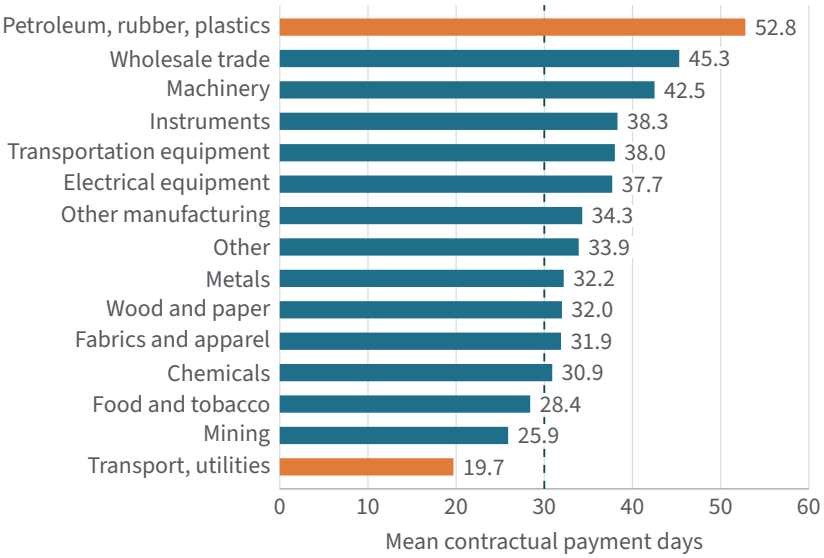


FIGURE 1.5

Average contractual payment term by industry group in 2,137 U.S. supply agreements. The dashed line marks net 30, the most common term; orange bars mark the shortest and longest average terms. Source: J. Yu (2022), "Getting the Banks on Board: Accounts Receivable Financing in the US," SSRN working paper 4307421, Table A3.

Trade credit of this kind is usually extended on **open account**: the seller ships and invoices, and the buyer pays on the agreed terms, with no bank instrument behind the promise. (The phrase is also used for informal credit granted without a signed agreement, common among small firms.) Open account is the cheapest way to settle a trade, but it puts the whole payment risk on the seller. When the parties do not know each other, or sit in different countries, sellers ask for more security (Table 1.3).

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TABLE 1.3

Payment methods and who bears the payment risk, from most secure for the seller (top) to most secure for the buyer (bottom). Source: adapted from OPIM 5115 Lecture 1.

Method	How it works	Security for
Cash in advance	The buyer pays before the goods ship.	Seller
Letter of credit	The buyer's bank promises to pay once the seller presents documents that comply with the credit.	Balanced
Documentary collection	Banks release the title documents to the buyer only against payment or a signed promise to pay, without guaranteeing either.	Somewhat balanced
Open account	The seller ships and invoices; the buyer pays on the agreed terms.	Buyer

Over recent decades world trade has moved down this table: from 1978 to 2013, open-account volume grew many times over while letter-of-credit volume grew far more slowly.[7] The shift saved paperwork and fees, but it took the bank out of the transaction and left sellers carrying the credit risk and the financing burden alone. Much of supply chain finance can be read as the industry's answer: a way to bring financing and risk protection back into open-account trade. Chapter 6 studies the upper rows of Table 1.3 and the road to open account.

7 Source: OPIM 5115 Lecture 1, slide 20; the chart does not state its original source or units.

1.5 Working Capital and the Cash Conversion Cycle

Trade terms decide when each payment happens, and inventory decisions decide how long goods sit before they are sold. Riverbend Bank wants the sum of all these decisions: how much money does Nutmeg have tied up in its supply chain, and for how long?

The first part of the answer is a stock of money. **Working capital** is the capital a business uses for its day-to-day operations, normally measured as current assets minus current liabilities: what the firm owns that will turn into cash within a year, less what it owes within a year. Supply chain finance focuses on its operating core, which this book calls **net working capital** (NWC). Other names are operating or trade working capital, and many finance texts use the phrase "net working capital" for current assets minus current liabilities. We define it as receivables plus inventory minus payables:

$$\text{NWC} = \text{AR} + \text{Inventory} - \text{AP} = \$800{,}000 + \$500{,}000 - \$400{,}000 = \$900{,}000.$$

Trent observes that finance professionals tend to watch the receivables and payables in this sum while supply chain professionals watch the inventory. The number belongs to both.

The second part of the answer is time: how many days does a dollar spend in each account? Nutmeg's inventory of $500,000 is one sixth of its

annual COGS, so a bottle or a gallon of concentrate spends about one sixth of a year, roughly 61 days, in the plant: the **days inventory outstanding** (DIO). Receivables are 16 percent of annual credit sales, so customers take about 58 days to pay: the **days sales outstanding** (DSO). Payables are 13 percent of annual COGS, so Nutmeg takes about 49 days to pay its suppliers: the **days payables outstanding** (DPO). Inventory is measured against COGS because both are recorded at cost, and receivables against sales because they are recorded at selling price. Strictly, payables arise from purchases rather than COGS; we use COGS because purchases are rarely reported directly, a choice Chapter 2 revisits. All three are averages over Nutmeg's customers and suppliers: Shoreline pays on day 60 and Quinnipiac is paid on day 30, while other partners run on their own schedules.

The **cash conversion cycle** (CCC) combines the three clocks. It measures the time between the moment a firm pays cash for its inventory and the moment it receives cash from selling the product: $\text{CCC} = \text{DIO} + \text{DSO} - \text{DPO}$. It is arguably the most important single metric in supply chain finance, because every instrument in this book changes one of its three terms.

EXAMPLE 1.3 Nutmeg's cash conversion cycle

Using a 365-day year and the figures in Table 1.1:

$$\text{DIO} = \frac{\text{Inventory}}{\text{COGS}} \times 365 = \frac{\$500{,}000}{\$3{,}000{,}000} \times 365 = 60.83 \text{ days},$$

$$\text{DSO} = \frac{\text{Receivables}}{\text{Credit sales}} \times 365 = \frac{\$800{,}000}{\$5{,}000{,}000} \times 365 = 58.40 \text{ days},$$

$$\text{DPO} = \frac{\text{Payables}}{\text{COGS}} \times 365 = \frac{\$400{,}000}{\$3{,}000{,}000} \times 365 = 48.67 \text{ days},$$

$$\text{CCC} = 60.83 + 58.40 - 48.67 = 70.56 \text{ days}.$$

Nutmeg's cash is tied up for about 71 days from the day it pays its suppliers to the day its customers pay it. The time from receiving inventory to collecting cash, the **operating cycle**, is DIO + DSO = 60.83 + 58.40 = 119.23 days. Suppliers finance the first 48.67 days of that trip; Nutmeg and Riverbend finance the remaining 70.56.

Figure 1.6 draws the same numbers as a timeline. Two cycles appear on it. The operating cycle runs from the arrival of inventory to the collection of cash. The cash cycle, which is the cash conversion cycle, runs only from the day cash goes out to suppliers to the day cash comes in from customers. If Nutmeg paid cash on delivery, the two would coincide. Buying on credit shortens the cash cycle by exactly the DPO.

We will use Figure 1.6 as a **game board** for the chapters ahead. It has four dates (inventory arrives, the supplier is paid, the product is sold, the customer pays) and three segments (DIO, DSO and DPO). Every decision in

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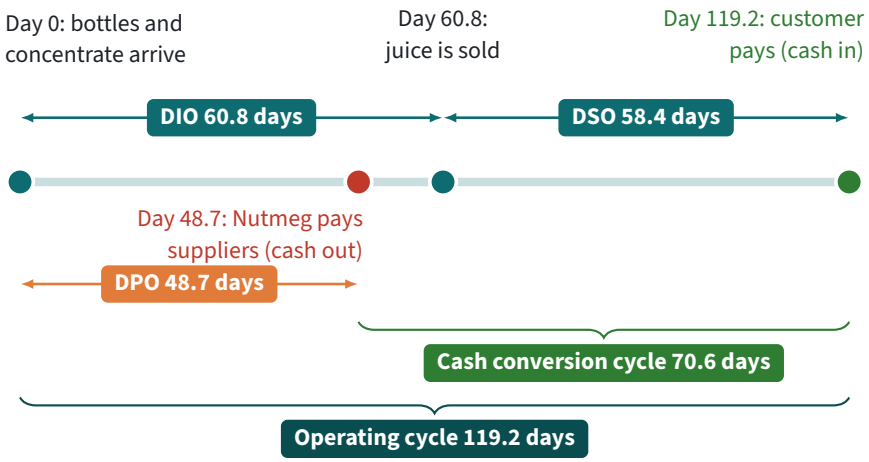


FIGURE 1.6

Nutmeg's cash conversion cycle as a timeline, the game board of this book. Suppliers finance the first 48.7 days of the operating cycle; Nutmeg and its bank finance the remaining 70.6 days.

the coming chapters moves one of those dates, changes who finances one of the segments, or changes whose credit sets the price of that financing. A rule of thumb follows directly: to shorten the cycle, pay suppliers later, hold less inventory and collect from customers faster.

Should Nutmeg then simply aim for the shortest cycle it can find? Look first at how widely the cycle varies across well-run companies (Figure 1.7). In 2012 Verizon's cycle was −72 days, because it paid its suppliers in about 116 days while collecting from customers in 36. Apple (−53) and Amazon (−11) also collected before they paid; in effect they borrowed from their suppliers. Grocers sell for cash; Kroger collected in 4 days, so modest supplier terms kept its cycle near zero. At the other end, Tiffany & Co., since acquired by LVMH, held inventory for 525 days and had a cycle of +486 days: it spent cash on its jewelry more than a year before customers paid for it.

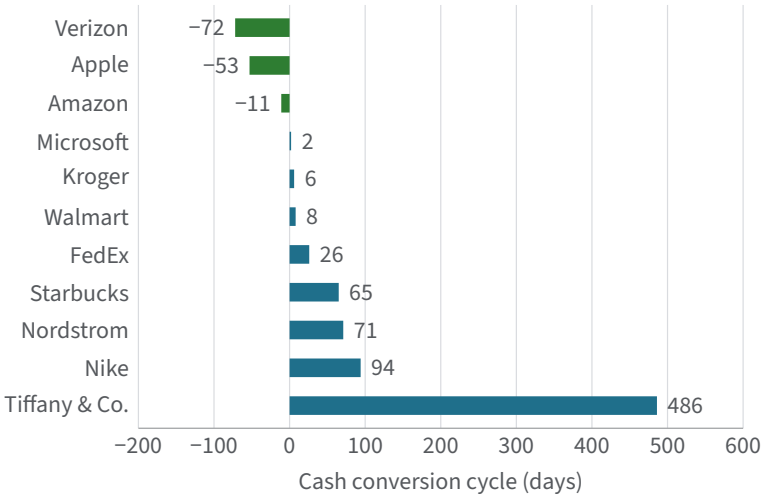


FIGURE 1.7

Cash conversion cycles of eleven companies in 2012. Green bars are negative cycles: customers pay before suppliers are paid. Source: J. Berk and P. DeMarzo, *Corporate Finance*, 3rd ed. (2014), Table 26.1 (Capital IQ data), as reproduced in OPIM 5115 Lecture 1.

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INSIGHT **The cycle reflects the business model**

A cash conversion cycle is not a score in which lower always wins. Jewelry needs long inventory, software licenses need no inventory, and grocers are paid at the register. Compare a firm's cycle with peers that share its business model, and ask which of the three clocks drives the difference.

Nutmeg cannot copy Verizon. Its customers are grocery chains with far more bargaining power than its own, and juice cannot be sold before it is bottled. Chapter 2 turns the cycle into a full diagnostic tool, with average balances, industry benchmarks and the link to financial performance. Here the important point is the one the Verizon bar raises: a negative cycle for one company usually means a long wait for someone else. If a shorter cycle is good, what stops every company from simply paying its suppliers later?

1.6 The Tug-of-War over Payment Terms

Every dollar a buyer adds to its payables is a dollar added to its supplier's receivables. The cash conversion cycle rewards both sides for pulling, but they pull on the same rope. So what happens when the stronger party pulls harder?

Picture the two sides of an invoice. The buyer thinks: I want to pay late, to preserve my cash, improve *my* working capital and use supplier financing instead of bank loans. The supplier thinks: I want to collect early, to meet my obligations, reduce *my* working capital and keep my financing costs down. Neither is wrong, and the outcome usually depends on bargaining power, which large buyers have in abundance.

The press has documented the pattern for years under headlines such as "Big Companies Pay Later, Squeezing Their Suppliers" (The New York Times, 2015). Diageo moved from 30-day to 60-day payment with no warning and no offsetting compensation; Johnson & Johnson and Tesco cited the global financial crisis when extending terms; and Walmart, according to Forbes, earned about $2 million a day on its payables at a 3 percent interest rate.[8] The trend had not reversed by 2021: the 1,000 largest U.S. companies by revenue took an average of 62.2 days to pay suppliers that year, up from 61.9 days in 2020 and higher than a decade earlier.[9]

What does a long extension do to the rest of the chain? The clearest illustration comes from the depths of the 2009 recession.

CASE IN POINT 1.2 **Anheuser-Busch InBev shifts the burden, 2009**

In January 2009 the brewer Anheuser-Busch InBev extended its payment terms from 30 days to 120 days. Its trade accounts and other deferred expenses payable rose from $4.833 billion at the end of 2008 to $5.657 billion at the end

8 Source: OPIM 5115 Lecture 1, slide 30, citing S. Strom, The New York Times, April 6, 2015; S. Ng, The Wall Street Journal, April 16, 2013; J. Murfin, Yale Insights, November 26, 2014; and Forbes. The Forbes figure is undated in the course source.

9 Source: Hackett Group survey, reported by K. Broughton, The Wall Street Journal, July 20, 2022.

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CASE IN POINT 1.2 *continued*

of 2009, freeing $824 million of working capital for the company. Suppose the smaller, less powerful suppliers who absorbed the change financed it at around 15 percent, while AB InBev's own cost of funds was around 3.5 percent. Then the suppliers incurred about $0.15 \times \$824$ million $= \$123.6$ million a year of extra financing cost, while AB InBev saved about $0.035 \times \$824$ million, or about $28.8 million. The chain as a whole lost roughly $94.8 million a year, and the suppliers paid about $4.29 for every dollar the buyer saved.[10]

10 Source: OPIM 5115 Lecture 1, slide 31, based on Anheuser-Busch InBev's 2008 and 2009 financial statements. The 15 and 3.5 percent rates are the lecture's illustrative assumptions, and the change in payables also reflects other changes in the business.

So what is the problem here? A term extension does not create cash; it moves the need for cash from the buyer's balance sheet to the supplier's, a move we call the **burden shift**. Whether the move creates or destroys value depends on one comparison: the rate at which the supplier must finance the extra receivables versus the rate at which the buyer would have financed the same amount. When a strong buyer borrows cheaply and its suppliers borrow expensively, the chain loses (Figure 1.8). Writing P for the annual purchases covered by the new terms, Δt for the extension in days, r_B for the buyer's cost of funds and r_S for the suppliers', the chain's net annual effect is

$$(r_B - r_S) \times \Delta\text{AP}, \qquad \Delta\text{AP} = P \times \frac{\Delta t}{365}. \tag{1.1}$$

For AB InBev,

$$(0.035 - 0.15) \times \$824 \text{ million} = -\$94.8 \text{ million a year}.$$

Shoreline's letter to Nutmeg is exactly this situation, on a smaller scale.

EXAMPLE 1.4 **Shoreline's 30-day extension**

Shoreline buys $3,650,000 of juice a year from Nutmeg and wants to move from net 60 to net 90. The annual cost of carrying a balance is the balance times the annual rate.

Steps 1 and 2. $\Delta t = 30$ days, so $\Delta\text{AP} = \$3{,}650{,}000 \times 30/365 = \$300{,}000$. Shoreline's payables to Nutmeg rise by $300,000, and so do Nutmeg's receivables from Shoreline.

Step 3. At its 4 percent cost of funds, Shoreline gains $0.04 \times \$300{,}000 = \$12{,}000$ a year.

Step 4. Nutmeg must borrow the $300,000 on its Riverbend line at 10 percent, which costs $0.10 \times \$300{,}000 = \$30{,}000$ a year, an amount equal to 6.7 percent of its $450,000 operating income.

Step 5. The chain's net effect is $(0.04 - 0.10) \times \$300{,}000 = -\$18{,}000$ a year. Nutmeg pays $2.50 for every dollar Shoreline saves.

Step 6. Nutmeg also needs Riverbend to raise its credit line by $300,000. If the bank declines, Nutmeg may have to cut inventory, delay its own suppliers or turn down orders.

On the game board of Figure 1.6, Shoreline's letter stretches Nutmeg's DSO segment by 21.9 days, from 58.4 to 80.3, because Shoreline buys 73 percent of Nutmeg's juice: 30 more days on 73 percent of sales is 21.9 days on all of them. Nutmeg's cycle grows by the same 21.9 days. Figure 1.8 shows how the chain's verdict on the extension depends on the rate at which Nutmeg carries that segment: every percentage point above Shoreline's 4 percent costs the chain another $3,000 a year.

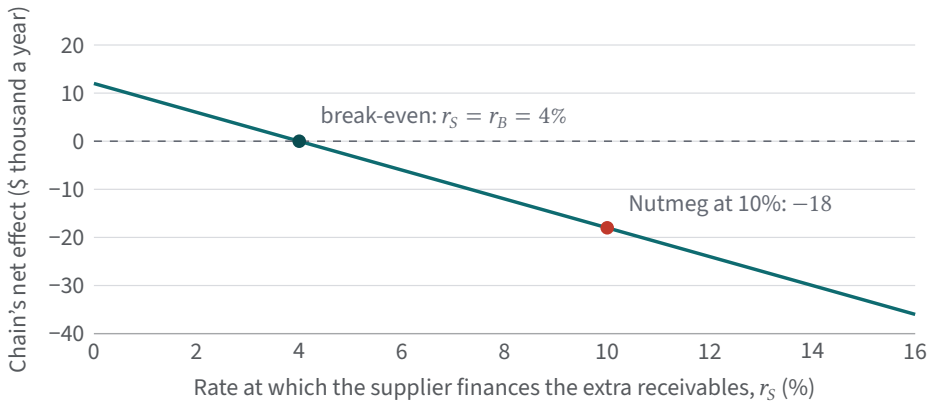


FIGURE 1.8

The burden shift on Shoreline's extension: the chain's net annual effect, $(r_B - r_S) \times \$300{,}000$, as a function of the supplier's financing rate. The chain loses whenever the supplier finances at more than Shoreline's 4 percent.

You might object that Nutmeg could simply raise its prices to recover the $30,000. Perhaps; but a supplier with that much power would probably not have accepted 90-day terms in the first place.

Exhibit 1.2 turns Equation (1.1) and the steps of Example 1.4 into a procedure you can reuse on any term extension.

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EXHIBIT 1.2 A procedure to quantify the burden shift of a term extension

Step 1. Measure the extension Δt (new term minus old term, in days) and the annual purchases P that the new term covers.

Step 2. Convert days into dollars: $\Delta\text{AP} = P \times \Delta t/365$. If two balance sheets are available, read ΔAP as the change in the buyer's payables. The suppliers' receivables rise by the same amount. In a spreadsheet, put P in B2, the old and new terms in B3 and B4, r_B in B5 and r_S in B6.

Step 3. Compute the buyer's annual gain, $r_B \times \Delta\text{AP}$, using its cost of short-term funds. Price a balance carried for a year as balance times annual rate. (An actual/360 loan would charge 365/360 times as much, about 1.4 percent more; the simplification does not change any comparison.)

Step 4. Compute the suppliers' annual cost, $r_S \times \Delta\text{AP}$, using the rate at which they finance the extra receivables.

Step 5. Compute the chain's net annual effect, $(r_B - r_S) \times \Delta\text{AP}$. A negative number is value destroyed.

Step 6. List what the arithmetic leaves out: supplier distress, price increases, weaker service, lost goodwill.

```
=B2*(B4-B3)/365*(B5-B6)
```

SPREADSHEET TIP Tracing the break-even line

With the inputs of Exhibit 1.2 in cells B2 to B6, put the chain's net effect in B8: `=B2*(B4-B3)/365*(B5-B6)`. Type supplier rates from 0 to 0.16 in steps of 0.01 in A11:A27, enter `=B8` in B10, select A10:B27 and choose Data, What-If Analysis, Data Table, with B6 as the column input cell. The table reproduces Figure 1.8 and changes sign exactly where r_S equals the buyer's rate in B5.

PITFALL Counting the buyer's cash as the chain's gain

A buyer that extends terms reports that it has "freed" working capital. For the chain nothing has been freed: the same dollars must now be financed by the suppliers, usually at a higher rate. A term extension moves working capital. It creates value only if the party that ends up financing the gap can do so more cheaply than the party that financed it before.

Why would sophisticated buyers destroy value in their own supply chains? The root cause usually lies inside the buyer, in the dissonance between the goals of its finance department and those of its supply management group. Finance is rewarded for improving working capital; supply management wants strong supplier relationships to secure quality, pricing, availability, delivery, responsiveness and new ideas. When a large buyer

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imposes new terms unilaterally, finance wins the internal argument and the buyer inherits risks it did not price: business continuity risk, supplier viability risk, material cost inflation and suppliers that stop going the extra mile.

1.7 Supply Chain Finance: A Definition and a Map

The tug-of-war destroys value whenever the weaker party finances the gap at a higher rate than the stronger party could. Is there a way to give the buyer its extra days without handing the bill to the supplier at the supplier's rate? That question, and the instruments that answer it, define a field.

The Global Supply Chain Finance Forum, whose members include the International Chamber of Commerce and the Bankers Association for Finance and Trade, defines **supply chain finance** (SCF) as "the use of financing and risk mitigation practices and techniques to optimise the management of the working capital and liquidity invested in supply chain processes and transactions."[11] The Forum adds that SCF is typically applied to open-account trade, that it is triggered by supply chain events, and that the finance provider's visibility of the underlying trade flows is a necessary component, which can be enabled by a technology platform.

11 Source: Global Supply Chain Finance Forum, *Standard Definitions for Techniques of Supply Chain Finance* (2016); also quoted in OPIM 5115 Lecture 1, slide 34.

Each phrase of the definition connects to something we have already built. *Financing and risk mitigation* means that SCF is more than lending: guarantees, insurance and hedges belong to it too. *The working capital and liquidity invested in supply chain processes* are the segments of the game board in Figure 1.6: inventory, receivables and payables. *Open-account trade, triggered by supply chain events* says where SCF operates: in the open-account world described in Section 1.4, where no letter of credit stands behind the payment, with financing that switches on at specific moments such as a purchase order, a shipment or an approved invoice. And *visibility* is the information flow of Figure 1.1: a bank finances only what it can see. The solutions under this umbrella fall into three broad families: receivables finance, payables finance and inventory finance.

Figure 1.9 draws an umbrella that organizes the field, and we will use it as the map of this book. Payables-centric techniques are led by the buyer and start from an invoice the buyer has approved. Receivables-centric techniques are led by the supplier and start from an invoice the supplier has issued. Other SCF techniques finance goods before an invoice exists: a purchase order, raw materials, stock in a warehouse. Related techniques are older or broader tools that sit next to SCF, such as letters of credit and loans secured by all of a firm's current assets. Chapter 7 gives each technique its standard market definition; the rest of the book takes the most important ones apart.

How big is this market? The global market for corporate supply chain finance reached an estimated $1.8 trillion in 2021, up 38 percent from 2020,

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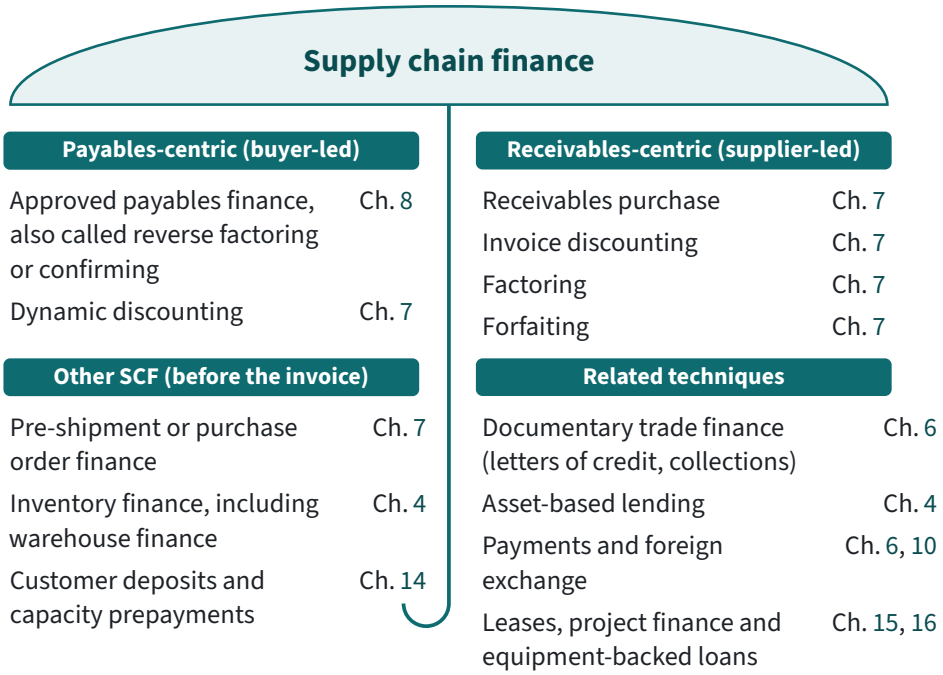


FIGURE 1.9

The supply chain finance umbrella as a map of this book. Each technique carries the chapter that treats it in depth; hidden debt (Chapter 9), risk (Chapter 10) and Parts V and VI cut across the whole map. Source: taxonomy adapted from OPIM 5115 Lecture 1, slide 35; chapter tags and the rows for Part VI added by the author.

according to BCR Publishing, a data provider. Demand tends to rise with interest rates, because the programs give suppliers a relatively cheap source of cash, and the bank's cut is set by the buyer's credit rating rather than the supplier's.[12] That last phrase contains the whole idea. Of all the techniques under the umbrella, the one that shows it most clearly is the payables-centric archetype in the upper left corner, so let us see how it works for Nutmeg.

12 Source: K. Broughton, "Companies Offer Supply-Chain Financing to Vendors as They Bulk Up on Inventory, Push Out Payment Terms," The Wall Street Journal, July 20, 2022. Market figures as of 2021.

1.8 Reverse Factoring: Borrowing a Buyer's Credit

Shoreline's letter did not stop at the 90-day terms. It also offered an early-payment program run with its bank. How can a bank pay Nutmeg early and still let Shoreline pay late, and why would anyone come out ahead?

The program is an example of **reverse factoring**, also called payables finance or approved payables finance. Ordinary factoring starts with the supplier, which sells its receivables to a factor; reverse factoring starts with the buyer, hence the name. Figure 1.10 follows one invoice through the program. Nutmeg delivers a large July order and invoices Shoreline $360,000 on day 0. On day 10 Shoreline approves the invoice. Approval is the key event: Shoreline confirms that the goods arrived as ordered and commits to pay the full $360,000 at maturity, whatever happens to Nutmeg in the meantime. From that moment Nutmeg can sell the invoice to the bank. The bank pays the face value less a discount for the 80 days until maturity at an annual rate of 5 percent, or $360,000 × 0.05 × 80/360 = $4,000, so Nutmeg receives $356,000 on day 10. Because the bank deducts its charge from the

face value up front, 5 percent is a discount rate; as an interest rate on the $356,000 Nutmeg actually receives it is $4,000/$356,000 × 360/80 = 5.06 percent. On day 90 Shoreline pays the bank $360,000.

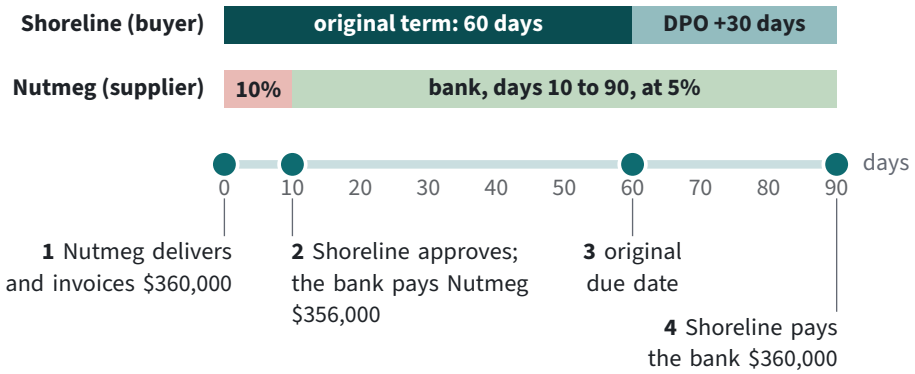


FIGURE 1.10
One invoice in Shoreline's reverse factoring program. Nutmeg finances days 0 to 10 itself at 10 percent (red) and is paid on day 10 instead of day 90 (its DSO falls); Shoreline pays on day 90 instead of day 60 (its DPO rises); the bank bridges the gap at a rate set by Shoreline's credit (green).

Why 5 percent, when Nutmeg's own bank charges 10? Because the bank is not lending to Nutmeg. Once Shoreline has approved the invoice, the bank's money comes back from Shoreline, so the bank prices Shoreline's credit: roughly Shoreline's own 4 percent cost of funds plus a margin. Who gains, and by how much? To answer, we compare three situations for the same invoice.

EXAMPLE 1.5 **Reverse factoring Nutmeg's July invoice**

Financing charges on a single invoice use a 360-day year, the money-market convention banks quote: cost = face value × rate × days / 360. (Working capital days and annual carrying costs use 365.) Nutmeg finances itself at 10 percent and Shoreline at 4 percent; the program rate is 5 percent. For simplicity we charge every rate on the invoice's face value.

Situation 1: net 60, no program. Nutmeg waits 60 days and finances the invoice on its credit line: $360,000 × 0.10 × 60/360 = $6,000.

Situation 2: net 90, no program. Nutmeg now waits 90 days: $360,000 × 0.10 × 90/360 = $9,000, or $3,000 more. Shoreline keeps $360,000 for 30 more days, worth $360,000 × 0.04 × 30/360 = $1,200 to it. The chain is $1,200 − $3,000 = −$1,800 worse off: the burden shift of Section 1.6 on one invoice.

Situation 3: net 90 with the program. Nutmeg finances the first 10 days itself, $360,000 × 0.10 × 10/360 = $1,000, and pays the bank's $4,000 discount for the remaining 80 days: $5,000 in total. Shoreline still saves $1,200.

Compared with situation 1, Nutmeg saves $6,000 − $5,000 = $1,000 and Shoreline saves $1,200, so the chain gains $2,200 on this invoice. Both parties win (Table 1.4).

Where did the $2,200 come from? Start from situation 2, in which the invoice is financed for 90 days, just as in situation 3. The 30 extra days by themselves cost the chain $1,800. What the program changes is the rate

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TABLE 1.4

Reverse factoring Nutmeg's $360,000 July invoice: annual rates of 10 percent (Nutmeg), 4 percent (Shoreline) and 5 percent (program); actual/360. Changes are measured against situation 1.

Situation	Nutmeg's cost	Shoreline's saving	Chain's change
1. Net 60, no program	$6,000	$0	$0
2. Net 90, no program	$9,000	$1,200	−$1,800
3. Net 90 with program	$5,000	$1,200	+$2,200

on 80 of the 90 days, 5 percent instead of 10. On $360,000 that is worth $(0.10-0.05)\times\$360{,}000\times 80/360 = \$4{,}000$, exactly the gap between Nutmeg's costs in situations 2 and 3. The $4,000 rate gain more than covers the $1,800 lost to the extension, which leaves the chain $\$4{,}000 - \$1{,}800 = \$2{,}200$ ahead of situation 1. The days destroyed value; the rate created it.

INSIGHT **The value lives in the rate gap**

Holding the terms fixed, the value a program creates on one invoice is face value × (supplier's rate − program rate) × days the bank finances / 360. For Nutmeg's July invoice that is $\$360{,}000 \times (0.10 - 0.05) \times 80/360 = \$4{,}000$. Moving days creates no value by itself; replacing an expensive rate with a cheap one does. The wider the credit gap between buyer and supplier, the larger the pie the program can share.

The program lets a supplier with weak credit raise cash on the strength of a buyer with strong credit: it sells an approved invoice rather than borrowing. We call this **credit arbitrage**; it is the central value proposition of supply chain finance. Credit markets supply the raw material: from 2005 to 2013, euro bond yields of BBB-rated issuers, the lowest investment grade (most small suppliers sit below it), stayed above those of AA-rated issuers, typical of large buyers, and the gap widened sharply in the 2008–2009 crisis.[13]

13 Source: OPIM 5115 Lecture 1, slide 40.

On the game board, the program moves two dates at once: Nutmeg's collection date jumps from day 90 to day 10, and Shoreline's payment date moves from day 60 to day 90. The segment between them still has to be financed. What changes is whose credit prices it.

You might ask why Shoreline bothered to extend its terms at all. With the program and the old 60-day terms, Nutmeg's cost would have been only $\$1{,}000 + \$360{,}000 \times 0.05 \times 50/360 = \$3{,}500$, and Shoreline would have gained nothing. The extension is how Shoreline claims part of the surplus that its own credit creates. The claim is not free: Nutmeg now pays the bank 5 percent on the 30 extra days, $\$360{,}000 \times 0.05 \times 30/360 = \$1{,}500$, to give Shoreline a saving of $1,200, so the chain keeps $2,200 of the $2,500 it would have gained at 60 days. How the pie is split, through terms, program rates and fees, is a matter of design and bargaining power, and the split explains why programs that look win-win-win on paper can go wrong. When Procter & Gamble paired a 30-day extension with exactly such a program in April 2013, The Wall Street Journal's headline read "P&G, Big Companies Pinch

Suppliers on Payments": even a win-win-win design can feel coercive to suppliers that cannot or will not join. Chapter 8 tells that story in full and studies program design in depth.[14]

When a program works, the supplier gets early, cheaper cash, the buyer gets longer payables and a steadier supply base, and the bank gets a short-term asset backed by a strong buyer. Every one of those benefits rests on two things that are easy to take for granted: that the bank keeps funding, and that everyone can see what the program really is.

14 Sources: OPIM 5115 Lecture 1, slides 30 and 41; B. C. Esty, E. S. Mayfield and D. Lane, "Supply Chain Finance at Procter & Gamble," Harvard Business School case 216-039 (2016).

1.9 Hidden Risks and the New Disclosure Rules

If reverse factoring lowers everyone's cost, what could go wrong? In 2020 two Wall Street Journal headlines gave the answer in compressed form: "Supply-Chain Finance Is New Risk in Crisis" (April 4) and "SEC Asks Boeing, Coca-Cola to Disclose More About Popular Financing Tool" (August 27).

The first risk is hidden leverage. After the program starts, Shoreline owes $360,000 on Nutmeg's July invoice for 90 days instead of 60, and the party it owes is now a bank. In economic terms Shoreline has borrowed from the bank for an extra 30 days. Yet companies have typically recorded such amounts as accounts payable rather than debt, which can make liquidity look stronger than it is: the program boosts working capital without adding to reported borrowing.[15] Until recently, companies were not even required to say whether they ran such a program.

15 Source: K. Broughton, The Wall Street Journal, July 20, 2022.

The second risk is dependence on the funder. Every benefit in Table 1.4 assumes that the bank keeps buying approved invoices. If Shoreline's funder withdrew, the 90-day terms would stay but the early payment would vanish, and every new invoice would fall back from situation 3 to situation 2 overnight: $9,000 of financing cost on every invoice like the July one instead of $5,000. Worse, as unsold invoices piled up over the next 80 days, Nutmeg would have to finance 90 days of Shoreline receivables instead of 10, an extra $800,000. The suppliers that a program helps most are the ones least able to absorb its sudden end. On the game board, the funder's exit hands the segment from day 10 to day 90 back to Nutmeg, one new invoice at a time.

Both risks materialized at once in March 2021.

CASE IN POINT 1.3 Greensill Capital and the new disclosure rules

Greensill Capital, a U.K.-based supply chain finance provider with 16 offices worldwide, financed billions of dollars of supply chain transactions, some through programs much like Shoreline's and, increasingly, some against *future receivables* that no buyer had yet approved, packaged into investment funds and protected by credit insurance. When its insurer refused to renew cover, the funds that bought its paper were frozen, and in March 2021 Greensill filed

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CASE IN POINT 1.3 *continued*

for insolvency. Its Germany-based bank was closed by regulators, investors lost billions, and companies that relied on its programs lost access to that funding overnight.

The collapse intensified regulatory scrutiny of the industry. In September 2022 the U.S. Financial Accounting Standards Board issued disclosure requirements for companies that use supplier finance programs.[16] They took effect for fiscal years beginning after December 15, 2022 (calendar 2023 for most companies), and the roll-forward followed one year later. Buyers must disclose the amount they have confirmed to the finance provider that is still outstanding at period end; describe the key terms, including the payment terms and any assets pledged or guarantees given; say where on the balance sheet the obligations appear; and present a roll-forward of the obligations during the year.

Coca-Cola's 2023 Form 10-K shows what such a disclosure looks like. "Our current payment terms with the majority of our suppliers are 120 days," it states. Two global financial institutions offer a voluntary program, and Coca-Cola and its suppliers agree on prices, quantities and payment terms regardless of whether a supplier joins. Obligations that Coca-Cola had confirmed to the banks under the program stood at $1,421 million at December 31, 2023, up from $1,351 million a year earlier, and are included in accounts payable.[17]

16 The board approved the rule in July 2022 and issued it as Accounting Standards Update 2022-04; Chapter 9 covers it in full.

17 Sources: OPIM 5115 Lecture 1, slides 44 to 47; The Coca-Cola Company, Form 10-K for 2023. Figures as of December 31, 2023.

Greensill teaches four lessons: supply chain finance is powerful but not risk-free, and it demands transparency about the underlying transactions, an honest assessment of risk, proper disclosure in the financial statements, and diversification so that no firm depends on a single funding source. Notice that none of these lessons condemns the instrument. Where Greensill financed genuine approved invoices, reverse factoring did what it was designed to do. The damage came from stretching the label to receivables that did not yet exist, from how little of it was visible, and from how much of it rested on one funding channel and its insurers. Chapter 9 tells the full Greensill story, explains when program payables should be treated as debt, and teaches how to read a disclosure note like Coca-Cola's.

THINK ABOUT IT

Coca-Cola stresses that its payment terms are the same whether or not a supplier joins the program. Why might that one sentence matter both to a supplier deciding whether to join and to an investor deciding whether the company's payables are really debt?

1.10 The Game Board: A Roadmap of the Book

We have met a dozen instruments, a handful of real companies and one fictitious juice bottler. Is there a single picture that holds them together and tells us where each chapter of this book fits?

There is, and we have already drawn it. The cash conversion cycle of Figure 1.6 has four dates and three segments, and every instrument in this chapter turned out to be a move on that board. A trade credit term sets the date on which the supplier is paid. An early-payment discount pays the buyer to move that date earlier: under 2/10 net 30 the supplier gives up 2 percent to be paid 20 days sooner. A term extension moves it later and hands the financing of the extra days to the supplier. Reverse factoring moves two dates at once, the supplier's collection earlier and the buyer's payment later, and puts a bank in between at the buyer's rate. Inventory finance and receivables finance, which later chapters develop, turn the DIO and DSO segments into cash before their time. Nutmeg's CFO can now read all three of the morning's messages on one board.

EXAMPLE 1.6 Three moves on Nutmeg's game board

Each move starts from the position the previous move left, and the scoreboard at the end tracks the position. The annual cost of carrying a balance is the balance times the financer's rate, as in Exhibit 1.2.

Move 1: Shoreline's extension. Nutmeg's collection date on \$10,000 a day of Shoreline sales moves 30 days later, so receivables rise by \$300,000 to \$1,100,000. Nutmeg finances the extra segment at 10 percent, \$30,000 a year (Example 1.4).

Move 2: joining Shoreline's program. The collection date on Shoreline invoices moves from day 90 to day 10. Nutmeg finances 10 days of Shoreline sales (\$100,000) itself, and the bank finances the other 80 days (\$800,000) at 5 percent. If the sale to the bank is a true sale, the invoices leave Nutmeg's balance sheet (Chapter 7 explains when that holds), and receivables fall to \$200,000 + \$100,000 = \$300,000. Carrying Shoreline's receivables now costs

$$\$100{,}000 \times 0.10 + \$800{,}000 \times 0.05 = \$50{,}000 \text{ a year,}$$

against \$60,000 at net 60 and \$90,000 at net 90 without the program.

Move 3: taking Quinnipiac's discount. Nutmeg's payment date on \$3,000 a day of bottle purchases moves from day 30 to day 10, so payables fall by $\$3{,}000 \times 20 = \$60{,}000$ to \$340,000: DPO falls and the cycle *lengthens*. Yet Nutmeg captures $0.02 \times \$1{,}095{,}000 = \$21{,}900$ of discounts a year and pays about $0.10 \times \$60{,}000 = \$6{,}000$ of extra interest, a net gain of about \$15,900.

			Days		
Position	AR (\$)	AP (\$)	DSO	DPO	CCC
Today	800,000	400,000	58.40	48.67	70.56
After move 1	1,100,000	400,000	80.30	48.67	92.46
After move 2	300,000	400,000	21.90	48.67	34.06
After move 3	300,000	340,000	21.90	41.37	41.36

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EXAMPLE 1.6 *continued*

Days use a 365-day year: DSO is receivables over $5,000,000 of sales and DPO is payables over $3,000,000 of COGS, each times 365. DIO stays at 60.83 days, so CCC = 60.83 + DSO − DPO.

The third move makes Nutmeg's cycle 7.3 days longer and Nutmeg richer: the credit it gives up costs 36.7 percent a year, and the credit it uses instead costs 10. The cycle is a means, not an end. The goal is to finance each segment of the board as cheaply and as safely as possible, and Exhibit 1.3 turns that goal into a procedure.

EXHIBIT 1.3 A procedure to read a financing instrument as a move on the game board

Step 1. Identify the flow the instrument rides on: goods in inventory, an invoice (the seller's receivable and the buyer's payable) or a trade document.

Step 2. Mark which date moves, for whom and by how many days: the supplier's collection date (DSO), the buyer's payment date (DPO) or the date inventory becomes a sale (DIO).

Step 3. Convert the days into dollars with the daily flow: sales for receivables, purchases or COGS for payables and inventory.

Step 4. Identify who finances the segment after the move and whose credit sets the rate.

Step 5. Price the move for each party at its own rate, and add the results for the chain.

Step 6. Ask who is left holding the segment if the financing disappears, and whether they can carry it.

Step 6 is the Greensill question. If Shoreline's funder walked away, early payment would stop overnight; over the next 80 days, as unsold invoices piled up, Nutmeg's DSO would climb from 21.9 back to 80.3 days and its credit line would have to carry an extra $800,000.

The same board organizes the rest of the book (Figure 1.11). Part I builds it: Chapter 2 measures the cycle from financial statements, and Chapter 3 asks what the inventory segment is worth. Part II studies the traditional moves: lending against inventory (Chapter 4), trade credit (Chapter 5) and cross-border payment (Chapter 6). Part III studies programs in which a funder bridges the supplier's collection and the buyer's payment (Chapters 7 and 8) and asks when the bridge becomes hidden debt (Chapter 9). Part IV shakes the board with commodity and disruption risk (Chapter 10) and plays every move at once in The Cool Connection, a team simulation of a

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manufacturing company (Chapter 11). Part V extends the board to deep tiers (Chapter 12) and green pricing (Chapter 13). Part VI follows it into the AI buildout: the commitments that finance scarce capacity (Chapter 14), the leases, bonds and power contracts that pay for the buildings (Chapter 15), and the loans and guarantees behind the chips (Chapter 16).

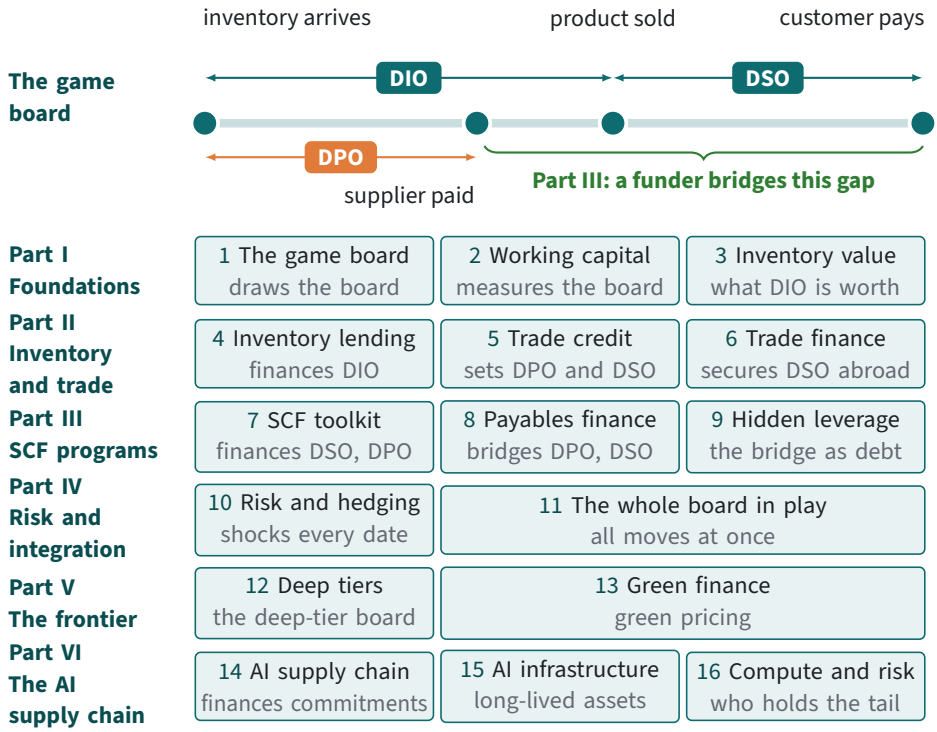


FIGURE 1.11
The book on the game board. Each tag gives a chapter and the move on the board that it studies. Part I builds the board; Part II covers the traditional moves on inventory, payment terms and cross-border collection; Part III covers programs in which a funder bridges the gap between paying suppliers and collecting from customers; Part IV adds risk and plays every move at once; Part V extends the board to deep tiers and green pricing; Part VI follows it into the AI buildout.

Trent's closing agenda for the discipline serves as the agenda for this book: appreciate the language of finance, apply its tools to decisions and risk, demonstrate the value of supply chain work in the indicators executives watch, and keep learning as instruments and technologies change. Nutmeg's CFO has already taken the first step: three routine messages turned out to be three moves on a board, each with a price, a winner and a risk.

CHAPTER SUMMARY

A supply chain is everyone who takes part, directly or indirectly, in filling a customer's order, and it carries three flows: goods move downstream, money moves upstream and information moves both ways. The flows do not move at the same speed. Goods arrive long before the money that pays for them, and every gap between the two has to be financed by someone: the seller who waits, the buyer who pays early, or a bank that steps in for a price.

Supply chain managers and executives speak different languages. Everything a supply chain does ends up on the balance sheet, the income statement or the cash flow statement. The evidence that markets punish

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supply chain glitches is strong, and the evidence that they reward supply chain excellence is suggestive. An improvement measured in turns or days persuades an executive only once it is translated into cash released and profit earned.

Most trade between firms runs on trade credit and open account. Terms such as 2/10 net 30 hide an expensive loan: a buyer that skips the discount borrows from its supplier at a high implied rate, and should pay early whenever it can borrow for less. Across borders and with unfamiliar partners, letters of credit, collections and cash in advance shift payment risk toward the buyer, but over recent decades trade has moved steadily toward open account.

Working capital is the stock of money tied up in operations, and the cash conversion cycle measures how long it stays there: days of inventory plus days of receivables minus days of payables. The cycle reflects the business model as much as efficiency, and it serves as the game board of this book, on which every instrument moves a date, changes who finances a segment, or changes whose credit sets the price.

Because one firm's payables are another firm's receivables, payment terms are a tug-of-war. When a powerful buyer extends its terms, the financing burden shifts to suppliers that usually borrow at higher rates, and the chain as a whole loses. The root cause often lies inside the buyer, where finance and supply management pursue conflicting goals.

Supply chain finance uses financing and risk mitigation to optimize the working capital invested in supply chain processes. Its umbrella covers payables-centric, receivables-centric and pre-invoice techniques, plus related tools such as letters of credit and asset-based lending. Reverse factoring, the archetype, lets a supplier sell approved invoices at a price set by its buyer's credit; the value it creates comes from that credit gap, and how the value is shared depends on design and bargaining power.

The same programs can hide leverage in accounts payable and leave suppliers dependent on a single funder. The collapse of Greensill Capital exposed both risks and led to new disclosure rules for supplier finance programs.

MANAGERIAL TAKEAWAYS

- Translate every supply chain improvement into cash released and annual profit before presenting it: units to days, days to dollars, dollars to a line on a financial statement.
- Treat an early-payment discount as a loan offer. Skipping 2/10 net 30 costs roughly 37 percent a year, so pay early whenever you can borrow for less.

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- Before imposing or accepting a payment-term change, price both sides: the buyer's gain at its rate and the supplier's cost at its rate. When the supplier's rate is higher, the chain loses.
- Benchmark the cash conversion cycle against firms with the same business model, and remember that a negative cycle usually means someone else is financing you.
- For any financing instrument, ask four questions: which date moves, who finances the segment, whose credit sets the rate, and who is left holding the segment if the funding stops.
- Judge a supply chain finance program by the credit gap it exploits, not by the days it moves; a program that only moves days creates nothing for the chain.
- Design programs that suppliers would join voluntarily, keep terms independent of participation, disclose the program and avoid relying on a single funder.

KEY TERMS

Burden shift	The transfer of working capital and its financing cost from a buyer to its suppliers when the buyer lengthens payment terms.
Cash conversion cycle	The number of days between paying suppliers for inventory and collecting cash from customers: DIO plus DSO minus DPO.
Communication challenge	The gap between the operational metrics supply chain managers report and the financial indicators executives act on.
Credit arbitrage	Creating value by financing a weaker firm's working capital at a rate based on a stronger firm's credit.
Days inventory outstanding	Inventory divided by COGS, times 365: how long goods wait to be sold.
Days payables outstanding	Accounts payable divided by COGS, times 365: how long a firm takes to pay.
Days sales outstanding	Receivables divided by credit sales, times 365: how long customers take to pay.
Early-payment discount	A price reduction for payment within a stated period, such as 2 percent within 10 days.
Game board	This book's name for the cash conversion cycle timeline, on which every instrument is a move.
Net working capital	Accounts receivable plus inventory minus accounts payable (this book's operating definition; many fi-

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	nance texts use the term for current assets minus current liabilities).
Open account	Settlement in which the seller ships and invoices and the buyer pays on agreed terms, with no bank instrument behind the payment.
Operating cycle	The time from receiving inventory to collecting cash from its sale: DIO plus DSO.
Perfect order	An order delivered complete, on time and undamaged.
Reverse factoring	A buyer-led program in which a bank buys a supplier's approved invoices at a discount priced on the buyer's credit; also called payables finance.
Supply chain	Everyone who takes part, directly or indirectly, in filling a customer's order.
Supply chain finance	The use of financing and risk mitigation practices and techniques to optimize the working capital and liquidity invested in supply chain processes and transactions.
Supply chain glitch	A disruption in matching supply with demand, such as a production delay or a shipment failure.
Supply chain management	The management of the two-way flows of materials, services, funds and information through a supply chain.
Three flows	The movement of goods downstream, money upstream and information in both directions.
Trade credit	A short-term arrangement in which a seller lets a buyer receive goods now and pay later.
Working capital	Current assets minus current liabilities; the capital used in day-to-day operations.

DISCUSSION QUESTIONS

1. Pick a product you bought this week and sketch its supply chain as three flows. At which link do you think money waits longest, and who finances that wait?
2. Nutmeg's plant manager raised inventory turns from 6 to 8 and the CFO did not react. Draft the two-sentence message that would have made the CFO react, and explain why it works.
3. The BusinessWeek comparison and the Singhal and Hendricks study both link supply chain performance to financial results. Which is more convincing evidence of cause and effect, and why?

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4 In 2012 Verizon's cash conversion cycle was −72 days and Tiffany's +486. Is either a sign of good or bad management? What would you need to know to decide?

5 A large retailer argues that extending its payment terms is fair because suppliers can always borrow. Using the burden-shift arithmetic, explain when that argument fails and when an extension could create value for the chain.

6 Why might a supplier refuse to join a reverse factoring program that would lower its financing cost? Consider the information it must share, the dependence it accepts and the precedent it sets.

7 One reading of Greensill is that the collapse came from the misuse of supply chain finance rather than from the tools themselves. Do you agree? Which of the four lessons (transparency, honest risk assessment, disclosure, diversification) would have protected a supplier most?

8 Rank the four payment methods in Table 1.3 by the risk they leave with the seller. Nutmeg then receives its first order from a distributor in Montreal it has never dealt with. Which method would you propose first, what might it cost the relationship, and how might supply chain finance let Nutmeg offer open account anyway?

9 Choose two techniques from Figure 1.9 other than reverse factoring, for example factoring and inventory finance. Use Exhibit 1.3 to read each one as a move on the game board: which date moves, who finances the segment afterward, and whose credit sets the price?

PROBLEMS

1.1 **Pine Hollow Hardware**, a regional distributor of tools and fasteners, reports these year-end balances: cash $80,000, accounts receivable $240,000, inventory $310,000, prepaid expenses $20,000, accounts payable $190,000, accrued wages $60,000 and a short-term bank loan of $100,000.

(a) Compute working capital (current assets minus current liabilities).

(b) Compute net working capital (receivables plus inventory minus payables).

(c) Pine Hollow uses $50,000 of cash to pay suppliers. Recompute both measures.

1.2 **Bluefin Seafood Supply** receives a $48,000 invoice from its packaging supplier on terms of 1/10 net 30.

(a) How much does Bluefin pay if it pays on day 10? On day 30?

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(b) What is the rough annual cost of skipping the discount (simplified formula, 360-day year)?

(c) Bluefin can borrow on a bank line at 8 percent a year. If it borrows the day-10 payment for 20 days to take the discount, how much interest does it pay (actual/360), and what is its net gain?

1.3 **Lakeshore Ceramics** reports average inventory of $600,000, COGS of $3,600,000, accounts receivable of $450,000, credit sales of $4,500,000 and accounts payable of $300,000. Using a 365-day year, compute its DIO, DSO, DPO, cash conversion cycle and operating cycle.

1.4 **Fairhaven Textiles** has COGS of $4,380,000 and accounts payable of $360,000. Its yarn supplier sells it $1,825,000 a year on terms of 2/10 net 40, and Fairhaven pays on day 40. Fairhaven borrows at 9 percent a year. Use a 365-day year for days, value payables at invoice amounts, assume any extra borrowing equals the fall in payables, and take the annual cost of carrying a balance as the balance times the rate.

(a) Compute Fairhaven's DPO.

(b) What is the rough annual cost of skipping the discount (simplified formula, 360-day year)?

(c) If Fairhaven starts paying the yarn supplier on day 10, by how much do its payables fall? What are its new DPO and the change in its cash conversion cycle?

(d) What discount does Fairhaven capture each year, what extra interest does it pay, and what is its net annual gain?

1.5 **Cobalt Bay Snacks** has COGS of $14,600,000 a year and turns its inventory 5 times a year. A new demand-planning system raises turns to 8. Cobalt Bay borrows at 9 percent a year.

(a) Compute average inventory before and after, and the cash released.

(b) Compute days of inventory before and after (365-day year).

(c) Compute the annual interest saved on the released cash.

1.6 **Zephyr Online Retail** has annual revenue of $9.125 billion and COGS of $7.3 billion. It holds 20 days of inventory, collects from card processors in 5 days and pays suppliers in 70 days.

(a) Compute Zephyr's cash conversion cycle.

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(b) Compute inventory and payables from daily COGS and receivables from daily revenue (365-day year), and Zephyr's net working capital.

(c) If revenue and COGS grow by 10 percent with the same days, how much cash does working capital release?

1.7 **Kestrel Home Stores** buys $2.4 billion of merchandise a year and extends its supplier terms from 30 to 60 days. Kestrel's cost of short-term funds is 3.5 percent; its suppliers finance receivables at 12 percent. Using a 365-day year, compute:

(a) the increase in Kestrel's payables;

(b) Kestrel's annual saving;

(c) the suppliers' annual financing cost;

(d) the chain's net annual effect.

1.8 **Tidewater Brewing** extended its supplier terms from 45 to 90 days during the year, and its trade payables rose from $2.10 billion to $2.62 billion. Assume its suppliers finance at 11 percent and Tidewater at 4 percent.

(a) How much working capital did Tidewater free?

(b) Compute Tidewater's annual saving, the suppliers' annual cost and the chain's net annual effect.

(c) If the entire increase in payables came from the extension, what annual purchases does it imply (365-day year)?

(d) At what supplier financing rate would the chain break even?

1.9 **Oriole Apparel** launches a reverse factoring program. Enrolled suppliers sell $50 million a year of approved invoices to Oriole's bank, each financed for 60 days on average. The program rate is 4.25 percent; the suppliers' own borrowing rate is 8.5 percent. Use actual/360 throughout.

(a) What discounts do the suppliers pay under the program each year?

(b) What would it cost them to finance the same receivables at their own rate, and what do they save?

(c) Oriole extends its terms by 30 days when the program starts, so invoices are financed for 90 days. What discounts do the suppliers now pay, how much do they save compared with financing 60 days at their own rate, and what is Oriole's annual gain at its 3.5 percent cost of funds?

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1.10 **Alder Creek Furniture** invoices **Summit Home Stores** $720,000. Summit approves invoices on day 15. Alder Creek finances itself at 9 percent, Summit at 3.5 percent, and Summit's reverse factoring program charges 4 percent. Use actual/360.

(a) Situation 1, net 45 with no program: what is Alder Creek's financing cost on the invoice?

(b) Situation 2, net 75 with no program: what is Alder Creek's cost, and what are the 30 extra days worth to Summit?

(c) Situation 3, net 75 with the program: Alder Creek finances days 0 to 15 itself and sells the approved invoice for the remaining 60 days. What is its total cost, and how much cash does it receive on day 15?

(d) Compared with situation 1, how much does Alder Creek save, and how much does the chain gain? How much does Alder Creek save compared with situation 2?

1.11 **Copperline Packaging** sells $7,300,000 a year on credit, with COGS of $5,110,000, average inventory of $700,000, accounts payable of $490,000 and accounts receivable of $1,000,000. Of the receivables, $450,000 is owed by its largest customer, **Larkspur Foods**, which buys $3,650,000 a year on net 45. Copperline borrows at 9 percent. Use a 365-day year, and take the annual cost of carrying a balance as the balance times the rate.

(a) Compute Copperline's DIO, DSO, DPO and cash conversion cycle.

(b) Larkspur moves to net 90. Compute Copperline's new receivables, DSO and cycle, and the extra annual carrying cost.

(c) Larkspur then offers reverse factoring: invoices approved on day 15 can be sold at 4.5 percent for the remaining 75 days, and sold invoices leave Copperline's balance sheet. Compute the new receivables, DSO and cycle, and the annual cost of carrying Larkspur's receivables. Compare it with the cost before the extension and with the cost after the extension without the program.

(d) A year later the bank behind Larkspur's program withdraws, and Larkspur keeps net 90. Once the unsold invoices have built up, what are Copperline's receivables, DSO and cycle? How much more must its credit line carry than under the program, and what does carrying Larkspur's receivables now cost a year?

1.12 **Godwit GPU Cloud** buys GPU servers for $600 million, pays the vendor on delivery and finances the payment on a credit line at 10 percent a year. Installation takes six months; then a customer rents the whole cluster for $15 million a month, invoiced at the end of

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each month of service and paid 30 days after each invoice. Use 30.4 days a month and interest of balance times rate times months/12, and assume the credit line stays fully drawn all year.

(a) Counting from delivery, on which days does service start, does the first invoice go out and does the first rent arrive? How long is Godwit's money out before any cash returns?

(b) How much interest accrues during installation, while the cluster earns nothing?

(c) Better planning cuts installation to four months. Repeat (a) and (b). How much shorter is the gap, and how much less do the idle months cost?

(d) How much more rent does Godwit earn in its first 12 months? Its CFO adds this to the interest saved in (c) and calls the sum the first-year gain. What is the gain, and why is the sum too high?

FURTHER READING

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CHAPTER 2

Cash Is a Supply Chain Decision

Financial Statements, Working Capital, and the Cash Conversion Cycle

In January 2026 the finance team of Charter Oak Confectionery Co., a chocolate and candy maker in Hartford, Connecticut, closed the books on a record year: revenue of $1.2 billion and net income of $81 million. Cash, however, rose by exactly $1 million. The head of procurement has a fix. Pay every supplier 26 days later, and the company keeps about $51 million that it would otherwise hand over. One of those suppliers, a family-owned cocoa processor that ships Charter Oak two-fifths of its output, is already drawing most of its bank line. Is the longer payment term free money, a loan in disguise, or a way to lose a critical supplier? To answer, we have to read the financial statements the way an operations manager should.

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LEARNING OBJECTIVES

1. Locate the supply chain footprint on the balance sheet, income statement and statement of cash flows, and explain why operating cash flow differs from net income.
2. Compute days inventory outstanding, days sales outstanding, days payables outstanding and the cash conversion cycle from financial statements, stating the denominators, averaging and day count used.
3. Convert working capital days into dollars of cash and free cash flow, and read industry benchmarks such as the J.P. Morgan Working Capital Index.
4. Compute current, quick and cash ratios and explain why liquidity and working capital efficiency can point in opposite directions.
5. Trace a supply chain decision through the ROTA-CL (DuPont) tree to its effect on margin, capital turnover, return and liquidity.
6. Assess a supplier's financial health with ratio analysis and the Altman Z-score, and explain why one firm's working capital gain can become its supplier's liquidity crisis.

2.1 Three Statements, One Supply Chain

Why should someone who negotiates cocoa contracts, schedules a molding line or runs a distribution center care how accountants arrange numbers? Chapter 1 argued that every supply chain decision eventually lands on a financial statement. The useful question is where it lands, because that is where the finance function will look for it. We answer it with the books of Charter Oak.

Charter Oak makes chocolate bars, hard candy and seasonal assortments for Halloween, the winter holidays and Easter. It buys cocoa products, sugar, milk powder and packaging, runs two plants in Connecticut, and ships to grocery chains, mass merchants and drugstores on 30-day terms. Its shares are listed; at the end of 2025 the market valued its equity at $600 million, about 1.6 times its book value. Two partners matter in this chapter: its largest customer, Elmstead Grocers, and its largest supplier, Cocoa Coast Ingredients, the family-owned cocoa processor of the opening. Table 2.1 shows Charter Oak's income statement for fiscal 2025 and its balance sheets at the end of 2025 and 2024.[1]

1 Charter Oak and its trading partners are fictitious. All of their figures are in millions of dollars unless stated.

The **balance sheet** is a snapshot of what the firm owns and owes at one moment: assets equal liabilities plus shareholders' equity, or 730 = 100 + 250 + 380 for Charter Oak at the end of 2025. Current assets are cash and items expected to be turned into cash, sold or used up within a year, or within the normal operating cycle if that is longer: cash, receivables (amounts customers owe for candy already delivered), inventories and prepaid items. A manufacturer's inventory has three layers. Charter Oak's raw materials

CHAPTER 3

Inventory on the Books

Valuation Choices and What Inventory Reveals

> *At times like these, gross margins get inflated, earnings accelerate, but cash flow is often negative as inventories build.*
>
> David Berman, quoted in *Barron's*, December 13, 2004

On the morning of April 11, 2005, David Berman, a hedge fund manager who had built his reputation by reading retailers' inventories, was looking at a valuation of Jos. A. Bank Clothiers that said buy. The numbers were handsome: sales up 24 percent in fiscal 2004, a gross margin that had risen every year since fiscal 1999, to 60 percent, and a share price of $34.37 that the model called too low. Berman's eye went to a different line. Since fiscal 1998 the company's sales had doubled, but its inventory had nearly tripled, and a suit now waited about 300 days in a store or warehouse before it sold. Was the margin real, or was part of it sitting in the stockroom?

This chapter explains how a pile of goods becomes a number on the balance sheet, how that number can flatter earnings, and how an investor, a lender or a supply chain manager can tell. We learn the mechanics on one overcoat at Pemberton Outfitters, a small fictitious retailer, then return to Jos. A. Bank to use them.

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LEARNING OBJECTIVES

1. Explain why firms hold inventory, what holding it costs, and which costs and which goods belong in the inventory account.
2. Compute cost of goods sold, ending inventory and gross profit under FIFO, LIFO (periodic and perpetual) and weighted average cost, show the range specific identification opens to a manager, and trace each method's effect on taxes and reported income.
3. Restate a LIFO company's statements to an as-if FIFO basis with its LIFO reserve.
4. Apply the lower of cost or net realizable value rule and show how the timing of markdowns moves profit between years.
5. Compute inventory turnover, GMROI and the three inventory signals investors watch, and use them to estimate excess inventory.
6. Separate the mechanical and revision effects of an inventory write-off on firm value.

3.1 Inventory as a Capital Commitment

Chapter 2 measured inventory as a clock: days inventory outstanding (DIO), the number of days a dollar of cost waits on a shelf before it becomes a sale. A shorter clock releases cash, which is why DIO is one of the three levers of the cash conversion cycle. Why, then, does any firm hold inventory at all? And once it does, why should an investor or a lender care about the particular number that the accountants print next to the word "Inventory"?

Firms hold inventory for five operating reasons. They want to meet customer demand promptly, because a shopper who finds an empty rack buys elsewhere. They buffer against uncertainty in both demand and supply. They capture economies of scale by buying, shipping and producing in large lots. They smooth production when demand is seasonal. And they sometimes hedge, buying ahead of an expected price increase. Each reason is a decision, and each has a price. A rule of thumb taught in operations courses puts the annual cost of holding inventory at about 30 percent of its value, counting the cost of the capital tied up, storage, insurance, obsolescence and depreciation.[1] A retailer with $10 million of inventory therefore bears roughly $3 million a year in holding costs, part of it the return the tied-up capital could have earned elsewhere. How much to order and when are questions for operations models; here we take the quantities as given and ask what the books say about them.

1 Source: OPIM 5115 Lecture 3. The 30 percent figure is a rule of thumb for U.S. manufactured goods; actual holding costs vary widely by product.

Pemberton Outfitters

Pemberton Outfitters is a specialty retailer in Portland, Maine, that sells one product: a heavy wool overcoat, offered through a single store and a

PART

II

Financing Inventory and Trade

Goods in a warehouse, an invoice waiting to be paid, a container on the ocean: each is cash in disguise. Part II studies the traditional tools that unlock it: inventory-based lending, trade credit, and the instruments of international trade finance.

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CHAPTER 4

Lending Against Inventory

Collateral, Borrowing Bases, and the Bank That Ages Cheese

It's a very physical business because it involves the physical movement of a product, it's an old fashioned but good approach.

Adolfo Bizzocchi, chief executive of Credem, quoted in Trichakis, Tsoukalas and Moloney (2015)

In the Parmigiano Reggiano district of northern Italy, a regional bank runs two warehouses with room for 440,000 wheels of the cheese (as of 2014). Custom machines roll down the aisles, turning and brushing the wheels while they age. Many of those wheels belong to small dairy cooperatives that have borrowed against them from the warehouse's owner, Credito Emiliano, which has accepted young cheese as collateral since 1953. Why would a bank go into the business of aging cheese?

Because for these borrowers the cheese is the only asset a lender might trust, and trusting cheese is harder than it looks. This chapter follows a fictitious cooperative, Collina Dairy, and its equally fictitious lender, Valdora Bank, from the day the loan is sized to the October morning when the covenant breaks: sell now, or wait?

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LEARNING OBJECTIVES

1. Explain why long-cycle producers cannot fund inventory with trade credit, cash-flow loans or fixed-asset loans, and why lending against the inventory fits.
2. Compute collateral value, borrowing base, availability and loan-to-value, including a retail "lesser of" formula based on net orderly liquidation value.
3. Explain how warehouses, warehouse receipts, collateral managers and lender-owned facilities reduce the risks of inventory collateral.
4. Test a collateral maintenance covenant and size the cure in collateral or cash.
5. Solve a liquidate-or-wait decision tree with a real option and report its credit risk.
6. Screen a product with the monitorability checklist, and describe how U.S. retail inventory loans are appraised, secured and liquidated.

4.1 Collina Dairy Cooperative

Chapter 3 valued inventory as an accountant does, at cost or at net realizable value. A lender asks a blunter question: if this borrower fails, what will the goods fetch, and how quickly? Few businesses depend on the answer more than a dairy whose product takes a year to make, because for that year the goods are almost all it owns.

Collina Dairy Cooperative is a fictitious cooperative of 20 family dairy farms in the hills of Emilia-Romagna (*collina* is Italian for hill). The members deliver milk twice a day, and the dairy turns about 6.7 million liters a year into 12,000 wheels of Parmigiano Reggiano of about 40 kilograms each, or 480,000 kilograms of cheese. The product carries a Protected Designation of Origin (PDO), so the rules are strict: the milk must come from the designated area and every wheel must age for at least 12 months. At 12 months the producers' consortium inspects each wheel, tapping it with a hammer, X-raying it and, where needed, sampling it with a screw needle; wheels that pass receive the fire brand, and wheels that fail are marked so they can never be sold as first grade.[1] Collina sells its wheels to wholesalers right after the inspection, and the wholesalers age the best of them further.

Table 4.1 collects Collina's numbers. At €8.50 per kilogram for 12-month cheese, revenue is 480,000 × 8.50 = €4,080,000 a year. Milk payments to members absorb €3,120,000, or 76.5 percent of revenue, and after processing, warehouse fees and overhead the cooperative keeps EBITDA of €200,000, a margin of 4.9 percent. The thin margin is by design: a cooperative exists to pay its members for their milk, so it passes its earnings through to the farms and holds almost nothing in reserve.

Now look at the timing. Collina pays its members at month end, about 15 days after delivery on average, cannot sell the cheese for 12 months, and waits about 30 days for wholesalers to pay. In the language of Chapter 2,

1 Source: the production rules and the role of the consortium (founded in 1934) are summarized from Trichakis, Tsoukalas and Moloney (2015), pages 2 to 4.

CHAPTER 5

Trade Credit

The Bank Inside the Supply Chain

Every December the melon trucks roll into Mercado Sol's distribution center outside São Paulo, and every December Helena Costa, the grocer's director of supplier finance, looks at the same line in her spreadsheet. Mercado Sol pays its largest melon grower, Fazenda Rio Verde, thirty days after each delivery. Some of its peers pay in ninety. Her bonus this year depends on the value she can create from new payment terms, and sixty more days would leave tens of millions of reais in Mercado Sol's account for much of the year.

Some two thousand kilometers to the northeast, the Rocha family finances the same melons with a bank line that costs 17.5 percent and must be paid off completely for a month every year. Every real that Mercado Sol keeps a little longer is a real that Rio Verde must borrow. Who should finance the melons between the field and the checkout counter? Is the answer simply a matter of who holds the power, or does it depend on what money costs on each side of the trade?

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LEARNING OBJECTIVES

1. Decode a credit term such as 2/10 net 30, price the forgone discount two ways, and decide whether to take it.
2. Explain why suppliers lend, and plan a seasonal supplier's cash budget and credit line.
3. Build a pro forma payables schedule and measure each party's gain and the deadweight loss of new terms.
4. Test a supplier's covenants and value it with a one-year discounted cash flow and economic value added.
5. Combine payment timing with price, and solve for the break-even discount.
6. Use the Modigliani-Miller checklist, bargaining power and contagion risk to set trade credit policy.

5.1 Mercado Sol, Fazenda Rio Verde, and a Proposal to Pay Later

Who pays for the weeks a melon spends between the field and the checkout counter? For Mercado Sol and its largest melon grower, the fine print of an invoice is about to become a negotiation.

Mercado Sol S.A. is a listed Brazilian grocer with about 900 stores and R$38 billion of annual sales. Its surplus cash earns the interbank base rate of 11.5 percent, and it can borrow at about 12.2 percent.[1] Fazenda Rio Verde Ltda. is a family-owned melon grower in Brazil's semi-arid northeast, run by Paulo Rocha. It expects to sell R$202.3 million of melons this year, 75 percent to Mercado Sol and the rest to small regional grocers. January, February, November and December carry 79 percent of revenue, while the next crop absorbs cash from June to November.

Rio Verde finances that gap with a revolving line of credit, the revolver of Chapter 4, at Banco Mandacaru, priced at the base rate plus 6.0 points, or 17.5 percent. The line is capped at R$95 million and carries a **cleanup provision**: it must be fully repaid for at least 30 consecutive days every year, so that the bank never becomes a permanent source of capital. Rio Verde keeps a minimum of R$2.0 million of cash, which earns the base rate, and plans R$20 million of irrigation and cold storage, R$10 million in each of April and May. Both firms pay 34 percent income tax. The Rocha family requires 24 percent a year on its capital, quoted like the bank line as 2 percent a month (26.8 percent once compounded). That is well above the bank's rate, because the family's wealth is tied up in one farm (Section 5.10 returns to this). The family has no more capital to put in, and it holds an offer of R$45 million for the land and equipment, with working capital to be bought separately at book value.

1 Mercado Sol, Fazenda Rio Verde, Banco Mandacaru and every person in them are fictitious. The structure of the example follows the Grupo Pão de Açúcar case cited in Further Reading; all numbers are original.

CHAPTER 6

Financing Trade Across Borders

Letters of Credit, Collections, and the Road to Open Account

> *For exporters, any sale is a gift until payment is received... For importers, any payment is a donation until the goods are received.*
>
> U.S. Department of Commerce, International Trade Administration, *Trade Finance Guide* (2008)

The email reaches Gulfport Protein Exports in New Orleans on a Monday morning. A food distributor in Ho Chi Minh City that Gulfport has never sold to wants four refrigerated containers of frozen chicken leg quarters, worth $120,000, and asks to pay 60 days after the ship arrives. Gulfport would earn about $6,000 on the order. If the distributor never pays, Gulfport loses twenty times that. The distributor has its own worry: if it pays first, what guarantees that the right chicken, at the right temperature, ever leaves Louisiana? The two firms sit half a world apart, under different laws and currencies, and someone has to move first. This chapter is about the instruments that let them trade anyway, and about what each one costs.

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LEARNING OBJECTIVES

1. Explain why distance, time and law magnify payment risk in cross-border trade, and place cash in advance, letters of credit, documentary collections and open account on the risk ladder.
2. Compute the all-in cost of a payment term for one shipment, for the exporter, the importer and the pair, from financing days, bank fees, insurance premiums and expected credit losses.
3. Describe how documentary collections and letters of credit work, the rules that govern them (UCP 600, URC 522, ISP98), and value a banker's acceptance.
4. Use the Incoterms 2020 rules to locate where cost and risk pass from seller to buyer and align them with the payment terms.
5. Choose payment terms for a new foreign customer, stress-test an existing customer for a currency shock, and design a collection response.
6. Explain why open account came to dominate cross-border trade and how export credit insurance, export credit agencies, factoring, forfaiting and supply chain finance support it.

6.1 Gulfport's First Order

Why can Gulfport not sell to Ho Chi Minh City the way it sells to a wholesaler in Houston? At home a supplier ships, invoices and waits 30 days, and the arrangement works because both parties share a court system, a currency and a web of credit bureaus. Chapter 5 studied that world of trade credit. Cross a border and each of those supports weakens at once. The tools that rebuild them are called **trade finance**: the payment terms and financing arrangements that decide who carries the risk of an international sale and who finances the goods while they travel.

Gulfport Protein Exports is a fictitious New Orleans exporter of frozen poultry and pork.[1] It sells about $520 million a year to customers in Asia, Latin America and the Caribbean at an operating margin of about 3 percent. Its receivables average $40 million, about 40 days of the $364 million it sells on terms other than cash in advance, funded on a revolving bank line that costs 8 percent a year. Last year 30 percent of its sales were paid in advance, 8 percent were settled through letters of credit, 7 percent through documentary collections and 55 percent on open account.

The new order is concrete. Saigon Lantern Foods, a fictitious distributor in Ho Chi Minh City, wants four 40-foot refrigerated containers of chicken leg quarters for $120,000, shipped by sea at Gulfport's cost. Gulfport can load 14 days after the contract is signed, and the 35-day voyage brings the cargo in on day 49. The distributor asks to pay 60 days after arrival, on day 109, by which time it expects to have resold the chicken. After product

1 Gulfport, its customers and their numbers are invented. The setting is modeled loosely on the Belco Global Foods case cited at the end of the chapter, but every figure differs.

PART

III

Supply Chain Finance Programs

Part III turns to the modern toolkit, in which a strong buyer's credit rating is lent to its suppliers through technology platforms, and asks when these programs create value, who captures it, and when they quietly become debt.

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CHAPTER 7

The Supply Chain Finance Toolkit

Standard Definitions, Credit Arbitrage, and Choosing an Instrument

On a Monday morning in Monterrey, the treasurer of Monterrey Precision Components opens four envelopes. Great Lakes Motors, the Detroit automaker that buys 60% of the company's output, will stretch its payment terms from 45 to 60 days and invites its suppliers to join a bank-run early payment program. A factor offers to buy the company's small aftermarket invoices. A German toolmaker quotes a $5 million machining center on two years of deferred payment. And Great Lakes has sent a $24 million purchase order for a new pickup truck, due in 90 days, while the company's credit line has $5 million of room. Four needs arrive at once. Each calls for a different instrument, and behind each one sits the same question: who should pay for the days?

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LEARNING OBJECTIVES

1. Classify any technique by its standard name, cluster, triggering event and leading party.
2. Compute who pays for the days under alternative terms and rates, and the size and split of the credit arbitrage gain.
3. Evaluate a dynamic discount offer against the buyer's return on cash and the supplier's cost of funds.
4. Calculate the cash flows and annualized cost of factoring, forfaiting and pre-shipment finance, and the cushion a securitization gives its senior investors.
5. Choose an instrument for a financing need and explain its legal foundations: assignment, true sale, perfection and electronic transferable records.
6. Describe who provides supply chain finance today, why programs stall, and where the toolkit's limits lie.

7.1 Why Supply Chains Need a Toolkit

Chapter 6 ended with a quiet revolution: most goods now cross borders on open account, with no letter of credit and no bank standing between buyer and seller. If banks have stepped out of the transaction, who finances the goods while they move, and on whose credit?

Who carries an open-account invoice?

On open account the seller ships, sends an invoice and waits. The Global Supply Chain Finance Forum cites an estimate that more than 80% of world export volume is settled by clean payment, while only about one third of world trade uses any formal trade financing.[1] Waiting is not free. Monterrey Precision Components (MPC) machines aluminum parts in Monterrey and sells about $120 million a year: $72 million to Great Lakes Motors (GLM) and $48 million to about 80 aftermarket distributors. It ships $6 million of parts to GLM every month on 45-day terms, so close to $9 million of its money ($6 million a month for a month and a half) always sits in GLM's payables ledger. MPC, a mid-sized manufacturer with a BB-type credit profile, funds that receivable on a bank line costing 12% a year. GLM, an AA-rated automaker, borrows short term at 5%. The financing has landed on the party that pays the most for money.

Why should GLM care? Because a cash-starved supplier is an operational risk. In 2016 Clark-Cutler-McDermott, a family-owned just-in-time supplier to General Motors, filed for bankruptcy; with no reserve inventory and no alternate source, GM ended up funding the supplier's factory to keep its own lines running.[2] A buyer that squeezes its suppliers' liquidity can pay for it twice.

1 Source: Global SCF Forum, *Standard Definitions for Techniques of Supply Chain Finance* (2016), pp. 17–18, citing UniCredit (2015) and the Bank for International Settlements.

2 Source: Strategic Treasurer, *2017 Supply Chain Finance Analyst Report*, p. 19, citing *Automotive News* (2016).

CHAPTER 8

Borrowing the Buyer's Credit Rating

Reverse Factoring and Payables Finance

In April 2013, Procter & Gamble wrote to its suppliers with two pieces of news. The first was unwelcome: P&G, a company rated AA- by Standard & Poor's, would take about 30 days longer to pay its invoices. The second was an offer. A supplier that joined P&G's new supply chain finance program could sell its approved P&G invoices to one of P&G's banks and be paid within 15 days, at a discount priced off P&G's credit rather than its own. More than a decade later the program is still running: at June 30, 2024, P&G owed participating suppliers about $5.6 billion.

Picture the treasurer of a Brazilian pulp producer whose P&G contract is up for renewal on the same two-part terms. The company is rated BB, borrows at 7.5 percent, and ships P&G a $20 million invoice every month. Is the offer a gift, a fair trade, or a polite way of asking the supplier to finance P&G? This chapter gives the treasurer the tools to answer.

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LEARNING OBJECTIVES

1. Explain why a highly rated buyer extends payment terms and how payables finance lets its suppliers borrow on the buyer's credit rating.
2. Measure a buyer's days payables outstanding three ways and estimate the cash a payment-term extension releases, on paper and in practice.
3. Trace a term extension through a supplier's receivables and cash conversion cycle, with and without early payment.
4. Price a payables finance invoice from a reference rate and a spread, and compute the supplier's break-even funding rate.
5. Use the rate-gap value chain to split the value of a financed invoice among supplier, funder and buyer, and identify who loses.
6. Design a program: segment suppliers across payment tools, choose funders, protect trade-payable status, and track performance.

8.1 Why Would a AA- Buyer Pay Later?

Chapter 7 introduced *payables finance*, also called reverse factoring, as one technique in the supply chain finance toolkit. This chapter takes it apart, starting with a puzzle. Procter & Gamble, the world's largest consumer products company when it launched its program, has carried Standard & Poor's AA- rating since 2001 and still held it at June 30, 2024. Few companies can borrow more cheaply. Why would such a company pay its suppliers later? And having decided to pay later, why would it arrange for banks to pay those same suppliers sooner?[1]

The decision grew out of a cost drive. In February 2012, chief executive Bob McDonald announced a five-year program to take $10 billion out of P&G's costs: about $6 billion from cost of products sold, $3 billion from overhead and $1 billion from marketing. That was close to 15 percent of the year's combined cost of products sold and selling, general and administrative expense. Purchasing had already pressed suppliers hard on price. Payment terms were the lever nobody had pulled.

Benchmarking supplied the argument. In an April 2013 letter to suppliers, chief purchasing officer Richard Hughes explained that months of external benchmarking had shown P&G's payment terms to be out of line with its peers. The letter gave no numbers. According to the case, P&G's internal measure, which counted payments to all suppliers, put its own payment time at about 45 days, against 75 to 100 days or more at other firms in the industry. Adding about 30 days would bring P&G to the bottom of that range. New suppliers were put on "net 75 globally, where legally allowed" from March 1, 2013, and existing suppliers moved as their multi-year contracts came up for renewal.[2]

1 Sources: Esty, Mayfield and Lane (2016), p. 1, for P&G's profile and rating history; P&G 2024 Annual Report for the 2024 rating.

2 Sources: Hughes letter to suppliers, April 2013, posted on P&G's supplier website; *Household & Personal Products Industry*, June 2013; Esty, Mayfield and Lane (2016), pp. 2 and 4.

CHAPTER 9

Hidden Leverage

Debt-Like Payables, Disclosure, and the Fall of Greensill

> *If banks stop these facilities, it's like cutting credit lines and we might see some companies run into heightened liquidity problems as a result.*
>
> Frédéric Gits, Fitch Ratings, quoted in *The Wall Street Journal*, April 4, 2020

On March 1, 2021, a court in Sydney refused to force an insurer to renew its cover on billions of dollars of loans arranged by Greensill Capital. Within hours, Credit Suisse froze four investment funds holding about $10 billion of Greensill-sourced assets. Within days, Greensill's German bank was closed by its regulator and the firm, valued at $3.5 billion to $4 billion by a SoftBank investment less than two years earlier, entered insolvency.

The assets in those funds were sold as some of the safest paper in finance: a large company's promise to pay for goods it had already received. How did a working capital tool become a systemic event? What were household-name companies carrying in their accounts payable that investors could not see? And what does such a program rest on that someone else can take away?

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LEARNING OBJECTIVES

1. Explain why a supplier finance program makes a buyer's payables behave like short-term debt, and compute leverage with and without the program balance.
2. Stress-test the three dependencies of a supplier finance program: the funder, the insurer and the verifiability of the receivable.
3. Describe how Greensill Capital departed from traditional reverse factoring, and separate the direct trigger of its collapse from the underlying causes.
4. Measure concentration, correlated-default and verification risk in a supply chain finance fund with concentration losses, expected loss and a stress test.
5. State what ASU 2022-04 and the IAS 7 and IFRS 7 amendments require, read a disclosure note, and compute the ratios an analyst needs.
6. Recommend governance safeguards for buyers, suppliers, funders, investors, insurers and regulators.

9.1 Northfield's Extra Thirty Days

A company that starts paying its suppliers in 90 days instead of 60 looks, on paper, like a company that has negotiated well. Its payables rise, its operating cash flow jumps, and no new debt appears. Chapter 8 explained why suppliers accept the longer terms: a funder pays them early at a rate set by the buyer's credit rating. That chapter left one question open. What does such a program do to the buyer's own risk, and who can see it? We answer with three fictitious companies whose numbers run through the whole chapter.

Northfield Consumer Brands, a packaged foods and household products company in Columbus, Ohio, reported fiscal 2025 revenue of $24.0 billion and EBITDA of $3.6 billion; its debt is rated A minus. It buys $14.6 billion of goods and services a year on credit. Until 2021 it paid suppliers in 60 days, the industry norm. That year it moved every supplier to 90 days and invited them to join a supplier finance program run by the *Meridian Working Capital Fund*, a nonbank funder that buys approved invoices and pays for them by selling short-term notes to pension funds, corporate treasurers and wealthy families. A single insurer covers the notes, which earns them an investment-grade rating. About 4 percent of Northfield's suppliers joined, but they are the large ones and account for 60 percent of its spend. One of them is *Harbor Packaging*, a carton maker that sells Northfield $180 million a year.[1]

The mechanics are those of reverse factoring, developed in full in Chapter 8. Harbor ships cartons and invoices Northfield. Within about 10 days Northfield matches the invoice to its purchase order and the delivery and confirms to Meridian that it will pay the full amount on day 90. Harbor may then sell the confirmed invoice to Meridian for its face value less

1 Northfield, Harbor and Meridian are fictitious. Their data set is used throughout the chapter.

PART

IV

Risk and Integration

Part IV widens the lens from financing to risk: why firms hedge, how to measure and manage commodity and disruption risk, and how all the levers of the book interact inside a single company run through a simulated turnaround.

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CHAPTER 10

Spend at Risk

Supply Chain Risk Management and Financial Hedging

The role of risk management is to ensure that a company has the cash available to make value-enhancing investments.

Kenneth A. Froot, David S. Scharfstein, and Jeremy C. Stein, 1994

On the first business day of July, the chief financial officer of Brightwater Home Care opened the board's budget pack and found one line circled in red. Brightwater does not buy plastic resin. It buys detergent bottles. Yet the price of the polyethylene inside those bottles had risen for three straight quarters, and the bottle makers passed every dollar of it through. The plan put fourth-quarter resin spend at $80 million. The board's question was blunt: how likely is a quarter above $92 million, and what would it cost to make that less likely?

In 2006 the supply managers of Unilever asked the same question about the same resin. This chapter gives you the tools they used and a few they lacked: a reason to hedge at all, a map of where supply risk hides, a way to measure the right tail of spend, the derivatives that cut it, and the controls that keep hedging from turning into speculation.

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LEARNING OBJECTIVES

1. Explain why hedging can create value when external finance is costly, and use the cash-gap test to decide how much of an exposure to hedge.
2. Distinguish recurrent risk from disruption risk, and rank supplier nodes by time to recovery, time to survive, and a risk exposure index.
3. Compute expected spend, its standard deviation, and the probability that spend exceeds a budget threshold.
4. Compare forwards, futures, swaps, and call options, and compute the expected cost, exercise probability, and maximum exposure of an option hedge and the basis risk of a cross-hedge.
5. Build and read a cost-risk frontier by Monte Carlo simulation, and explain how operational hedges and pass-through shift it.
6. Design authorization, limit, and monitoring controls that keep a hedging program from becoming speculation.

10.1 Brightwater Home Care

Most of this book has treated risk as credit risk: will the buyer pay, will the supplier survive, is a financing program quietly a loan (Chapters 5 and 9)? Yet for many companies the largest swings in cash flow come from the price and availability of what they buy. Why should a company that makes laundry detergent lose sleep over the price of crude oil?

Brightwater Home Care, Inc., of Stamford, Connecticut, makes laundry detergent, dish liquid, surface cleaners, and body wash, sells them through grocery, drug, mass, and club retailers across the United States, and earns about $4.8 billion of revenue a year.[1] Nearly every product leaves its plants in a bottle blow-molded from high-density polyethylene (HDPE) by one of two bottle makers.

Here is the catch. Brightwater's contracts price each bottle at its resin content times a monthly published resin index, plus a fixed conversion fee. The bottle makers buy the resin; Brightwater pays for it. Brightwater therefore carries the price risk of about 200,000 metric tons of HDPE a year without ever signing a resin contract. Unilever's supply managers were in exactly this position in 2006: their bottle suppliers passed the cost of resin through in the price of the containers.[2] The index, in turn, moves with energy. HDPE is made from ethylene, which crackers produce from natural gas liquids and from naphtha refined out of crude oil, so resin follows oil and gas, though loosely. Figure 10.1 traces the chain.

At the plan price of $1,600 per ton, 200,000 tons cost $320 million a year, or $80.0 million a quarter: the line circled in red. For the fourth quarter, Brightwater's planners expect to need 50,000 tons, but over the past three years their quarterly volume forecasts have missed with a standard deviation

1 Brightwater and all of its figures are fictitious, chosen to keep the arithmetic transparent at the scale of a mid-sized consumer-goods maker.

2 Source: Kleindorfer and Yücesan (2009), INSEAD case INS055, p. 5.

CHAPTER 11

Running the Financial Supply Chain

Integrated Decisions in The Cool Connection

On a Wednesday evening in February 2026, two teams of graduate students opened identical dashboards, and each became the new management board of The Cool Connection, a century-old Dutch maker of shampoo, hair gel, conditioner and skin cream. The company they inherited had lost €3.25 million in its last quarter on €27 million of revenue. Its bank had fined it €2.5 million for missing a promised current ratio. Its shareholders were lending it emergency cash at 35 percent a year, its warehouses were overflowing, and its customers had docked €3.3 million from their invoices because deliveries kept falling short of what had been promised.

No single function had caused those losses, and no single function could end them. This chapter follows two teams through the first four of the game's six quarters, prices the instruments of the last two, and shows how the levers of the previous ten chapters pull on one another inside one company.

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LEARNING OBJECTIVES

1. Explain how purchasing, sales, supply chain and finance decisions connect through the Promise, Deliver, Finance chain and roll up into return on investment.
2. Compute realized prices, sales bonuses or penalties and break-even contract indices.
3. Size stock parameters, warehouse space and in-transit stock in pallets, euros and financing need.
4. Test a working capital move against a current ratio covenant and choose among a bank loan, supplier credit and emergency borrowing.
5. Price guarantees, credit limits, supplier financing, consignment and credit insurance as funding decisions.
6. Use an AI assistant as an analyst: state the rules it cannot know, verify its arithmetic and document the decision.

11.1 The Company You Inherit

The first ten chapters took the financial supply chain apart one lever at a time: the current ratio in Chapter 2, payment terms in Chapter 5, Incoterms and bank guarantees in Chapter 6, payables finance in Chapter 8. Each analysis held everything else constant. Real companies never do. A longer payment term raises a supplier's price, a higher service promise needs more stock, more stock needs space and financing, and the financing is bound by a covenant someone promised the bank. So how do the levers interact when four managers pull them at once?

Public companies are a poor laboratory for that question, because their statements aggregate thousands of decisions we never see. A business simulation turns the problem inside out. The decisions are logged, the results arrive every quarter down to the euro, and two teams start from the same position. This chapter uses **The Cool Connection** (TCC), a web-based simulation published by Inchainge in the Netherlands, and follows two teams of graduate students who ran it in the spring of 2026.[1]

11.1.1 The business

TCC makes four personal care products: CoolClassic Lemon (a shampoo), CoolGuy (a hair gel), CoolCondition (a conditioner) and CoolCare (a cream, the only product in a jar). Five suppliers provide the components: bottles and jars from Quick Pack, a foaming agent from Ether Co., an additive called liposome from Double Dense, preservatives from Forever Fresh and perfume from Eau du monde. A mixer and a single bottling line make the products, and an inbound and an outbound warehouse hold them, each with 5,000

1 All TCC figures come from the simulator's exports and debriefs for two teams in the Spring 2026 section of OPIM 5115 at the University of Connecticut, called Team A and Team B, in euros per 13-week quarter. The game and its rules are Inchainge's product; the descriptions here are our own.

PART

V

The Frontier

Part V examines two forces reshaping supply chain finance: shared ledgers that carry an anchor buyer's credit to the deepest tiers, and sustainability-linked pricing and carbon markets that attach a price to outcomes no market prices.

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CHAPTER 12

Trust at Scale

Blockchain and Deep-Tier Supply Chain Finance

It requires both technology and business model changes that I think make it more challenging to drive.

Kathryn Guarini, chief information officer of IBM, on enterprise blockchain, as quoted in The Wall Street Journal, December 2022

In a mixing plant in eastern Connecticut, the owner of Riverside Rubber sells this month's invoices to a lender at an 18 percent discount rate, because payroll cannot wait 90 days. Riverside's compound becomes tires at Thames Tire, the tires are mounted on wheels at Ridgeline Wheels, and the wheels roll out under vans built by Constitution Motors, which borrows at about 6 percent. Constitution's bank would gladly lend against Constitution's promise to pay. It has never heard of Riverside. Neither has Constitution, which sees no further than its first tier and could not say which vans carry Riverside's compound if a batch went bad.

This chapter asks whether a shared ledger can carry trust, and with it a large buyer's credit, across that gap, and it builds the deep-tier cascade: the arithmetic of splitting one payment obligation so that a supplier three tiers away borrows at the anchor buyer's rate.

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LEARNING OBJECTIVES

1. Explain what a blockchain is, the coordination problem it solves, and how the trilemma and permissioning shape supply chain ledgers.
2. Decide whether a supply chain problem needs a blockchain, and whether a traceability problem needs a shared ledger.
3. Diagnose why blockchain projects stall using information complexity and supply chain buy-in, and name the barrier breakers that move them.
4. Explain why payables finance stops at tier 1 and how a divisible, transferable payment obligation carries an anchor buyer's credit deeper.
5. Compute every party's proceeds, savings and cash flows in a deep-tier cascade, and the pass-through margin that keeps the benefit moving.
6. Distinguish true deep-tier finance from look-alikes and assess its legal, credit-leakage and domino risks.

12.1 The Blockchain Puzzle

Why would a supply chain manager care about a technology best known for speculative coins? The evidence pulls in two directions. Crypto assets boomed: bitcoin passed $100,000 in December 2024.[1] Enterprise supply chain projects, meanwhile, have mostly stalled. In December 2022 *The Wall Street Journal* reported that Maersk and IBM were shutting down TradeLens, their blockchain platform for container shipping, and that Walmart's effort to track groceries on a blockchain had, in four years, added just one item beyond leafy greens: green bell peppers.[2]

To resolve the puzzle we must separate the technology from the hype and ask a narrower question: when does a shared ledger create value that a conventional database cannot? A second question earns the chapter its place in a finance book: can the same ledger carry a large buyer's credit to small suppliers it has never met? We follow two threads: Walmart's program to trace leafy greens, from public facts, and a fictitious automotive chain with a complete data set, so that we can compute every dollar.

Constitution Motors builds light commercial vans in Hartford, Connecticut. It is rated single A, and a bank will discount its approved invoices at about 6 percent a year. Each month Constitution approves roughly $1,000,000 of invoices from Ridgeline Wheels, its tier-1 supplier of wheel assemblies for its cargo-van line, and pays them 90 days after approval. This month's assemblies contain $400,000 of tires from Thames Tire in New London, Connecticut, and the tires contain $150,000 of compound from Riverside Rubber, a 60-person compounder with $9 million of annual revenue. Each tier pays its supplier when it is paid itself, on day 90 (Figure 12.1).

Look at the bottom row of Figure 12.1. The cost of money rises from 6 to 18 percent over three tiers. Each rate is what a lender charges, actual/360, to

1 Source: Reuters, "Bitcoin storms above $100,000 as Trump 2.0 fuels crypto euphoria," December 5, 2024.

2 Source: I. Bousquette, "Blockchain Fails to Gain Traction in the Enterprise," *The Wall Street Journal*, December 15, 2022.

CHAPTER 13

Financing the Green Supply Chain

Sustainability-Linked Finance and Carbon Markets

> *The art of government … is the art of so ordering life that self-interest prompts what justice demands.*
>
> William Temple, Archbishop of Canterbury (1940s), quoted in Kaplan, Ramanna and Roston (2023)

The treasurer of Duisburg Consumer Brands AG has two reports on her desk. The sustainability report says the company is responsible for two million tons of greenhouse gas a year, and that only about one ton in thirteen comes from its own plants, vans and purchased power; the rest is made in other people's factories and kitchens. The treasury report says that every week the company's supply chain finance program pays about four hundred of those factories early, at a price the treasurer negotiates. Could that price buy a greener supply base?

Across the Atlantic, Housatonic Foods will cut its own emissions by 30,000 tons by 2027 and still have 50,000 it cannot yet eliminate. A broker offers carbon credits from $2.50 to $150 a ton. Which is the bargain? This chapter is about attaching money to outcomes that markets do not price on their own, and about why the answer depends on who verifies the outcome.

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LEARNING OBJECTIVES

1. Distinguish Scope 1, 2 and 3 emissions and explain why a buyer's footprint and ESG risk sit mostly in its supply base.
2. Explain how sustainable supply chain finance changes one parameter of payables finance, and price an early payment under an ESG margin grid.
3. Evaluate a supplier's saving and break-even volume and the buyer's cost under different funding arrangements.
4. Design a sustainability-linked program that resists greenwashing, and compute the cost per genuinely improved euro and per ton of supplier emissions cut.
5. Distinguish allowances, offset credits, and compliance and voluntary carbon markets, and decompose market data into price and volume.
6. Diagnose additionality, baseline, permanence and leakage failures, and compute a credit's cost per real ton.
7. Record purchased removals as E-assets against E-liabilities, netting only earned and durably stored tons and reversing netting on impairment.

13.1 Where the Emissions Live

Chapter 12 showed how a shared ledger can carry an anchor buyer's credit to suppliers several tiers away. This chapter asks what else that credit can carry. The question begins with measurement. When a consumer goods company promises to cut "its" emissions, which emissions does it mean, and how many can its managers actually touch?

We begin with Duisburg Consumer Brands AG, a fictitious maker of detergent and shampoo with eleven plants in Europe.[1] It reports its greenhouse gases in tons of **carbon dioxide equivalent** (CO_2e), converting each gas at its global warming potential. The Greenhouse Gas Protocol, the dominant reporting standard, sorts the total into three scopes. **Scope 1 emissions** come from sources the company owns or controls, such as boilers, vans and refrigerant leaks. **Scope 2 emissions** arise where the electricity, steam and heat it buys are generated. **Scope 3 emissions** are everything else in the value chain: upstream, in making and moving purchased goods, and downstream, in the use and disposal of its products.

Figure 13.1 maps Duisburg's footprint. Its plants and vans emit 90,000 tons a year and its electricity another 60,000. Purchased materials, from surfactants and bottles to the adhesives on its labels, embody 1,350,000 tons, and households heating water to wash with its detergents account for most of the 500,000 tons downstream. The total is

$$90{,}000 + 60{,}000 + 1{,}350{,}000 + 500{,}000 = 2{,}000{,}000 \text{ t } CO_2e,$$

1 All figures for Duisburg Consumer Brands, Rheinland Adhesive Components and Housatonic Foods are illustrative. The companies are fictitious.

PART

VI

Financing the AI Supply Chain

Artificial intelligence runs on chips, memory, buildings and power that must be ordered years before anyone knows how much computing the world will buy. Part VI follows the money behind that wait: the commitments and deposits that finance scarce capacity, the leases, bonds and power contracts that pay for the buildout, and the lenders, vendors and savers who hold the risk if the revenue arrives late.

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CHAPTER 14

Building Ahead of Demand

Bottlenecks, Capacity Commitments, and Working Capital in the AI Supply Chain

…our packaging capacity is so tight that now it limits my customers' growth.

C. C. Wei, TSMC, second-quarter 2026 earnings call, July 16, 2026

On September 30, 2026, Micron Technology, one of the three companies that sell most of the world's DRAM chips, told investors something unusual about its customers. Under 26 multi-year supply agreements, some running into 2031, they had committed $32 billion of financial support, "the vast majority" of it cash deposits, and $12.3 billion had arrived in the quarter just ended. Micron did not need the money. It held $68.3 billion more cash than debt.

Why would buyers lend billions to a supplier with more cash than it could spend? Artificial intelligence runs on chips, memory, buildings and power that take years to add, and in 2026 every layer of that chain was committing money long before anyone knew how much computing the world would buy. Who pays for the years between an order and the output? How much should a firm commit before demand shows up, and who is left holding the commitment if it does not?

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LEARNING OBJECTIVES

1. Map the AI factory's physical chain, locate its binding constraints and lead times, and measure concentration at a chokepoint.
2. Size a capacity commitment with the critical fractile when the commitment itself must be financed.
3. Value a deposit-backed take-or-pay agreement with a price collar for the buyer and for the supplier.
4. Map who finances whom along a multi-tier chain, and compute the working capital a mid-chain integrator must fund.
5. Read a supplier's working capital, purchase commitments and write-downs for overbuilding signals against thresholds set in advance.
6. Test whether falling unit prices expand demand, and locate the layer of the stack that earns the profit needed to repay the buildout.

14.1 The AI Factory as a Supply Chain

Why should a book about invoices, inventories and payment terms end inside a data center? NVIDIA's chief executive, Jensen Huang, calls such facilities AI factories. An **AI factory** takes in electricity and turns out tokens, the units of text, code and images that AI models generate. Like any factory, it has suppliers with long lead times, inventory that can spoil, customers who may pay late, and a funding gap between the day cash goes out and the day it comes back.

> *Every token is electrons moving, heat being managed, and energy being converted into computation.*
>
> JENSEN HUANG, NVIDIA (MARCH 2026)

Huang describes the industry as a **five-layer stack**: energy, chips, infrastructure (land, buildings, power delivery and networking), models and applications. His central claim is that "every successful application pulls on every layer beneath it." The same system can be read as three interlocking supply chains. The physical chain runs from wafer to die to package to memory to server to rack; the data chain from scraping to licensing to labeling; the model chain from pretraining to deployment to inference. A failure in any one of them stops output.

How large is this factory? In its quarter ended July 26, 2026, NVIDIA alone reported revenue of $96.2 billion, 106 percent more than a year earlier, and $89.0 billion of it, 93 percent, came from data centers. Huang put the investment in one gigawatt (GW) of AI computing capacity at about $60 billion, double the roughly $30 billion of five years earlier, and NVIDIA's own

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revenue opportunity in a gigawatt built on its next platform, Vera Rubin, at about $40 billion.[1] If both claims describe a Vera Rubin gigawatt, about 40/60 = 67 percent of its cost buys NVIDIA's systems and the rest pays for buildings, cooling, power equipment and everything else; on the current Grace Blackwell platform, about $25 billion per gigawatt by NVIDIA's account, the share is nearer 25/60 = 42 percent. A gigawatt is now the unit in which this industry budgets, and every unit must be ordered years before it runs.

1 Source: NVIDIA, results for the second quarter of fiscal 2027 and earnings call, August 26, 2026. The per-gigawatt figures are company claims, not audited amounts.

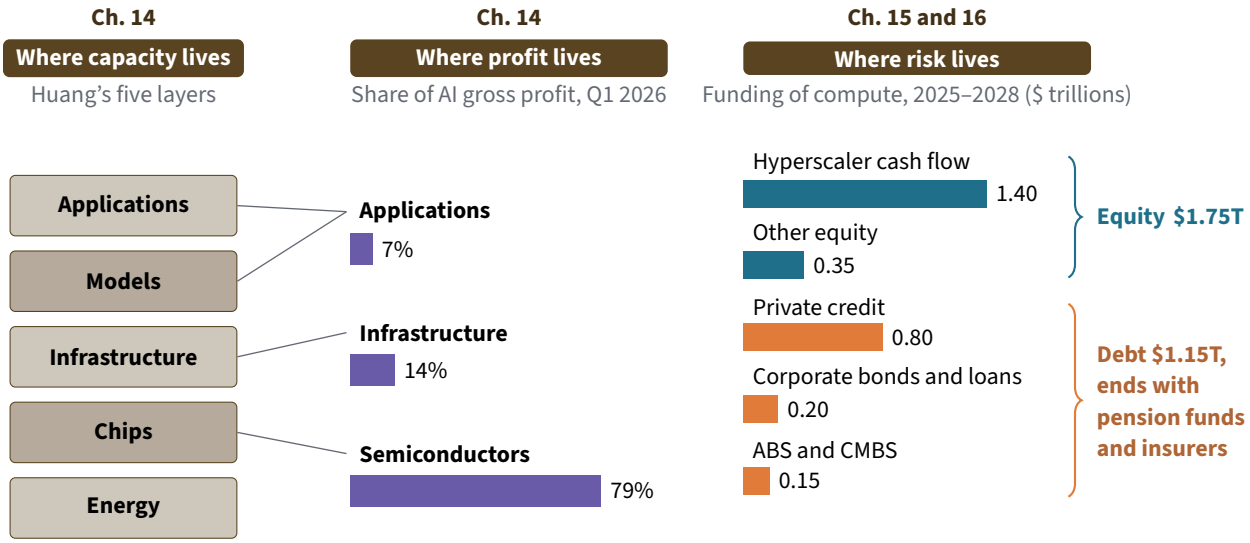


FIGURE 14.1 Three lenses on one system: the map of Part VI. Other equity is private equity, venture and sovereign funds. Sources: Huang (2026); Agrawal (2026), as presented in OPIM 5115 Lecture 13; Morgan Stanley Research (August 2025), as restated by the Financial Stability Board (May 6, 2026) and the Structured Finance Association (July 23, 2026), with the split of the external $1.5 trillion from Van Nieuwerburgh (2026) and MUFG (2025).

Figure 14.1 is the map of Part VI: Huang's layers show where capacity lives, Apoorv Agrawal's economic layers where profit lives, and the capital stack, which Stijn Van Nieuwerburgh asks us to follow, where risk lives. The lenses disagree: the profit is earned in chips, the debt is raised for buildings and servers, and the risk ends with savers who own neither. This chapter works with the first two lenses; Chapters 15 and 16 follow the capital stack and the risk.

We follow four real companies and one fictitious one. NVIDIA, the dominant designer of AI accelerators, sits in the middle of the physical chain: it buys wafers and packaging from Taiwan Semiconductor Manufacturing Company (TSMC) and memory from Micron, SK hynix and Samsung, and sells systems to the companies that run AI data centers. GE Vernova, a maker of gas turbines, stands for the power equipment that every campus needs. The largest buyer of all this equipment is the **hyperscaler**, one of the few

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giant cloud and platform operators (Amazon, Microsoft, Alphabet, Meta and Oracle). Next comes the **neocloud**, a specialist such as CoreWeave or Nebius that buys GPUs and rents their computing time to AI labs under long contracts (Chapter 16).

Redfish Rack Systems, Inc., a fictitious company in Houston, Texas, assembles liquid-cooled GPU racks for data centers. It buys GPU boards from a chip supplier on 30-day terms, buys memory under a one-year supply agreement that requires a deposit and limits the price to a band, and sells finished racks to two hyperscalers on 60-day terms; one of them has asked for 90. Redfish's chief financial officer must settle two questions before fiscal 2027 begins. How many memory sets should the company commit to buy, knowing that it must pay for every one and deposit part of the price up front? And can it afford to give Hyperscaler A the 90 days it wants? Table 14.1 collects the plan. Redfish is small enough to compute with, and it sits exactly where the chain's financing changes direction: its supplier wants money early, and its customers want to pay late.

TABLE 14.1

Redfish Rack Systems, Inc.: data for fiscal 2027 (all figures fictitious).

Item	Value
Rack demand	Normal, mean 2,000 racks, standard deviation 500 racks
Price and cost per rack	Price $3.50M; cost of goods sold $3.20M (GPU boards $2.60M, memory set $0.35M, other $0.25M); gross margin $0.30M (8.6%)
Memory agreement	Commit to Q memory sets for the year, one per rack, take-or-pay (Redfish pays for every committed set, used or not); price equals the market price kept within a band: floor $0.30M, ceiling $0.42M (reference price $0.35M)
Deposit	25% of committed value at the reference price, paid at signing, credited against fourth-quarter deliveries
Unused or missing memory	An unused set resells at 70% of the reference price (loss $0.105M); a rack short of memory loses its $0.30M margin (the spot market is sold out)
Customers	Two hyperscalers, half of sales each, on 60-day terms; Hyperscaler A asks for 90 days and offers its payables finance program at 5.2%
Funding	Revolving credit line at 9%; the memory supplier's own cost of funds is 6%
Working capital policy	DIO 45 days, DSO 60 days, DPO 30 days (365-day year, ending balances)

Where, then, does this chain bind first?

14.2 Where the Chain Binds

If customers will deposit billions with a supplier that does not need the money, money is not what the chain lacks. So why can it not simply build faster when demand surges? To answer, we need three facts about each stage of the physical chain: what runs out first, how long it takes to add more, and how many firms can supply it.

Start with the chips. A frontier accelerator such as NVIDIA's is a processor die and several stacks of high-bandwidth memory (HBM) joined by advanced packaging, and most are packaged by TSMC with a process known as CoWoS. In July 2026 TSMC's chief executive, C. C. Wei, described that packaging capacity in the words that open this chapter, and TSMC raised its 2026 capital budget to $60 to $64 billion from $52 to $56 billion, putting 10 to 20 percent of it into advanced packaging, testing, mask making and related uses. Trade reports put TSMC's CoWoS capacity at 120,000 to 140,000 wafers a month in 2026, with the gap between supply and demand narrowing from about 20 percent to about 10 percent.[2]

Memory binds next. Micron reported in September 2026 that most of its 2027 HBM supply was already agreed, while SK hynix was still negotiating its 2027 volumes and prices in July. Neither could add capacity quickly. Micron's next Idaho fab is due to produce its first wafers in mid-2027 and its New York fab in 2030; SK hynix's new Yongin plant opens in early 2027. A memory fab costs tens of billions of dollars and takes two to three years to build, so the supply for 2027 was largely fixed by decisions made in 2024 and 2025.

Power binds longest. A large power transformer took about 128 weeks to deliver in 2025 and a generator step-up transformer about 144 weeks, and Wood Mackenzie estimated a 30 percent shortfall of supply.[3] GE Vernova, which makes gas turbines for power plants, said in July 2026 that it expected to end the year with at least 125 GW of turbines under contract, "making us mostly sold out through '30." A new connection to a congested grid takes four to seven years.[4]

The input that limits output when every other input is available is the **binding constraint**. In 2024 it was the GPU itself and its packaging; by 2026 the scarcity had spread down the bill of materials to memory and out to power. Chapter 10's resilience metrics make the danger plain: at these stages the time to recover from losing a supplier is measured in years, far longer than any buyer's time to survive without one (Chapter 10). Figure 14.2 lays the stages side by side, with the instrument that finances each stage ahead of demand.

Lead time tells us how long a shortage lasts. Concentration tells us how many firms it depends on. The standard measure is the **Herfindahl-Hirschman index** (HHI): square each firm's market share in percentage

2 Sources: TSMC, second-quarter 2026 earnings call, July 16, 2026; Economic Daily News, citing institutional investors, as reported by TrendForce, June 15, 2026 (trade estimates, not TSMC figures).

3 Source: OPIM 5115 Lecture 13 (April 2026), citing Wood Mackenzie (2025). We found no 2026 update of these lead times.

4 Sources: GE Vernova, second-quarter 2026 earnings call, July 22, 2026 (an expectation, not the June 30 position); Energy Institute at Haas, blog post of October 5, 2026.

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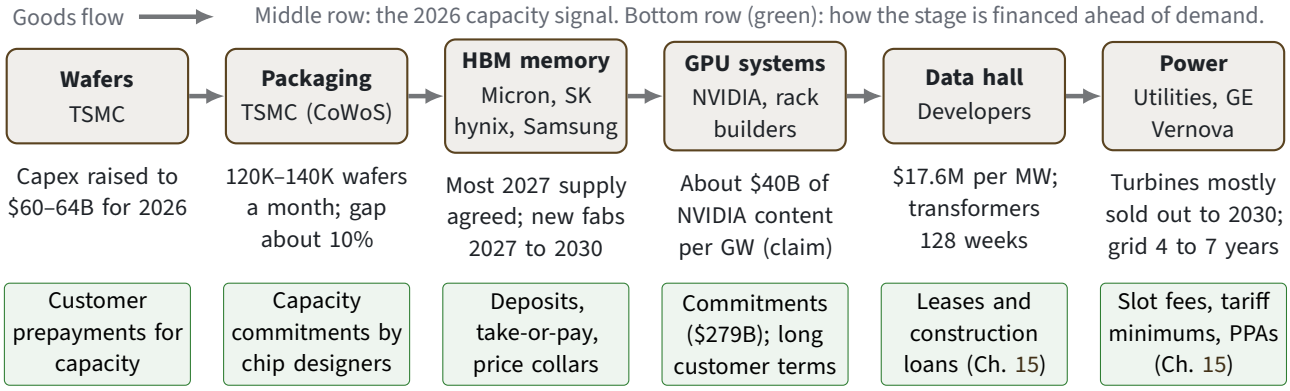


FIGURE 14.2 The physical chain of an AI data center, from wafer to power. The middle row gives the capacity signal for 2026 and, where our sources have it, the time to add more; the bottom row names the instrument that pays for capacity before it is used. Sources: TSMC (July 16, 2026); Economic Daily News via TrendForce (June 15, 2026); Micron (September 30, 2026); NVIDIA (August 26, 2026; company claim); Cushman and Wakefield (2026); Wood Mackenzie (2025) via OPIM 5115 Lecture 13; GE Vernova (July 22, 2026; an expectation); Energy Institute at Haas (October 5, 2026).

points and add the squares. Take the market for data center GPUs. A study of the AI supply chain by economists at the Bank for International Settlements reports that in 2023 NVIDIA earned 92 percent of the revenue, AMD 3 percent and all others 5 percent.[5] Treating the others as many small firms,

$$\text{HHI} = 92^2 + 3^2 = 8{,}464 + 9 = 8{,}473,$$

or 8,498 if the others were a single firm. A monopoly scores $100^2 = 10{,}000$, and n firms of equal size score $10{,}000/n$, so dividing 10,000 by the index gives the equivalent number of equal-sized firms: $10{,}000/8{,}473 = 1.2$.

Take the two stages whose coverage we can measure. On the trade estimates, TSMC's advanced packaging covered roughly 90 percent of 2026 demand; on Wood Mackenzie's, large power transformers covered about 70 percent of need in 2025. Coverage decides what binds: a campus that needed both would be held to about 70 percent of its plan until transformers caught up. Concentration and lead time decide what binds longest. Data center GPUs, with an HHI of 8,473, and HBM, sold by three firms from fabs that take two to three years to build, are the stages where losing one supplier would stop the chain for years.

You might object that new suppliers can enter. They can, but entry takes at least as long as the lead time. A stage that is both concentrated and slow to expand is a *chokepoint*: a shortage there lasts for years, and losing one supplier stops the chain. Exhibit 14.1 turns these distinctions into a routine.

5 Source: L. Gambacorta and V. Shreeti, "The AI supply chain," BIS Papers No 154 (March 2025), Graph 2, from 2023 and early-2024 shares. The HHI calculations are ours.

EXHIBIT 14.1 A procedure to locate the binding constraint in a multi-tier chain

Step 1. List the stages from raw input to delivered service. For each, record the lead time to add capacity (weeks) and the capacity available in the delivery year as a percentage of what the chain needs.

Step 2. Compute each stage's HHI from supplier shares (squared percentage shares, summed) and the equivalent number of equal firms, 10,000/HHI.

Step 3. The binding constraint is the stage with the lowest coverage: the chain cannot deliver more than that percentage of its need.

Step 4. Flag as chokepoints the stages with fewer than four equivalent firms (HHI above 2,500) and a lead time above a year; these bind longest and fail together if one supplier stops.

Step 5. For each binding stage and chokepoint, list the alternatives (second source, imports, redesign) and the time each takes. Where no alternative is faster than the lead time, commit early.

A chain whose longest lead times run from two to seven years cannot sprint when demand surges. It can only commit early, and an early commitment is a bet on demand that nobody can yet see. How large a bet should a firm make?

14.3 How Much Capacity to Commit?

If the chain cannot sprint, every buyer must decide how much capacity to reserve before it knows how much it will need. A firm that wants power and chips in 2028 must reserve transformers, grid capacity and GPUs today, before it knows how many customers it will have; developers were preleasing 70 to 90 percent of new data center capacity before breaking ground (as of early 2026). How much should a firm commit? The decision is a single bet on uncertain demand, the setting of the **newsvendor model** of operations management.

Let C_u be the cost of committing one unit too little (the margin, net of the capacity's own cost, lost on a unit of demand that cannot be served) and C_o the cost of committing one unit too much (what an idle unit loses). Expected profit is highest at the commitment Q^* for which the probability that demand D stays at or below Q^* equals the **critical fractile**, $C_u/(C_u + C_o)$. If demand is normal with mean μ and standard deviation σ, then $Q^* = \mu + z\sigma$, where $\Phi(z)$ equals the critical fractile (Appendix A reviews the normal distribution).

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EXAMPLE 14.1 Two hyperscalers, one forecast, two capex plans

Two stylized hyperscalers face the same forecast of AI demand for 2028: normal with mean 10 gigawatts (GW) and standard deviation 3 GW. The first nets $3.0 million per megawatt-year (MW-year) of served demand and carries an idle megawatt for $1.2 million a year. Its critical fractile is $3.0/(3.0 + 1.2) = 0.714$, so $z = 0.566$ and

$$Q^* = 10 + 0.566 \times 3 = 11.7 \text{ GW}.$$

The second has thinner margins and expensive, debt-funded idle capacity: $C_u = \$1.2$ million and $C_o = \$3.0$ million. Its critical fractile is $1.2/4.2 = 0.286$, so $z = -0.566$ and $Q^* = 10 - 0.566 \times 3 = 8.3$ GW. Same forecast, a 3.4 GW difference in plans.

The example explains why capex guidance diverged across firms reading the same forecasts. In late 2025 Meta's Mark Zuckerberg defended front-loading capacity because, at worst, Meta could slow construction and grow into what it had built.[6] Zuckerberg's argument is a claim that C_u far exceeds C_o. A firm with thinner cloud margins that funds capex with debt faces a smaller C_u and a larger C_o, and should commit less.

6 Paraphrased from remarks summarized in the OPIM 5115 slides (2026).

You might object that the weakest-rated hyperscaler, Oracle, spent more, not less: its capex for the fiscal year ended May 31, 2026 reached $55.7 billion, above its own guide of about $50 billion. The model says what a firm's numbers justify, not what it will do (Chapter 15 shows how the credit market priced the gap). A plan built on it should change when the forecast does: when Oracle and OpenAI decided not to expand one Stargate site in Abilene, Texas, from 1.2 GW toward 2 GW, they were revising a commitment as demand information arrived.[7]

7 Sources: Oracle, fourth-quarter and fiscal 2026 results (June 10, 2026) and third-quarter release (March 10, 2026); Van Nieuwerburgh (March 2026), citing Oracle and OpenAI disclosures.

So far a commitment has cost nothing until the capacity is used or left idle. In 2026 it rarely worked that way. A **capacity commitment** is a promise made today to buy capacity or output later, on terms fixed today, and a buyer of memory had to back it with a deposit, financed whether or not the memory was ever used. How does that cost change the bet?

Go back to Redfish (Table 14.1) and consider one more memory set. If rack demand exceeds the commitment, the set lets Redfish build a rack it would otherwise lose: $C_u = \$0.30$ million. If demand falls short, Redfish must still pay about the $0.35 million reference price and can resell the set for 70 percent of it, losing $C_o = 0.35 \times 0.30 = \$0.105$ million. Either way, the set's share of the deposit, $0.25 \times 0.35 = \$0.0875$ million, sits with the supplier for about a year, financed on Redfish's 9 percent line: $c_f = 0.0875 \times 0.09 = \$0.007875$ million, or $7,875, paid whether or not the set is used. An extra set earns C_u with the probability that demand exceeds the commitment, loses C_o with the probability that it does not, and costs c_f for certain. Setting the

expected gain to zero gives the central model of this chapter:

$$\Pr(D \le Q^*) = \frac{C_u - c_f}{C_u + C_o} = \frac{0.30 - 0.007875}{0.30 + 0.105} = \frac{0.292125}{0.405} = 0.721. \qquad (14.1)$$

The financing cost lowers the gain from a used set by c_f and raises the loss on an unused set by c_f, so it appears in the numerator but leaves the denominator, $0.292125 + 0.112875 = 0.405$, unchanged.

EXAMPLE 14.2 Redfish's memory commitment

Without the deposit, Redfish's critical fractile would be $0.30/0.405 = 0.741$, so $z = 0.646$ and

$$Q^* = 2{,}000 + 0.646 \times 500 = 2{,}323 \text{ sets.}$$

With the deposit's carrying cost, Equation (14.1) gives 0.721, so $z = 0.587$ and

$$Q^* = 2{,}000 + 0.587 \times 500 = 2{,}293 \text{ sets.}$$

The financing cost removes 30 sets and raises the chance of running short of memory from 25.9 to 27.9 percent. Committing 2,293 sets means taking memory worth $2{,}293 \times 0.35 = \$802.6$ million at the reference price and paying a deposit of $2{,}293 \times 0.0875 = \$200.6$ million at signing.

Now suppose memory gets scarcer. A hyperscaler contract adds a \$0.15 million penalty for every rack delivered late, so $C_u = \$0.45$ million, and an unused set resells at 85 percent of the reference price, so $C_o = 0.35 \times 0.15 = \$0.0525$ million. The fractile becomes $(0.45 - 0.007875)/(0.45 + 0.0525) = 0.880$, so $z = 1.174$ and $Q^* = 2{,}000 + 1.174 \times 500 = 2{,}587$ sets, 294 more than before. Scarcity raises the cost of being short and lowers the cost of being long, and both push the commitment up.

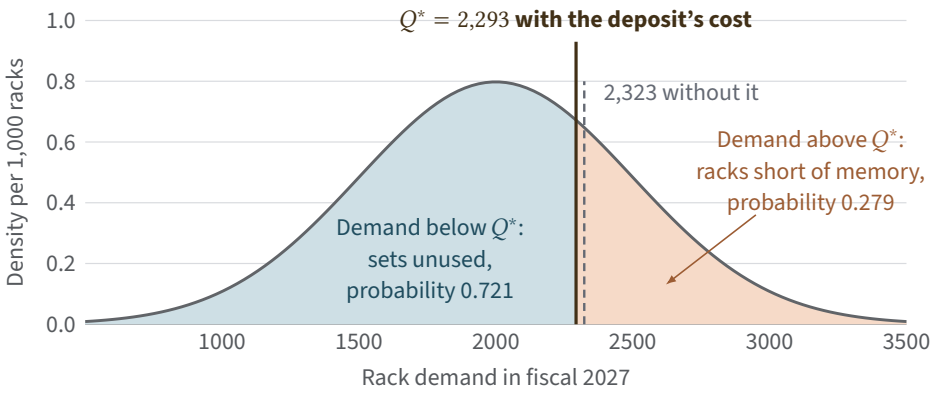


FIGURE 14.3
Redfish's memory commitment, with rack demand normal (mean 2,000, standard deviation 500). The dashed line is the commitment without the deposit's financing cost. Source: Table 14.1 (fictitious).

Figure 14.3 shows why the deposit matters so little to Redfish: its margin is large relative to the financing cost, so the commitment moves by only 30 sets. Setting Equation (14.1) equal to 0.5 shows that Redfish would commit exactly its mean demand only if the financing cost per set reached $C_u - 0.5(C_u + C_o) =$

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$0.30 - 0.2025 = \$0.0975$ million, about 12 times today's \$7,875. For a firm with thin margins and dear credit, like the second hyperscaler of Example 14.1, the same deposit takes a much larger bite. Exhibit 14.2 collects the steps.

EXHIBIT 14.2 A procedure to size a capacity commitment, including the cost of financing it

Step 1. Describe demand in the commitment period as a distribution (for a normal forecast, its mean μ and standard deviation σ), in the units of the commitment (racks, sets, megawatts).

Step 2. Compute the underage cost C_u: the margin lost on a unit of demand that cannot be served, plus any penalty for late delivery.

Step 3. Compute the overage cost C_o: what an unused committed unit loses (the price paid minus its resale or redeployment value).

Step 4. Compute the financing cost per committed unit c_f: the deposit or prepayment per unit times your cost of funds times the time it is outstanding, plus any reservation fee that will not be refunded.

Step 5. Compute the critical fractile $(C_u - c_f)/(C_u + C_o)$. Below 0.5, commit less than mean demand; if c_f is at least C_u, do not commit on these terms.

Step 6. Find $Q^* = \mu + z\sigma$ with $\Phi(z)$ equal to the fractile, and carry the deposit, Q^* times the deposit per unit, into the working capital plan.

SPREADSHEET TIP A commitment that responds to its terms

Put C_u, C_o, c_f, μ and σ in cells B2 to B6. Then `=NORM.INV((B2-B4)/(B2+B3),B5,B6)` returns Redfish's 2,293 sets. A one-variable data table over B4 shows how the commitment shrinks as the deposit or the interest rate rises.

But why could Redfish's memory supplier demand a deposit at all? Why did the makers of a chip long treated as a commodity hold that much power in 2026?

14.4 When Suppliers Hold the Power: The Memory Supercycle

Memory chips have long been the boom-and-bust business of the semiconductor industry. In each of four earlier cycles, from the personal computer boom of the 1990s to the pandemic boom of 2020 and 2021, buyers double-ordered, makers added fabs, and the shortage turned into a glut within about two years. The last bust shrank the memory market from about \$167 billion in 2021 to about \$95 billion in 2023, a fall of $1 - 95/167 = 43$ percent.[8]

8 Sources: Yole Group, *Status of the Memory Industry 2025* (May 2025); the cycle history is from the author's research materials (2026), citing industry analysts.

Experienced buyers had learned to wait out a shortage. So why did buyers in 2026 sign contracts years ahead and hand over cash, instead of waiting for the next glut?

Part of the answer is how AI uses memory. High-bandwidth memory stacks many layers of DRAM dies, and analysts estimate that each gigabyte of HBM uses the wafer area of three to four gigabytes of ordinary DRAM. When the memory makers shifted wafers to HBM for AI accelerators, the supply of ordinary DRAM for servers, notebooks and phones shrank just as AI servers wanted more of it, and, as Section 14.2 showed, new fabs could not arrive before 2027. Industry analysts call such an episode a **memory supercycle**: a price rise driven by a lasting shift in demand, long and deep enough that buyers stop waiting for the glut and start paying for assured supply.

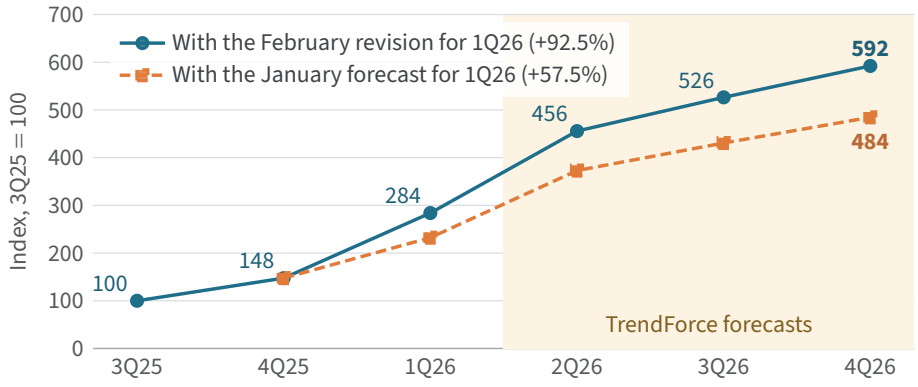


FIGURE 14.4

Conventional DRAM contract prices, compounded from TrendForce's quarterly changes (midpoints of its ranges). Only the fourth-quarter 2025 rise is reported as actual; the first-quarter 2026 rise is TrendForce's revision of February 26, 2026, and the later quarters are its forecasts of March 31, September 7 and September 30, 2026. The dashed line uses TrendForce's January 5 forecast for the first quarter. Sources: TrendForce press releases; index and compounding are ours.

The price signal was large and fast. TrendForce, a market research firm, estimated that contract prices for conventional DRAM rose 45 to 50 percent in the fourth quarter of 2025 and 90 to 95 percent in the first quarter of 2026, and it forecast rises of 58 to 63, 13 to 18 and 10 to 15 percent for the next three quarters. Compounding the midpoints, memory that cost $100 in the third quarter of 2025 would cost

$$100 \times 1.475 \times 1.925 \times 1.605 \times 1.155 \times 1.125 = \$592$$

by the end of 2026, if the forecasts hold (Figure 14.4). The forecasts themselves moved fast. In early January, TrendForce expected the first-quarter rise to be 55 to 60 percent; seven weeks later it put it at 90 to 95, a revision of 35 points within one quarter. On the January forecast the end-2026 index would have been about 484, not 592. Chapter 10 shows how buyers without a liquid futures market tried to hedge this exposure. Here we ask where the money went.

It went upstream. Industry DRAM revenue reached $154.73 billion in the second quarter of 2026, 154.73/53.58 = 2.9 times the $53.58 billion of two

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quarters earlier, and three firms earned $39.4 + 24.9 + 23.3 = 87.6$ percent of it: Samsung, SK hynix and Micron. Even treating every other seller as negligible, the market's HHI is at least $39.4^2 + 24.9^2 + 23.3^2 = 2{,}715$, the equivalent of 3.7 equal-sized firms. What did the price rise do to the producers? SK hynix's operating margin went from 41 percent in the second quarter of 2025 to 76 percent a year later. Micron's revenue for its quarter ended September 3, 2026 was $54.2 billion, up 379 percent in a year, at a GAAP gross margin of 86.8 percent.[9]

Who paid? Everyone downstream, in proportion to how badly they needed the chips and how little they could pass the cost on.

CASE IN POINT 14.1 The squeeze passes down the chain

At the bottom of the chain, makers of notebooks and phones could not pass the full cost on. CPUs, DRAM and solid-state drives made up about 68 percent of the bill of materials of a $900 notebook in the third quarter of 2026, and TrendForce expected notebook shipments to fall 9.4 percent for the year.

The richest buyers paid too. In April 2026 Microsoft said that its expected capital spending of about $190 billion for calendar 2026 included about $25 billion, 13 percent, "from the impact of higher component pricing." (In July it restated the expectation at about $175 billion after extending the useful lives of its data centers, which moves future leases out of capex; it said the plan was otherwise unchanged.) A hyperscaler that commits to build a data center commits to buy its memory at whatever the market charges, unless it has signed for that memory in advance.[10]

Redfish shows why the contracts of the next section exist. A 45 percent rise in the price of its $0.35 million memory set, the low end of TrendForce's estimate for the fourth quarter of 2025 alone, would add $0.45 \times 0.35 = \$0.1575$ million to the cost of a rack, more than half of its $0.30 million margin.

The supplier captures the rent, and the buyers with the least bargaining power are rationed out or pay the most. With TrendForce's estimates and forecasts putting the end-2026 price near six times its level of the third quarter of 2025, and no new fab due before 2027, a buyer that needs supply can pay the spot price and hope, or sign for years ahead and pay for the privilege. What do those contracts look like?

14.5 Contracts for Scarce Capacity

If prices can rise that fast and capacity is sold years ahead, how does a buyer secure supply, and what does it give up in return? In 2026 the answer was a ladder of contracts. Each rung guarantees the buyer more supply, and each moves more of the supplier's risk onto the buyer or onto a third party.

9 Sources: TrendForce, February 26 and September 7, 2026; SK hynix results releases (2025 and 2026); Micron Technology, results of September 30, 2026 (its non-GAAP gross margin was 87.0 percent).

10 Sources: TrendForce, September 4 and October 8, 2026 (forecasts); Microsoft, fiscal third- and fourth-quarter 2026 earnings calls, April 29 and July 29, 2026.

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At the bottom, a spot purchase leaves the supplier to decide how much capacity to build and the buyer to pay whatever the market charges. A purchase order fixes the price and quantity of one delivery. A **slot reservation agreement** goes further: the buyer pays a fee to hold a place in the supplier's production schedule, often years ahead, and converts it into a firm order later.

Redfish's memory agreement shows the next rung. Redfish must pay for every set it commits, used or not, which makes the agreement a **take-or-pay contract**. The price of a set follows the market between a floor of $0.30 million, which protects the supplier, and a ceiling of $0.42 million, which protects Redfish: a **price collar**. And Redfish pays a quarter of the committed value up front as a **customer deposit**, cash a buyer pays in advance and gets back later. Together they make a **deposit-backed take-or-pay agreement**, the contract Micron signed 26 times. Above that rung, a third party stands behind the buyer: a *capacity backstop*, a supplier's promise to buy capacity its customer cannot sell, such as the take-or-pay floors NVIDIA offers some neoclouds, or credit enhancement, such as the residual value guarantee NVIDIA gave for a campus leased to an OpenAI affiliate in Ohio (Chapter 16). Figure 14.5 draws the ladder.

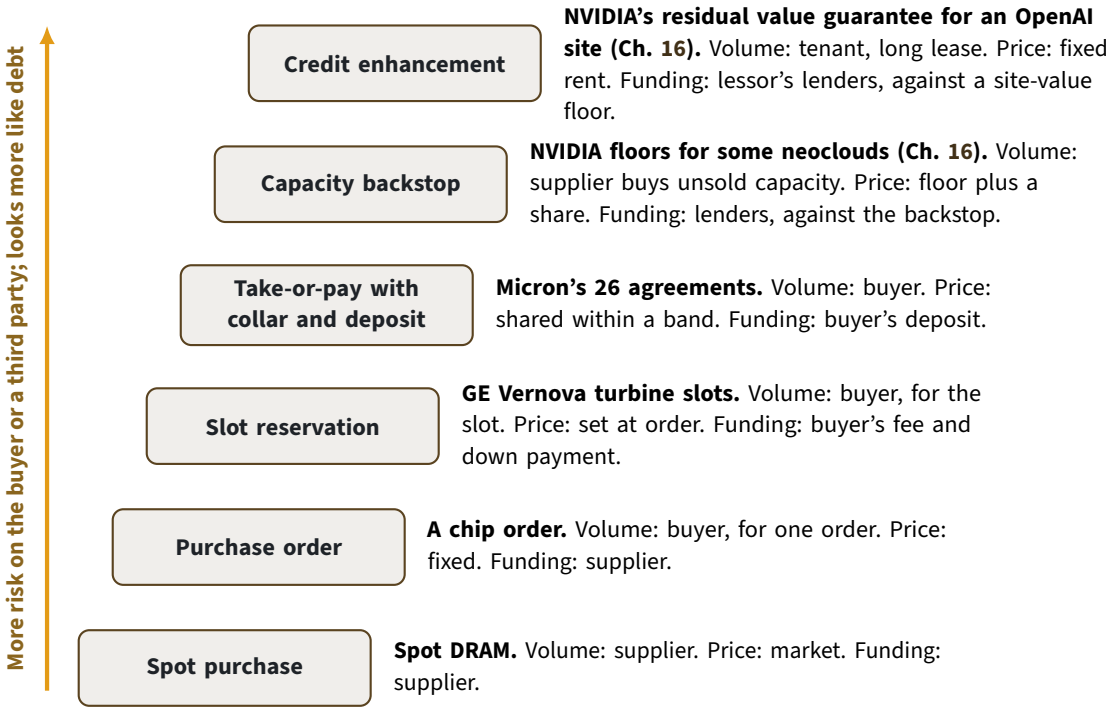


FIGURE 14.5 The contract ladder for scarce capacity. At the top the promise is fixed, dated and hard to escape, like debt. Examples as of mid-2026. Sources: Micron (September 30, 2026); GE Vernova (July 22, 2026); NVIDIA (August 17 and 26, 2026).

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Why climb? Each rung buys a place in line when the line is years long. And a supplier accepts a ceiling at the peak of a shortage because the floor, the take-or-pay promise and the deposit protect it against the glut that every earlier cycle produced. The question Chapter 9 asked of payables, when a commercial promise becomes debt in substance (Chapter 9), returns at every rung. Two later ladders continue this one: Chapter 15 ranks a buyer's obligations by how close they sit to recognized debt, and Chapter 16 ranks the support a supplier gives its customers, with the top two rungs here as its third and fourth.

Micron's agreements, whose deposits Chapters 1 and 5 met in passing, show the middle rungs in detail (Table 14.2). By September 30, 2026, it had signed 26 strategic customer agreements, expected to bring in more than 35 percent of its revenue through 2030. They are take-or-pay; about three quarters of the expected revenue under them comes with a pricing framework, mostly a floor and a ceiling. Remaining performance obligations under the agreements with a defined pricing framework, counted at committed volumes and floor prices, were about $150 billion, and Micron claimed that "even at floor prices" its margins would be "meaningfully above any prior cycle peak margins." SK hynix reported long-term agreements with "around 10 customers."

TABLE 14.2 Three contracts for scarce capacity ($ billions). Sources: Micron, fiscal Q3 and Q4 2026 prepared remarks and results (June 24 and September 30, 2026); NVIDIA, Form 10-K for fiscal 2026 and Forms 10-Q for the quarters ended April 27, 2025 and July 26, 2026; GE Vernova, second-quarter 2026 results and earnings call (July 22, 2026).

Contract	Term and price	Buyer's prepayment and how it is reported	As of
Micron strategic customer agreements (26)	Through 2030, some into 2031; take-or-pay; mostly floor and ceiling, largest ceilings at the 2Q26 market price	$32 committed, mostly cash deposits; $12.7 held; in financing cash flows, returned late in the term	Sep 30, 2026
NVIDIA supply and capacity agreements (NVIDIA as buyer)	Multi-year; premiums, deposits, long-term supply agreements and capacity commitments; terms not disclosed	Prepaid about $4.2 (Apr 27, 2025, latest disclosed); $279 of supply commitments, reported only in the notes	Jul 26, 2026
GE Vernova slot reservation agreements	Turbine slots for delivery to 2030 and beyond; first-half 2026 orders priced more than 20% above 4Q25	Reservation fees and down payments; contract liabilities up $13.7 in the first half	Jul 22, 2026

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EXAMPLE 14.3 Micron's deposits against its capex

In its quarter ended September 3, 2026, Micron received $12.3 billion of customer deposits and spent $10.8 billion on capital expenditures, so its customers financed $12.3/10.8 = 114$ percent of the quarter's capex. The $32 billion of commitments equals $32/150 = 21$ percent of the roughly $150 billion of remaining performance obligations. Read the perimeters before the ratio: the $32 billion covers all 26 agreements, while the $150 billion counts only those with a pricing framework, and only at floor prices, so the financing behind each dollar of contracted revenue is less than 21 cents.

Micron reports the deposits in financing activities, as it would a loan, so they do not flatter free cash flow. And Micron, with $68.3 billion more cash than debt and a rating of BBB+ or the equivalent from all three major agencies, is the case Chapter 5 called the harder one: between parties whose money costs about the same, a deposit creates no value. The buyers were paying for allocation.

How should one such contract be valued, by the buyer and by the supplier? Redfish's agreement, a Micron rung in miniature, lets us compute it.

EXAMPLE 14.4 Redfish's collar in three markets

Redfish commits 2,293 sets (Example 14.2). If the market price of a set averages $0.28 million over the year, below the floor, Redfish pays $0.30 million, or $2{,}293 \times (0.30 - 0.28) = \$45.86$ million more than spot buyers. At $0.35 million it pays the market. At $0.49 million it pays the $0.42 million ceiling and saves $2{,}293 \times (0.49 - 0.42) = \$160.51$ million; in a spike to $0.60 million it saves $2{,}293 \times 0.18 = \$412.74$ million. At every price the collar moves money between the two parties and creates none in cash; whatever value it creates comes from sharing risk, as Chapter 10 explains.

The deposit is different. Redfish pays $200.6 million at signing and finances it at 9 percent, a cost of $200.64 \times 0.09 = \$18.06$ million for the year. The supplier, whose money costs 6 percent, values the same cash at $200.64 \times 0.06 = \$12.04$ million. Moving the financing to the more expensive balance sheet destroys $6.02 million (the cost-of-capital lens of Chapter 5): the deposit is the price of the place in line, not a loan the supplier needed. Read against the collar, the ceiling repays the deposit's carry only if the market price ends above $0.42 + 18.06/2{,}293 = \$0.428$ million.

Finally, the take-or-pay promise. If demand comes in at 1,500 racks, Redfish must still pay for 2,293 sets and resell the 793 it cannot use, losing $793 \times 0.105 = \$83.3$ million at the reference price. The collar protects Redfish against high prices; nothing in the contract protects it against low demand.

The table below adds the pieces for each party ($ millions; net adds the deposit's carry).

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EXAMPLE 14.4 *continued*

	Redfish (9%)		Supplier (6%)		
Market price of a set	Collar	Net	Collar	Net	Joint
\$0.28 million	−45.86	−63.92	+45.86	+57.90	−6.02
\$0.35 million	0.00	−18.06	0.00	+12.04	−6.02
\$0.49 million	+160.51	+142.45	−160.51	−148.47	−6.02
\$0.60 million	+412.74	+394.68	−412.74	−400.70	−6.02

Every price row sums to the same −6.02 million: the collar only moves money between the parties, and the deposit destroys the same amount in every market. At 1,500 racks Redfish's loss grows to about \$101 million; the supplier's then depends on what else it could do with the unused sets.

Exhibit 14.3 turns the example into a routine. Notice that it asks for scenarios rather than a single forecast: the value of a collar lies in the prices nobody expects.

EXHIBIT 14.3 A procedure to value a deposit-backed take-or-pay agreement with a price collar

Step 1. Record the committed quantity, the floor and ceiling, the deposit and when it is repaid, and each party's cost of funds.

Step 2. Choose market price scenarios: below the floor, inside the band, above the ceiling, and a spike.

Step 3. For each scenario, find the price paid, the market price kept between floor and ceiling, and the buyer's gain against spot, (market price − price paid) × committed quantity; the supplier's gain is its negative.

Step 4. Value the deposit: deposit × cost of funds × years outstanding, at the buyer's rate (its cost) and at the supplier's rate (its value). The difference is value created or destroyed by moving the financing.

Step 5. For low-demand scenarios, compute the take-or-pay exposure: unused committed units × (price paid − resale value).

Step 6. Add the pieces for each party and scenario, and sign only if the worst plausible case is affordable.

```
=MIN(MAX(Spot,Floor),Ceiling)
```

THINK ABOUT IT

Redfish's memory supplier offers to lower the ceiling from \$0.42 million to \$0.39 million if Redfish raises its deposit from 25 to 40 percent of committed

THINK ABOUT IT *continued*

value. At a 9 percent cost of funds, what would you have to believe about next year's memory prices to accept?

Redfish now owes its supplier money before the first set arrives, and its customers want to pay later. Who, along the whole chain, ends up financing whom?

14.6 Who Finances Whom? Working Capital Along the Chain

In Chapters 5 and 8 the strong buyer stretched its suppliers' payment terms, and the suppliers financed the gap, sometimes by borrowing on the buyer's credit rating (Chapters 5 and 8). Redfish faces the opposite pattern on one side of its business and the familiar one on the other. Who finances whom along the AI chain, what decides the direction, and how much is left for the firm in the middle to fund?

Start upstream, where buyers pay early. Micron's customers financed its capacity (Section 14.5), and Case in Point 14.2 shows a turbine maker's customers doing the same. NVIDIA, in turn, pays its own suppliers early. As Chapter 1 noted, its annual report says it has "paid premiums, provided deposits, and entered into long-term supply agreements and capacity commitments," and its prepaid supply and capacity agreements came to about $4.2 billion at April 27, 2025, the latest amount it disclosed. The pattern reaches the hyperscalers too: in its quarter ended August 31, 2026, Oracle's customers prepaid $11.4 billion toward hardware, 11.4/28.5 = 40 percent of that quarter's $28.5 billion of capex (Chapter 15 returns to Oracle).[11]

CASE IN POINT 14.2 GE Vernova's slot reservations

GE Vernova's gas turbine business ran on slot reservation agreements in 2026. Gas turbines in backlog or under slot reservations rose from 100 GW at March 31, 2026 to 116 GW at June 30. In the second quarter it booked orders for 12.1 GW of gas turbines while shipping 3.3 GW, 12.1/3.3 = 3.7 times as much.

The cash came first. Contract liabilities, where customer down payments sit, rose by $13.7 billion in the first half of 2026, more than the company's $9.9 billion of free cash flow (a non-GAAP measure). As Chapter 5 computed, free cash flow would have been negative without that increase, and management described its gas turbine expansion from 20 GW to 30 GW of annual output as "all funded by customer down payments." A slot reservation turns trade credit around: instead of the supplier waiting to be paid, and perhaps selling the receivable to a factor, the customer pays early and finances the supplier's capacity.[12]

11 Sources: NVIDIA, Form 10-K for fiscal 2026 and Form 10-Q for the quarter ended April 27, 2025; Oracle, first-quarter fiscal 2027 results, September 10, 2026.

12 Source: GE Vernova, second-quarter 2026 results, presentation and earnings call, July 22, 2026.

Downstream, sellers wait. NVIDIA, a seller to the hyperscalers and neoclouds, let its days sales outstanding rise from 45 to 60 in one quarter of 2026 by giving certain investment-grade customers terms of 90 days up to one year. It also offers some neoclouds take-or-pay floors and has written guarantees (Chapter 16). Yet even NVIDIA collects some cash early: its customer advances rose to $2.8 billion at July 26, 2026, from $160 million six months before.

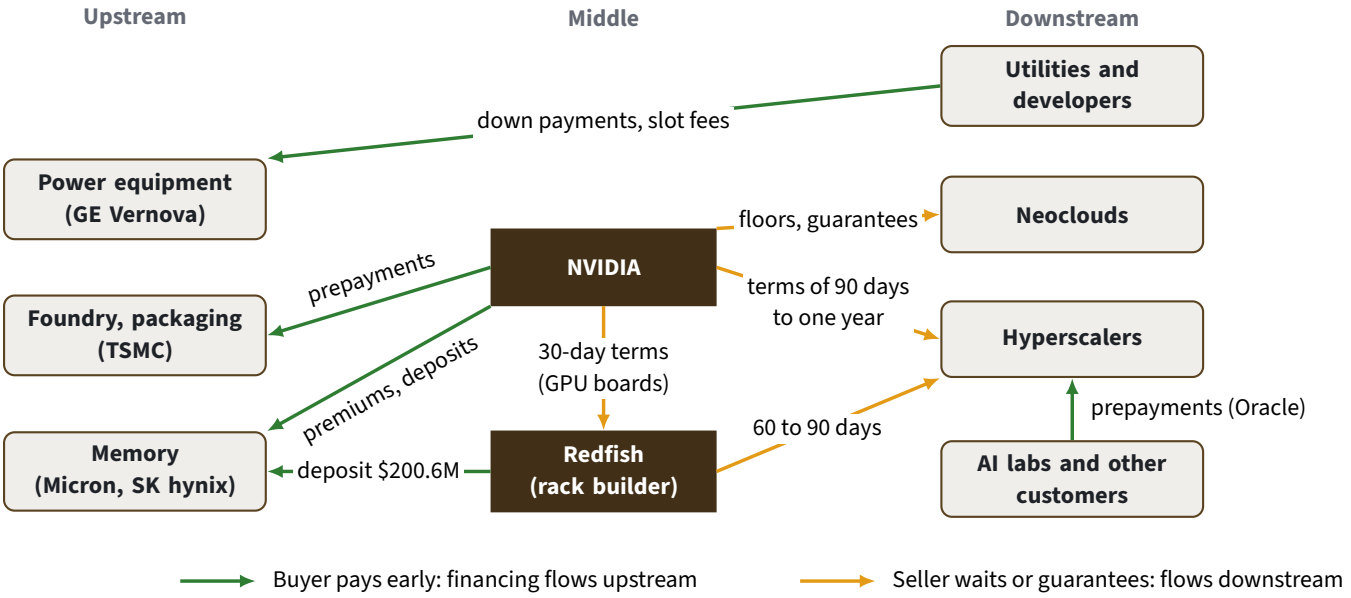


FIGURE 14.6 The financing-direction map of the AI chain in 2026. The firm in the middle can face both directions at once. Sources: GE Vernova (July 22, 2026); Micron (September 30, 2026); NVIDIA Form 10-K (February 2026) and 10-Q (July 26, 2026); Oracle (September 10, 2026); Redfish (fictitious).

Figure 14.6 is the **financing-direction map** of the chain: for every link, an arrow showing which party finances the other and with what instrument. Green arrows run upstream, toward the scarce suppliers. Amber arrows run downstream, from sellers that grant terms or stand behind their customers. Two features stand out. Financing does not follow credit quality: Micron and NVIDIA are both cash-rich, yet Micron collects deposits from its customers while NVIDIA lends to its customers for up to a year. And the firm in the middle can be squeezed from both sides.

> **INSIGHT** **Bargaining power, not credit rating, sets the direction of financing**
> Chapter 5 read a deposit through two lenses: the cost of capital says who can lend most cheaply, and bargaining power says who actually lends. Along the AI chain the second lens decides: the scarce party gets paid early, and the party that wants volume pays late. Before you ask which firm in a chain is creditworthy, ask which one is scarce.

How much does the squeeze cost a firm like Redfish? Example 14.5 puts numbers on it.

EXAMPLE 14.5 Redfish's working capital

At its mean demand of 2,000 racks, Redfish's revenue is $2{,}000 \times 3.50 = \$7{,}000$ million and its cost of goods sold $2{,}000 \times 3.20 = \$6{,}400$ million. With a 365-day year,

$$\text{Inventory} = 6{,}400 \times 45/365 = \$789.0 \text{ million},$$
$$\text{Receivables} = 7{,}000 \times 60/365 = \$1{,}150.7 \text{ million},$$
$$\text{Payables} = 6{,}400 \times 30/365 = \$526.0 \text{ million}.$$

Net working capital is $789.0 + 1{,}150.7 - 526.0 = \$1{,}413.7$ million, and the memory deposit adds \$200.6 million, so Redfish must finance \$1,614.3 million. On its 9 percent line that costs \$145.3 million a year, $145.3/600 = 24.2$ percent of its \$600 million gross profit. In days, the cash conversion cycle is $45 + 60 - 30 = 75$ days, and the deposit adds $200.6/(6{,}400/365) = 11.4$ days of cost of goods sold.

Now Hyperscaler A, with half of Redfish's sales, asks for 90 days. Its receivables rise from $3{,}500 \times 60/365 = \$575.3$ million to $3{,}500 \times 90/365 = \$863.0$ million, an extra \$287.7 million that costs $3{,}500 \times 0.09 \times 30/360 = \$26.25$ million a year.[13] But A also offers its payables finance program at 5.2 percent. Selling A's approved invoices at a discount of $0.052 \times 90/360 = 1.3$ percent costs $3{,}500 \times 0.013 = \$45.5$ million a year. Carrying the same 90-day receivables on the credit line would cost $3{,}500 \times 0.09 \times 90/360 = \$78.75$ million, and even today's 60-day receivables cost $3{,}500 \times 0.09 \times 60/360 = \$52.5$ million. Through the program, Redfish can grant 90 days and still come out \$7.0 million a year ahead of financing 60 days itself.

13 As in Chapter 8, charges for a number of days use actual/360; the year-long carrying costs above are simply balance times rate.

Notice where Redfish sits: it lends \$200.6 million upstream to a supplier whose money costs 6 percent, and more than a billion dollars downstream to customers strong enough to set a 5.2 percent program rate. The firm with the dearest money in the chain finances both neighbors unless it finds a cheaper balance sheet to carry the days. Exhibit 14.4 turns the map into a routine.

EXHIBIT 14.4 A procedure to map who finances whom along a chain

Step 1. Draw the chain from raw input to end customer. On each link, record payments made early (deposits, prepayments, reservation fees) and credit granted (days of terms, revenue floors, guarantees).

Step 2. Convert each into dollars. Days paid early times daily purchases is lent upstream; DSO times daily sales is lent downstream (365-day year); deposits count at their balance.

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EXHIBIT 14.4 *continued*

Step 3. For each firm, add its inventory to what it lends both ways (receivables and deposits), and subtract the credit it receives from its own suppliers. The result is the working capital it must fund; cost it at the firm's own rate.

Step 4. Record who is scarce and who has alternatives on each link, since bargaining power sets the direction.

Step 5. Look for a cheaper balance sheet: a customer's payables finance program, a sale of receivables, or a supplier that will take a smaller deposit for a higher price. Compare each at the parties' actual rates.

Step 6. Mark where credit risk ends up: the firm holding the largest receivables or deposits from the weakest party.

The largest firm in the middle of the real chain is NVIDIA, which buys ahead from its suppliers and sells on long terms to its customers. What does its working capital say about the buildout?

14.7 Reading the Supplier: Is NVIDIA Overbuilding?

Every GPU in Redfish's racks sat in NVIDIA's inventory first, and every memory chip NVIDIA has committed to buy will sit there too. If the buildout were running ahead of demand, the supplier's working capital should be among the first places to show it: unsold chips piling up, customers paying more slowly, write-downs creeping in. In late February 2026, after NVIDIA reported fiscal 2026 revenue of $215.9 billion, the *Wall Street Journal* ran a column headlined "Why Nvidia's Huge Numbers Don't Settle the Latest AI Fears." Can the tools of Chapter 2 settle them?

Begin with the snapshot an analyst could take that spring, for the quarter ended January 25, 2026. The bull case: quarterly revenue was 73 percent higher than a year earlier, while inventory shrank relative to revenue and customers paid faster than before the boom. The bear case: inventory had more than doubled to a record $21.4 billion, and three customers each owed more than 10 percent of receivables, 56 percent between them. The analysis behind the January column of Table 14.3 concluded that NVIDIA was "not yet overbuilding": its working capital ratios had held at pre-boom levels while quarterly revenue grew roughly twelvefold, and the one amber light was the growth spread, revenue growth minus inventory growth, at $73 - 112 = -39$ percentage points.

Six months later the picture had moved in three ways. First, the balance sheet grew faster than sales. At July 26, 2026, inventory was $31.6 billion: raw materials had tripled in six months, from $3.8 billion to $11.3 billion,

TABLE 14.3 NVIDIA's overbuilding signals. Pre-AI averages cover the quarters before fiscal 2024. The January column is the course analysis of the 10-K, on a method slightly different from the July column's 91-day, ending-balance calculation from the 10-Q (NVIDIA itself reports a DSO of 60). Thresholds for commitments and receivable concentration were set after July and apply from the October 2026 quarter. Sources: Xu, "Is NVIDIA Overbuilding?" (OPIM 5115, 2026); NVIDIA Form 10-K for fiscal 2026 and Form 10-Q and CFO commentary for the quarter ended July 26, 2026.

Signal	Pre-AI average	Jan 25, 2026	Jul 26, 2026	Warning
Days inventory outstanding (DIO)	111	113	119.3	above 140
Days sales outstanding (DSO)	58	51	59.6	above 70
Days payables outstanding (DPO)	56	52	56.9	
Cash conversion cycle (days)	113	112	122.0	
DPO/DIO	0.50	0.46	0.48	
Inventory / quarterly revenue	0.49	0.31	0.33	above 0.50
Growth spread (percentage points)		−39	−5	below 0
Inventory provisions ($M, quarter)		322	784	above 500
Accrued excess purchase obligations ($B)		2.7	2.1	
Supply commitments / trailing revenue		0.44	0.92	up 50% in a year
Customers owing 10% or more of receivables		3 (56%)	5 (about 70%)	4 or more

consistent with buying memory ahead of the Vera Rubin ramp, while finished goods fell from $8.8 billion to $6.9 billion. Receivables were $63.1 billion and payables $15.1 billion. Second, the receivables grew riskier: five customers each owed 10 percent or more, about 70 percent between them, and Chapter 2 traced about $15.5 billion of them to terms of up to a year (Chapter 2). Third, and largest, the exposure moved off the balance sheet. A **purchase obligation** is a commitment to buy goods or capacity later on terms fixed today; when it cannot be cancelled freely, the buyer bears the loss if demand falls, yet the obligation itself appears only in the notes, and only a loss judged probable is accrued on the balance sheet (Chapter 3). NVIDIA's accrued liability for excess purchase obligations fell from $2.7 billion to $2.1 billion in the six months, while its supply and capacity commitments, $95.2 billion in January, reached $279 billion in July, up from $119 billion in April, an increase NVIDIA said was "primarily related to the procurement of memory" (Figure 14.7). Of the $279 billion, $92 billion falls due in the rest of fiscal 2027 and $87 billion and $88 billion in the two years after.[14]

INSIGHT **A purchase obligation is a put written to the supplier, on volume**

A non-cancellable commitment to buy obliges the buyer to take the goods whatever they are worth when they arrive. On price it is a forward purchase: if market prices rise, the buyer gains, as Redfish does above its ceiling. On volume

14 Sources: NVIDIA, Form 10-Q for the quarter ended July 26, 2026 and CFO commentary, August 26, 2026; earlier quarters from the corresponding filings. Balances as of each quarter end.

INSIGHT *continued*

it behaves like a put written to the supplier: if demand holds, the commitment is simply inventory paid for later and earns nothing beyond the units used; if demand vanishes, the buyer pays for goods it cannot use. NVIDIA learned this in April 2025, when an export license requirement turned its H20 chips and the purchase obligations behind them into a $4.5 billion charge (Chapter 3). Like any written put, the volume side pays nothing extra when demand is strong and costs most when it collapses.

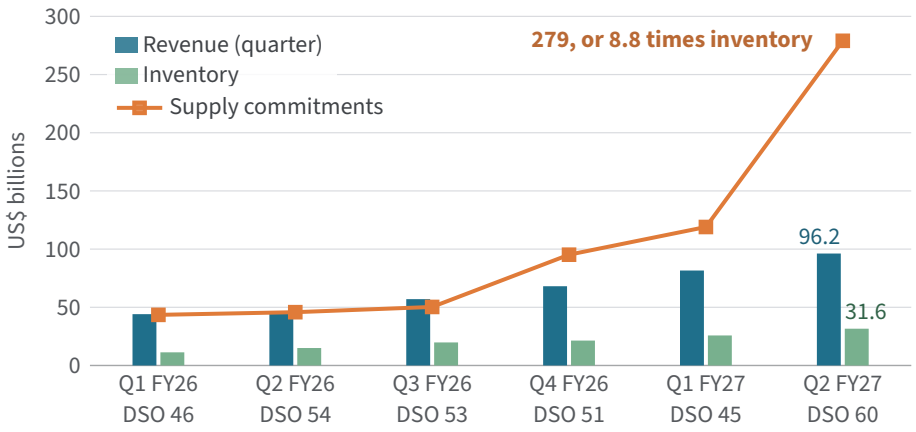


FIGURE 14.7

NVIDIA by quarter, fiscal 2026 to the second quarter of fiscal 2027 (quarters end in late April, July, October and January), on one scale: supply and capacity commitments are now almost three times a quarter's revenue; NVIDIA attributed their latest rise primarily to memory. Sources: NVIDIA quarterly filings and CFO commentaries, May 2025 to August 2026; DSO as reported by the company.

What did management say? "We expect to grow revenue by approximately 70% in fiscal 2028. This is a supply-constrained outlook," chief financial officer Colette Kress told analysts in August 2026, adding that customers' forecasts pointed to growth "doubling next year." The same call guided gross margin down from 75.0 percent toward 71 to 72 percent in the fourth quarter of fiscal 2027. Commitments of this size are the choice of a firm that expects demand to outrun supply, and a liability if it does not.

Example 14.6 reads the July numbers against the spring list and the new signals, and Exhibit 14.5 turns the reading into a routine.

EXAMPLE 14.6 NVIDIA's signals, January and July 2026

Net working capital rose from $21.4 + 38.5 - 9.8 = \$50.1$ billion in January to $31.6 + 63.1 - 15.1 = \$79.6$ billion in July, up 59 percent while quarterly revenue rose 41 percent, from $68.1 billion to $96.2 billion. With July cost of revenue of $24.08 billion and a 91-day quarter,

$$\text{DSO} = \frac{63.06}{96.22} \times 91 = 59.6, \qquad \text{DIO} = \frac{31.58}{24.08} \times 91 = 119.3,$$

and $\text{DPO} = 15.06/24.08 \times 91 = 56.9$, so the cycle reached $59.6 + 119.3 - 56.9 = 122.0$ days, about 10 days longer than in January, part of it a change of method. Every day ratio is still below its threshold, and DPO/DIO, at 0.48, is close to its pre-AI 0.50. The January amber light is still on by the rule, but it has dimmed:

EXAMPLE 14.6 *continued*

revenue grew 106 percent in a year and inventory $31.6/15.0 - 1 = 111$ percent, a growth spread of about −5 points. The quarter's inventory provisions of $784 million crossed the $500 million threshold set in the spring, as they would have a year earlier ($886 million). As a share of revenue, though, they fell from 1.9 to 0.8 percent: scale thresholds to the firm.

The new signals had no spring threshold, so we judge them and say so. Commitments to trailing revenue rose from $95.2/215.9 = 0.44$ in January to $279/303.0 = 0.92$ in July, more than doubling in six months, and commitments are now $279/31.6 = 8.8$ times inventory. If 3 percent of them turned into excess purchases with no resale value, the charge would be $0.03 \times 279 = \$8.4$ billion, more than all of fiscal 2026's $7.2 billion of provisions, H20 included. Both new signals would be amber on the thresholds set this edition, which apply from the next quarter. Classification: amber on judgment, because the balance sheet looks healthy while the exposure has moved into commitments and into receivables owed by five customers.

EXHIBIT 14.5 A procedure to set an overbuilding watch list for a supplier

Step 1. Each quarter, compute the cash conversion cycle (Chapter 2) and the inventory signals of Chapter 3: inventory to quarterly revenue, DPO/DIO, the growth spread and provisions as a share of revenue.

Step 2. Read the notes: supply commitments by maturity, accrued losses on excess purchase obligations, customer advances, receivable concentration and stated payment terms.

Step 3. Compute commitments to trailing twelve-month revenue and to inventory, and the receivables in excess of a normal-terms DSO.

Step 4. Add market signals, such as memory price momentum (Section 14.4).

Step 5. Fix the thresholds before looking at the latest quarter, and write down which signal would change your mind.

Step 6. Classify the supplier green, amber or red, and re-run the list every quarter.

PITFALL Commitments are the new inventory

An analysis that stops at the balance sheet misses most of the exposure. In July 2026 NVIDIA held $31.6 billion of inventory and had committed to buy $279 billion more. Its main commitments table totaled $366 billion, with supply the largest item; a further $56 billion sat in a separate table, and guarantees could reach $108.5 billion. Analysts' sums of about $530 billion mix these perimeters. Read each note, and say which perimeter you mean.

Is NVIDIA overbuilding? The July numbers do not say so: the company calls its outlook supply-constrained, and the inventory signal that worried

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analysts in the spring has dimmed. But a verdict is a snapshot, this one dated July 26, 2026, and the watch list now carries amber lights that the spring list did not measure. Each depends on one question: will demand keep showing up? (Chapter 16 analyzes the guarantees.)

14.8 Does Demand Show Up? The Jevons Test

Every commitment in this chapter, from Redfish's 2,293 memory sets to NVIDIA's $279 billion, is a bet that demand will arrive. Yet the price of what an AI factory sells keeps falling. Shouldn't falling prices shrink revenue and make overbuilding more likely? The answer depends on how much more computing customers buy when it gets cheaper.

The **Jevons paradox** is the observation that when greater efficiency makes a resource cheaper, total spending on it can rise, because use grows more than in proportion to the fall in price. Whether it holds is a question about the price elasticity of demand: the percentage change in quantity for a 1 percent change in price. When the elasticity is above 1, a lower price raises total spending; below 1, it lowers it. The first great test for AI came in early 2025 (Case in Point 14.3).

CASE IN POINT 14.3 DeepSeek and the Jevons paradox

In January 2025 the Chinese lab DeepSeek released an open reasoning model that matched a leading U.S. model at a fraction of the cost, and on January 27 NVIDIA lost $589 billion of market value in a day. Yet capacity additions of 15 to 18 GW a year continued through 2025. The price of inference fell from about $20 per million tokens in late 2022 to $0.40 in late 2025, while total inference spending rose 320 percent: cheaper computing expanded the market, as more efficient steam engines raised Britain's coal consumption in Jevons's day. Lenders should reject the intuitive model in which cheaper means less demand, at least for now. The flip side matters for collateral: when the price of a token falls 50-fold in three years, the hourly rent that secures a GPU loan can fall too (Chapter 16).[15]

15 Sources: OPIM 5115 Lecture 13 (April 2026), citing a16z and market data; figures as of early 2026.

EXAMPLE 14.7 Is AI demand price-elastic?

Between late 2022 and late 2025 the price of a million tokens fell from about $20 to about $0.40, a price ratio of $0.40/20 = 0.02$, or a 50-fold fall. Total inference spending rose 320 percent, a spending ratio of 4.2. Because spending is price times quantity, the quantity ratio is

$$\frac{\text{spending ratio}}{\text{price ratio}} = \frac{4.2}{0.02} = 210 :$$

customers bought 210 times as many tokens. For changes this large, measure the elasticity in logarithms (a constant-elasticity measure; the midpoint

EXAMPLE 14.7 *continued*

formula of introductory economics gives a smaller number for changes this large):

$$\text{elasticity} = \frac{\ln 210}{\ln 50} = \frac{5.35}{3.91} = 1.37.$$

Demand was elastic, so cheaper tokens raised total spending. Had the elasticity been 0.5, the same 50-fold fall would have multiplied spending by $50^{0.5} \times 0.02 = 0.14$, a fall of 86 percent, and every capacity commitment in this chapter would look reckless.

The historical elasticity says demand has grown with falling prices. Do the forecasts and order books say it will keep growing? In 2026 most pointed the same way. A McKinsey model reproduced by Van Nieuwerburgh in March 2026 put global data center demand at 82.3 GW in 2025 and 219.0 GW in 2030, growth of 21.6 percent a year. At June 30, 2026, Google Cloud reported a backlog of $513.9 billion, Amazon unrecognized performance obligations of about $496 billion, mostly for AWS, and Microsoft a commercial remaining performance obligation of $678 billion, each on its own definition. And NVIDIA said its customers' forecasts pointed to its growth "doubling next year."[16]

You might worry that forecasts count the same demand twice: a developer that shops one data center to five utilities appears in five queues. Grid Strategies forecast in November 2025 that U.S. peak demand would grow by 166 GW in the five years to 2030, found that about 90 GW of that growth was linked to data centers, and judged the data center forecasts likely overstated by roughly 25 GW. In Texas, the grid operator's queue of large loads reached about 474 GW by August 2026, about 90 percent of it data centers and more than five times the state's record peak.[17] Chapter 15 shows how utilities began to charge large customers whether or not they used the power, partly to filter out such phantom demand.

What does this mean for a commitment? Because $Q^* = \mu + z\sigma$, the commitment moves one for one with the forecast's mean: every 100 racks of phantom demand in Redfish's forecast add about 100 sets that are likely to go unused. Elastic demand raises the true mean, but duplicated requests raise only the forecast's, so a careful buyer strips them out before it signs. Uncertainty works differently. When the critical fractile is above 0.5, as Redfish's 0.721 is, z is positive, so a wider forecast raises the commitment: with $\sigma = 600$ racks instead of 500, $Q^* = 2{,}000 + 0.587 \times 600 = 2{,}352$ sets, while expected profit falls. If Redfish judged 200 racks of its forecast to be phantom and widened σ to 600, it would commit $1{,}800 + 0.587 \times 600 = 2{,}152$ sets, 141 fewer than today. A volatile forecast calls for terms that make a wrong commitment cheaper, such as a smaller deposit or a cancellation right, rather than for a smaller Q^*.

16 Sources: Van Nieuwerburgh (March 2026), McKinsey Data Center Demand Model (a forecast); Alphabet, Amazon and Microsoft quarterly filings for June 2026; NVIDIA earnings call, August 26, 2026.

17 Sources: Grid Strategies, national load growth report (November 2025); Utility Dive, August 5, 2026.

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Suppose demand does show up. Somebody must still earn enough from it to repay what the chain has spent. Who?

14.9 Who Earns the Money to Repay It?

Every deposit, commitment and receivable in this chapter is money spent ahead of revenue, and every dollar of it must eventually be repaid out of someone's profit. So where in the stack is the profit earned? Apoorv Agrawal of Altimeter Capital sizes the AI economy in three layers (the middle lens of Figure 14.1): semiconductors (chips and memory), infrastructure (cloud capacity, including the neoclouds) and applications (the model labs and the software built on them). Table 14.4 sets his early-2026 numbers beside the mature cloud-software stack.

TABLE 14.4 Annualized revenue and gross profit by layer, AI stack (first quarter of 2026) versus the cloud-software stack (US$ billions; shares in percent). Source: Agrawal (April 2026), as presented in OPIM 5115 Lecture 13.

	AI stack			Cloud-software stack		
Layer	Revenue	Gross profit	GP share	Revenue	Gross profit	GP share
Applications	60	20	7	600	470	70
Infrastructure	75	40	14	300	160	24
Semiconductors	300	225	79	80	40	6
Total	435	285	100	980	670	100

The two stacks are mirror images, a pattern we call **value chain inversion**. In cloud software the applications layer earns 70 percent of gross profit and chips 6 percent. In AI, chips earn $225/285 = 79$ percent and applications $20/285 = 7$ percent. The sellers of picks and shovels, above all NVIDIA, capture most of the profit.

You might argue that the applications layer is growing fastest and will soon close the gap. In percentages it is: its revenue rose from $5 billion in 2024 to $60 billion, up 1,100 percent, while semiconductor revenue rose 300 percent, from $75 billion to $300 billion. In dollars the story reverses. Applications added $55 billion and chips $225 billion, $225/55 = 4.1$ times as much. Fast percentage growth on a small base cannot close a large dollar gap quickly.

Why is the profit so concentrated upstream? Markups alone cannot explain it. If each layer simply resold what it bought, revenue would grow at every step toward the customer; in Table 14.4 it shrinks, from $300 billion of chips to $60 billion of applications, because chips are capital goods, bought

mostly out of hyperscaler cash flow and borrowing ahead of the end-user revenue they are meant to earn over several years. Markups do explain what the end user pays. Each layer marks up what it buys from the layer below, and the markups compound, a problem known as **double marginalization** that raises the end price and shrinks the market. With illustrative gross margins of 73 percent for chips, 55 percent for infrastructure and 33 percent for applications (Table 14.4 itself implies 75, 53 and 33 percent), and each layer pricing at its cost divided by one minus its margin, a chip that costs $10,000 to make is marked up three times on its way to the end user:

$$\begin{aligned} \text{Chip maker:} \quad & 10{,}000/(1-0.73) = \$37{,}037 \\ \text{Infrastructure:} \quad & 37{,}037/(1-0.55) = \$82{,}305 \\ \text{Application:} \quad & 82{,}305/(1-0.33) = \$122{,}843 \end{aligned}$$

The end price is $122{,}843/10{,}000 = 12.3$ times the production cost, although the chip maker's own gross profit, $27,037, is the smallest of the three layers'. The example assumes every chip is resold all the way to a paying end user. In 2026 most were not yet, which is why the chip layer's share of actual gross profit in Table 14.4 is 79 percent rather than the 24 percent of this example. An application with a 33 percent margin still pays about 67 cents of every revenue dollar upstream.

The textbook response to double marginalization is vertical integration, and the hyperscalers are applying it with their own chips. Google's TPUs, Amazon's Trainium, Microsoft's Maia and Meta's MTIA all aim to recapture NVIDIA's margin. If a hyperscaler buys a custom chip at a 50 percent margin instead of 73 percent, the end price in our example falls from $122,843 to $66,335, by 46 percent. But diversification by brand is not diversification by supply chain. TSMC fabricates most of these chips, and Broadcom is the design partner for several of them, so the hedge shares NVIDIA's single points of failure. The second-sourcing logic of Chapter 10 applies one tier down: watch the bill of materials, not the brand.

We can now answer the chapter's opening questions. Micron's customers lent to a supplier that did not need the money because a deposit buys a place in line and makes a commitment costly to abandon (Section 14.5). Redfish should commit 2,293 memory sets, 30 fewer than if the deposit cost nothing, and it can grant Hyperscaler A its 90 days if it sells those invoices through A's program (Example 14.5). If demand does not come, the loss stays with whoever signed the take-or-pay promise: Redfish for its sets, NVIDIA for its $279 billion of commitments.

What does this mean for whoever finances the buildout? Because chips earn over several years, the chip layer's annual revenue can exceed the applications layer's in any buildout, even in a chain that will pay for itself. The gap here is too wide for that to be the whole story: semiconductor revenue in Table 14.4 is $300/60 = 5$ times application revenue, and even

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spread over a five-year life, one year's chips cost 300/5 = $60 billion a year, all of the applications layer's revenue. Part of the return arrives as the hyperscalers' own advertising and search revenue, which the applications layer does not count; the rest of the difference is paid, for now, by hyperscaler cash flow, bonds, leases and private credit. The profit sits with the chip makers, which fund themselves mostly from that profit; the borrowing happens in data centers and neoclouds, whose rent is paid by an applications layer that earns 7 percent of the profit. The inverted value stack is financed by the capital stack. Chapter 15 follows that money: who lends it, in what form, and where each form shows up.

CHAPTER SUMMARY

The AI buildout is a supply chain before it is a technology: a stack of energy, chips, infrastructure, models and applications, read through three lenses that show where capacity, profit and risk live.

The binding constraints are physical. Advanced packaging, high-bandwidth memory, transformers, turbines and grid connections take months to years to expand, and several come from a handful of firms; a stage that is both concentrated and slow is a chokepoint. A chain that cannot sprint must commit early, and an early commitment is a bet.

The newsvendor model sizes that bet. When the commitment requires a deposit, the cost of financing it enters the numerator of the critical fractile, because it is paid on every committed unit, used or not.

Scarcity gave suppliers power. The memory supercycle nearly tripled conventional DRAM contract prices between the third quarter of 2025 and the first quarter of 2026, on TrendForce's estimates, with further rises forecast, and moved the profit to three memory makers. Buyers climbed a ladder of contracts. A collar moves cash between the parties at every price and creates value only by sharing risk; a deposit placed on the dearer balance sheet destroys value and is the price of a place in line; and the take-or-pay promise leaves the buyer holding the risk of low demand.

Along the chain, bargaining power rather than credit quality sets the direction of financing, and a firm in the middle can be squeezed from both sides unless it borrows a stronger party's credit. NVIDIA's balance sheet looked healthy in mid-2026 while its exposure moved into purchase obligations and concentrated receivables: commitments are the new inventory. Demand has so far grown faster than prices have fallen, but forecasts double-count and move quickly. And the profit that must repay the buildout sits mostly with the chip makers, while the borrowing happens in the layers below: the inverted value stack is financed by the capital stack.

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MANAGERIAL TAKEAWAYS

- Map any AI asset to its layer, chain and stage before you lend against it or commit to buy it. Ask what binds, how long it takes to fix, and how many firms can supply it.
- Size every capacity commitment with the critical fractile, and put the cost of financing it in the numerator: it is paid on every committed unit, used or not.
- When a supplier holds the power, expect to finance it. Price a deposit at both parties' cost of funds; if moving the financing destroys value, treat the deposit as the price of allocation and bargain it against the ceiling.
- Value a take-or-pay agreement with a collar in at least three markets and one demand shortfall: the collar protects against price, but nothing protects against low demand.
- In the middle of a chain, look for a cheaper balance sheet to carry the days, such as a customer's payables finance program.
- Read the commitments note with the balance sheet, divide commitments by trailing revenue, and fix your thresholds before the quarter: commitments are the new inventory.
- Strip duplicated requests out of a demand forecast before you size a commitment. Above a fractile of 0.5, a wider forecast raises the commitment; guard against volatility with terms, such as a smaller deposit or a cancellation right, not with a smaller order.
- Before lending to an application-layer firm, compute how much of each revenue dollar it pays upstream; revenue size is not profit capture.

KEY TERMS

AI factory — A data center built to turn electricity into AI output, measured in tokens, at industrial scale.

Binding constraint — The input that limits output when every other input is available.

Capacity commitment — A promise made today to buy capacity or output later, on terms fixed today.

Critical fractile — Underage cost, less any per-unit financing cost, divided by the sum of underage and overage costs; the optimal commitment covers demand with this probability.

Customer deposit — Cash a buyer pays a supplier in advance and receives back later.

Deposit-backed take-or-pay agreement — A multi-year take-or-pay contract with a price collar, secured by a customer deposit.

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Double marginalization	The compounding of markups as a product passes through several firms.
Financing-direction map	A diagram of a chain showing, link by link, which party finances the other and with what instrument.
Five-layer stack	Energy, chips, infrastructure, models and applications, in Huang's view of AI.
Herfindahl-Hirschman index (HHI)	The sum of squared market shares in percentage points; 10,000 for a monopoly and $10{,}000/n$ for n equal firms.
Hyperscaler	One of the few largest cloud and platform operators, such as Amazon or Microsoft.
Jevons paradox	A fall in the price of a resource that raises total spending on it, because use grows more than in proportion.
Memory supercycle	A memory price rise driven by a lasting shift in demand, long enough that buyers sign multi-year contracts and prepay for supply.
Neocloud	A specialist that owns GPUs and rents computing time to AI labs under long contracts.
Newsvendor model	A one-period model that sets capacity before demand is known by balancing the cost of too little against the cost of too much.
Price collar	A price that follows the market between a floor and a ceiling.
Purchase obligation	A commitment to buy goods or capacity later on terms fixed today, reported in the notes.
Slot reservation agreement	A contract in which a buyer pays to hold a place in a supplier's production schedule.
Take-or-pay contract	A contract under which the buyer pays for its committed quantity whether or not it uses it.
Value chain inversion	A profit split in which the upstream layer earns most of the gross profit.

DISCUSSION QUESTIONS

1. Is the chip layer's capture of most AI gross profit an equilibrium or the peak of a cycle? Name three indicators you would track monthly to tell the difference.
2. Oracle and OpenAI decided not to expand one Stargate site from 1.2 GW toward 2 GW, while Oracle's capex overshot its own guide. Do these moves refute or confirm the critical fractile view of capex plans?

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3 Micron collected deposits while holding $68.3 billion more cash than debt. As a buyer's chief financial officer, what would you want in return for a deposit, and when would you refuse one?

4 NVIDIA's watch list had no threshold for commitments to trailing revenue until that ratio had more than doubled, and its $500 million threshold for inventory provisions flagged a quarter in which provisions had fallen from 1.9 to 0.8 percent of revenue in a year. What does an analyst lose by setting thresholds after seeing the data, and how would you set them in advance for a supplier one-tenth NVIDIA's size (Exhibit 14.5)?

5 Should analysts and rating agencies treat a supplier's purchase obligations like inventory, like debt, or like neither? What disclosure would let you decide?

6 Memory had no liquid futures market in 2026, and three firms earned 88 percent of DRAM revenue. Would a futures market have made deposits and collars unnecessary? Who would have sold the futures?

PROBLEMS

14.1 **Grackle AI** cuts its price from $5.00 to $1.25 per million tokens. Over the same period, customers' spending with Grackle rises from $2.0 billion to $5.6 billion a year.

(a) Compute the quantity ratio and the log elasticity of demand, the logarithm of the quantity ratio divided by the logarithm of the price fall.

(b) Is demand elastic? What would spending have been after the price cut if the elasticity had been 0.8?

14.2 **Bittern Memory** sells a buyer 1 million memory modules under a one-year take-or-pay agreement with a reference price of $100, a floor of $85 and a ceiling of $120 per module. The buyer pays a deposit of 20 percent of committed value at the reference price, returned at year end without interest; its cost of funds is 6 percent.

(a) Compute what the buyer pays and its gain or loss against buying at spot if the market price averages $70, $100 or $150.

(b) Compute the deposit's cost to the buyer for the year.

(c) Above what market price does the ceiling's saving exceed the deposit's cost?

14.3 **Wagtail Notebooks** sells a notebook for $900. At the start of the year its bill of materials is $700, of which memory is $150. Memory contract prices then rise 47.5 percent in one quarter and 92.5 percent in the next.

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(a) Compute the memory price index after the two quarters (start = 100) and the new memory cost per notebook.

(b) Compute the new bill of materials and Wagtail's gross profit per notebook at the old price.

(c) What price would restore the original gross margin percentage?

14.4 **Dunlin Compute** plans to deploy 4,000 GPU racks next year and maps four stages of its supply chain.

- Substrates: suppliers' shares 60, 25 and 15 percent; capacity 110 percent of Dunlin's need; 30 weeks to add capacity.
- HBM memory: shares 50, 30 and 20 percent; capacity 92 percent of need; 104 weeks.
- Power transformers: shares 25, 20, 15, 10, 10, 10 and 10 percent; capacity 70 percent of need; 128 weeks.
- Rack assembly: twenty suppliers with 5 percent each; capacity 130 percent of need; 8 weeks.

(a) Compute each stage's HHI.

(b) Which stage is the binding constraint, and how many racks can Dunlin deploy?

(c) If imports lift transformer capacity to 100 percent of need, which stage binds, and how many racks can Dunlin deploy?

(d) Using the rule of Exhibit 14.1, which stage is a chokepoint?

14.5 **Egret Turbines** sells 20 GW of turbines a year at $1,000 per kW. A customer pays 10 percent when it books a slot four years before delivery, 20 percent when it converts the slot into a firm order two years before delivery, and 70 percent at delivery. In a steady state, bookings, orders and deliveries are 20 GW a year, and each payment is received during the year it falls due.

(a) Compute Egret's contract liabilities (cash received for turbines not yet delivered) at the end of a year.

(b) Compute the cash Egret collects in a steady-state year.

(c) This year, bookings for delivery in four years jump to 30 GW. By how much do cash collected and contract liabilities rise this year?

(d) A customer with 1 GW due for delivery in two years, which converted its slot this year, walks away and forfeits what it has paid. How much does Egret keep?

14.6 **Orrin Semiconductor** makes an accelerator for $12,000 and prices it at a 70 percent gross margin; **Pennant Cloud** prices compute at a 50 percent margin and **Quillon AI** its application at 30 percent. Each firm prices at its cost divided by one minus its margin.

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(a) Compute the price at each layer and the ratio of the end price to the production cost.

(b) What share of the end price is Orrin's gross profit, and how much of each revenue dollar does Quillon pay upstream?

(c) Pennant switches to a custom chip with the same $12,000 cost, bought from a design partner at a 40 percent margin. Compute the new end price and its percentage change.

14.7 **Curlew Systems**, a rack integrator, expects revenue of $2.4 billion and cost of goods sold of $2.16 billion, with DIO of 40 days, DSO of 55 days and DPO of 35 days (365-day year), and it has paid a $60 million memory deposit. Its credit line costs 8 percent.

(a) Compute inventory, receivables and payables, and the total Curlew must finance including the deposit.

(b) Compute the annual cost of financing that total (balance times rate).

(c) A customer with 40 percent of sales asks for 90 days instead of 55. Compute the extra receivables and their annual cost (actual/360).

(d) The same customer offers its payables finance program at 5.5 percent (discount = rate × days/360). Compute the annual discount on its invoices at 90 days and the saving against carrying its 90-day receivables on the credit line.

14.8 **Flicker Semiconductor** reports supply commitments of $150 billion, trailing twelve-month revenue of $200 billion and inventory of $18 billion. A year earlier, commitments were $40 billion and trailing revenue $130 billion. Of today's commitments, $60 billion fall due next year.

(a) Compute commitments to trailing revenue now and a year earlier, and commitments to inventory now.

(b) Under a watch-list rule that flags a rise of more than 50 percent in a year in commitments to trailing revenue, is the signal amber?

(c) Demand falls short, and 5 percent of the commitments due next year become excess. If the excess units can be resold for 40 percent of their cost, what charge does Flicker take?

14.9 **Winslow Hyperscale** must reserve 2029 capacity now. Demand is normal with a mean of 6 GW and a standard deviation of 2 GW. Each MW of unmet demand costs $2.4 million a year in lost margin, net of the capacity's own cost; each idle MW costs $1.6 million a year.

(a) Compute the critical fractile and the capacity to reserve.

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(b) A take-or-pay power contract raises the cost of an idle MW to $2.4 million without changing the lost margin. Recompute the capacity.

(c) Return to the original costs. Higher interest rates add $0.8 million a year to the cost of financing every reserved MW, used or idle. Recompute the capacity.

14.10 **Vireo Semiconductors** reports trailing revenue of $120 billion, cost of goods sold of $36 billion, inventory of $13.5 billion, receivables of $18.0 billion, payables of $4.2 billion and latest quarterly revenue of $33.0 billion. Revenue grew 40 percent and inventory 90 percent. Before the boom DIO averaged 120 days, DSO 55 days and DPO/DIO 0.42. Warning thresholds are DIO above 140, DSO above 70 and inventory to quarterly revenue above 0.50. A signal is amber if it has moved more than 10 percent against its pre-boom average or if the growth spread is negative.

(a) Compute DIO, DSO, DPO and the cash conversion cycle (365-day year).

(b) Compute inventory to quarterly revenue, DPO/DIO and the growth spread.

(c) How many thresholds are breached, and which signals are amber?

(d) Vireo's notes show supply commitments of $54 billion, against $20 billion a year earlier. Compute commitments to trailing revenue at both dates (trailing revenue a year earlier was 120/1.40). Is the signal amber under a rule that flags a rise of more than 50 percent in a year?

FURTHER READING

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Micron Technology (2026). Fiscal fourth quarter 2026 earnings call prepared remarks, September 30, 2026.

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CHAPTER 15

Financing the AI Factory

Capex, Leases, Bonds, and Power for the AI Buildout

Financing architecture increases leverage and makes risks opaquer, while preserving financial flexibility, high equity multiples and credit ratings for the hyperscalers.

Stijn Van Nieuwerburgh, *Data Centers: Financing the AI Buildout* (March 2026)

On October 16, 2025, Beignet Investor LLC sold $27.3 billion of notes paying 6.581 percent and due in 2049. Beignet exists to hold the 80 percent stake that funds managed by Blue Owl Capital took in Hyperion, the campus Meta is building in Richland Parish, Louisiana, to train and run its artificial intelligence models. Meta develops the campus, will be its only tenant, keeps a fifth of it, and guarantees part of its value if it walks away in the first 16 years of operation. Within days the notes traded near 110 cents on the dollar. On October 7, 2026, the same notes changed hands near 91 cents. Meta's rating had not moved, and in July Quartz reported that Meta would expand the campus to 5 gigawatts.

What had changed? Hyperion is one deal in a wave in which five companies pour most of their operating cash flow, and in Oracle's case more than all of it, into buildings, chips and power. Who supplies the rest of the money, in what form, and where does each form show up on the books, or fail to?

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LEARNING OBJECTIVES

1. Compute capex intensity, the external funding need and economic capex with partner-funded capacity added back, and place a firm on the funding pecking order.
2. Estimate the installed cost of an AI data center and match the financing of each layer to its life.
3. Read hyperscaler bond terms and estimate what wider spreads and longer maturities cost.
4. Analyze an own-versus-lease joint venture and test a lease for hidden debt under the "reasonably certain" and "probable" thresholds of lease accounting.
5. Compare construction-loan terms by loan-to-cost and spread, and size a stabilized data center's securitized debt with coverage and loan-to-value tests.
6. Price a campus's power: capacity charges, tariff minimums, power purchase agreements and the premium for speed.

15.1 How Big Is the Bet?

Chapter 14 ended on an uncomfortable split: most of the profit in the AI supply chain goes to the chip makers, while the spending happens one layer down, in the buildings, servers and power that the hyperscalers buy. Can the hyperscalers pay for all of it out of their own cash? For most of the past decade they could: their cloud and advertising businesses threw off more operating cash than their data centers consumed.

Figure 15.1 shows how quickly that changed. On one consistent series, the five largest spenders (Amazon, Alphabet, Meta, Microsoft and Oracle) spent $90 billion on capital expenditures (capex) in 2020, $90/242 = 37$ percent of their $242 billion of operating cash flow (OCF). By 2025 they spent $415 billion, $415/578 = 72$ percent of OCF, and the March 2026 estimate for 2026 was $655 billion, or 90 percent. The ratio of capex to revenue or to OCF is called **capex intensity**. Technology firms historically ran capex at 5 to 15 percent of revenue. A lender watches the cash flow ratio more closely, because it shows how much of the firm's own cash the program consumes before a dollar is returned to shareholders.

A five-firm total hides how differently the firms stand. Table 15.1 uses each firm's own definitions for its latest fiscal year. Two things stand out. First, the jump: in a single year every firm's capex rose by 14 to 72 percentage points of OCF. Second, the spread: Alphabet still spent 55.5 percent of its operating cash on capex, while Oracle spent 174 percent, $1.74 for every dollar its operations produced. Summed across fiscal years ending between December 2025 and June 2026, the five spent $467.0 billion against OCF of $634.9 billion, $467.0/634.9 = 73.6$ percent, more than the $415 billion

CHAPTER 16

Securitizing Compute

Neocloud Debt, Vendor Finance, and Systemic Risk

The terms of such deals are typically poorly disclosed, with risks of the same asset being pledged multiple times.

Bank for International Settlements, *Annual Economic Report 2026*, Chapter I (June 2026)

In March 2026 CoreWeave, a specialist cloud that rents out NVIDIA GPUs, arranged an $8.5 billion delayed-draw loan at SOFR plus 2.25 percent, and Moody's rated the loan A3. In April it sold its own unsecured notes at 9.75 percent. In August it arranged $2.6 billion more at SOFR plus 5.50 percent. Same company, same kind of chips, three prices. A week after the August loan, NVIDIA disclosed residual value guarantees, capped at $105 billion, for the landlord of an OpenAI campus built to run NVIDIA hardware: if OpenAI stopped paying, NVIDIA would cover the gap between a guaranteed minimum value and what the landlord recovered by re-leasing or selling.

What is a lender lending against when it finances a chip that its maker plans to surpass within a year or two? What is that chip worth after the successor ships? And when the supplier of the chips stands behind its own customers, who holds the loss if the revenue arrives late?

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LEARNING OBJECTIVES

1. Explain the neocloud business model and identify whose credit stands behind a GPU loan.
2. Compute the cost and coverage of contract-backed GPU debt and of a GPU-backed securitization.
3. Value GPU collateral on its three lives, and stress-test a securitization at a successor launch.
4. Classify vendor support from extended terms to guarantees, and test an arrangement for circularity and wrong-way risk.
5. Trace a contagion path, and read the official sector's warnings against the telecom and Greensill precedents.
6. Apply a due-diligence routine to an AI infrastructure credit, and judge the AI tools now used inside supply chain finance.

16.1 The Neocloud Business Model

Chapter 15 followed the money into the buildings: the bonds, leases and joint ventures that pay for shells, cooling and power gear that last for decades. The chips inside are a different asset. A hall full of GPUs costs more than the hall itself, and the market expects a better chip within a year or two. Who finances the chips, and why has a new kind of company appeared to do it?

Often the answer is a *neocloud*, the specialist GPU cloud we met in Chapter 14. A neocloud buys GPUs, installs them in leased or owned halls, and rents their computing time to AI labs, hyperscalers and trading firms under contracts that run for years. In the language of Chapter 15, it is the own-versus-lease decision applied to chips. Microsoft or Meta could buy the GPUs itself; by renting them from a neocloud it can book a service contract (unless it controls identified GPUs, which the lease rules of Chapter 15 treat as a lease), while the neocloud carries the asset and the debt.

CoreWeave at scale

CoreWeave shows the model at scale. Its revenue backlog, the contracted revenue not yet earned, rose from $66.8 billion at the end of 2025 to about $104 billion at June 30, 2026, before more than $25 billion of new commitments signed early in the third quarter. Of its $103.7 billion of remaining performance obligations (contracted revenue as the accounts measure it, Chapter 15), 41 percent was due within 24 months. OpenAI had committed up to $22.4 billion, and Meta up to $14.2 billion plus about $21 billion more.[1]

Growth on this scale is bought with debt, and the income statement shows the price. In 2025 CoreWeave earned $5.131 billion of revenue, lost $1.167 billion, and booked $1.229 billion of net interest expense: 24.0 cents

1 Sources: CoreWeave, Form 10-K for 2025, Q2 2026 release and Form 10-Q, and release of April 9, 2026.

END OF SAMPLE

The Financial Logic of Supply Chains

Cash, Credit, and Risk Across Firms

An Introduction to Supply Chain Finance,
from Inventory and Invoices to AI Infrastructure

The full book

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- An Instructor's Solutions Manual with worked solutions to every problem and teaching guides for the course cases.

This sample

The front matter, the opening pages of every part and chapter, and Chapters 1 and 14 in full. Page numbers match the full book.

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